

A Sharper Explicit Bound on the Subtour-LP Integrality Gap for Metric TSP

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Abstract

Karlin, Klein, and Oveis Gharan introduced a randomized better-than-3/2 approximation algorithm for metric TSP [KKO21] and subsequently established the corresponding improvement in the integrality gap of the subtour-elimination LP [KKO22], with an explicit constant $\varepsilon > 1.00000 \cdot 10^{-36}$. Gurvits, Klein, and Leake subsequently improved the certified saving to $2.18000 \cdot 10^{-34}$ [GKL24]. By localizing the matching discount, sharpening the electrical, rank-probability, and conditional parity estimates, and mixing additive cutwise credits before combining them with separated repair costs in a layered slack construction, we obtain a randomized polynomial-time $(3/2 - \varepsilon)$ -approximation for every fixed $0 < \varepsilon < \varepsilon_*$, where $\varepsilon_* > 1.63305 \cdot 10^{-18}$, and consequently the subtour-elimination LP has integrality gap at most $3/2 - \varepsilon_*$. The classical worst-case integrality-gap lower bound is $4/3$ [Wil90].

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1 Introduction

The traveling salesperson problem is among the oldest and most extensively studied problems in combinatorial optimization; formulations of the routing question go back to the nineteenth century [ABCC07, Chapter 1]. The modern polyhedral theory began with the cutting-plane work of Dantzig, Fulkerson, and Johnson [DFJ54]. Held and Karp subsequently developed the linear-programming lower bound now called the subtour-elimination or Held–Karp relaxation [HK70]. Metric TSP remains APX-hard [KLS15]. In particular, it is NP-hard to approximate within $123/122$. Thus the central algorithmic question has long been how closely a polynomial-time algorithm can approach the optimum.

The classical $3/2$ -approximation was discovered independently by Christofides and Serdyukov in the 1970s [Chr76, Ser78]. Their method combines a minimum spanning tree with a minimum-cost parity correction on its odd-degree vertices. Wolsey later showed that the same $3/2$ guarantee holds relative to the subtour LP [Wol80], which also established the long-standing upper bound $3/2$ on its integrality gap. The widely studied $4/3$ conjecture predicts that the true integrality gap is $4/3$. A related conjecture of Schalekamp, Williamson, and Zuylen asserts that the largest gap is attained by a half-integral vertex of the subtour polytope [SWZ14]; this made half-integral instances a particularly important testing ground for progress below $3/2$.

Substantial improvements were obtained earlier for restricted metrics. Euclidean TSP admits polynomial-time approximation schemes due independently to Arora [Aro98] and Mitchell [Mit99]. For graph metrics, Oveis Gharan, Saberi, and Singh introduced a randomized rounding method based on maximum-entropy spanning trees and obtained the first $(3/2 - \epsilon_0)$ -approximation [GSS11]. Mömke and Svensson then gave a combinatorial 1.461-approximation [MS11]; Mucha improved the factor to $13/9$ [Muc12], and Sebő and Vygen reached $7/5$ [SV14]. This graphic-TSP line demonstrated that parity correction could be made cheaper than half the optimum, while the maximum-entropy approach suggested a possible route for general metrics.

For instances with a half-integral optimum subtour-LP solution, [KKO20] gives a 1.49993-approximation. For general metric TSP, [KKO21] proves the first randomized approximation with ratio $3/2 - \epsilon$ and gives the explicit scale $\epsilon > 1.00000 \cdot 10^{-36}$. The subtour-LP analysis in [KKO22] proves that its integrality gap is strictly below $3/2$ and also gives a better-than- $3/2$ result for the 2-edge-connected multi-subgraph problem. The conditional-expectation method in [KKO23] derandomizes the max-entropy algorithm.

Gurvits, Klein, and Leake later revisited the probabilistic part of the KKO analysis using new capacity bounds for real stable polynomials [GKL24]. Their estimates raised the controlling common-event probability from $2 \cdot 10^{-10}$ to $1.5 \cdot 10^{-9}$ [GKL24, Section 4.2] and yielded the explicit approximation constant $\epsilon = 2.18000 \cdot 10^{-34}$ [GKL24, Corollary 4.6]. This refinement preserved the max-entropy algorithm and the combinatorial framework while sharpening the quantitative probability bounds used in its analysis.

1.1 Our result

We state our result as follows.

Theorem 1.1 (Informal version of Theorem 4.1). *For some absolute constant $\epsilon > 1.63305 \cdot 10^{-18}$, there exists a randomized polynomial-time algorithm that, on every metric TSP instance, returns a tour whose expected cost is at most $(3/2 - \epsilon) \text{OPT}_{\text{LP}}$. Consequently, the integrality gap of the subtour-elimination LP is at most $3/2 - \epsilon$.*

Organization. Section 2 reviews the KKO and GKL frameworks and summarizes the refinements that yield our improved constant. Section 3 introduces the notation, records the max-entropy

Table 1: Upper and lower bounds on the integrality gap of the subtour LP for metric TSP. Part (a) compares the certified savings ϵ in the upper bound $3/2 - \epsilon$. Part (b) records the classical lower bound, approached by a family of graphic metric instances.

(a) Upper bounds: certified savings below $3/2$.

Year	Authors	Reference	Certified lower bound on ϵ
2022	Karlin, Klein, and Oveis Gharan	[KKO22]	$> 1.00000 \cdot 10^{-36}$
2024	Gurvits, Klein, and Leake	[GKL24]	$\geq 2.18000 \cdot 10^{-34}$
2026	This work	Theorem 1.1	$> 1.63305 \cdot 10^{-18}$

(b) Lower bound on the integrality gap.

Year	Authors	Reference	Integrality-gap lower bound
1990	Williamson	[Wil90, p. 35, Figure 2.3]	$4/3 \approx 1.33333$

reduction and its finite-precision guarantee, and gives the conversion from slack vectors to tours. Section 4 establishes the probability, payment, and parity-correction estimates, constructs the layered slack vector, and proves the main result of this paper.

2 Technique overview

Sections 2.1–2.3 review the KKO21 framework, explain GKL24’s probability refinement, and summarize how our retuned probability, payment, and repair estimates yield the improved constant.

2.1 The KKO21 framework

Step 1: Max-entropy tree and parity correction. Here KKO21 refers to the max-entropy framework of Karlin, Klein, and Oveis Gharan [KKO21]; we also use its subtour-LP extension [KKO22]. Start with an optimum subtour-LP solution x^0 and perform the standard vertex-splitting reduction, introducing a zero-cost edge $e_0 = \{u_0, v_0\}$ of value one. After deleting e_0 , the remaining vector x lies in the spanning-tree polytope. The algorithm samples a maximum-entropy spanning tree T with marginals x , lets O be the set of odd-degree vertices of T , and augments T by a minimum-cost O -join. The zero-cost edge e_0 is used only in the LP reduction and the fractional-join bookkeeping. Thus the tree has expected cost $c(x)$, and the problem is to construct a feasible fractional O -join of expected cost strictly below $c(x)/2$; see [KKO22, Sections 2 and 6].

Step 2: Structure of near-minimum cuts. After a small coordinatewise decrease from $x/2$, only cuts whose x -value is extremely close to two can violate an odd-cut constraint. KKO therefore focus on the η -near-minimum cuts. Crossing components of these cuts admit polygon representations, while the remaining relevant cuts can be organized into a laminar hierarchy. This approximate-uncrossing description is what prevents one edge from being charged by an unbounded number of near-minimum cuts. It also separates cuts crossed on both sides, cuts crossed on at most one side, and the degree, triangle, and outer-polygon cuts appearing in the hierarchy. The extension in [KKO22] replaces charging to edges of an optimum tour by repair sets extracted from the polygon representation. Consequently the entire comparison is against $c(x)$, which is the key step needed for an integrality-gap bound rather than only an approximation guarantee.

Step 3: Happiness events and payment. The quantitative heart of the method is a random signed slack vector. A hierarchy edge bundle is declared *good* when one of several local tree events

has a uniform positive probability. These events say, for example, that prescribed boundary sets contain exactly one or two tree edges; they are called happiness events. On a happiness event, KKO decrease the slack of the corresponding bundle. A max-flow matching distributes the burden of an upward boundary among good horizontal bundles, and the payment theorem proves that every good edge receives a negative expected contribution. The common probability p is the minimum of six local happiness estimates, so the payment coefficient ϵ_P is proportional to p .

Step 4: Feasibility repair and parameter balance. Negative slack cannot be used without repairing the odd cuts on which it may create a deficit. KKO construct nonnegative repair vectors for cuts crossed on both sides and for the one-sided polygon configurations. Their support is chosen so that each relevant odd cut receives enough positive slack and every edge has bounded expected repair. In the notation of [KKO22, Theorem 6.1], the signed vector contributes a decrease of order $\epsilon_P \beta x_e$, whereas the repair costs order $\eta \beta x_e$. Choosing η proportional to ϵ_P makes the net saving quadratic in ϵ_P , hence quadratic in p . This architecture—max-entropy tree, near-minimum-cut hierarchy, happiness events, matching and payment, and bounded-congestion repair—is the part retained by all subsequent work.

2.2 The GKL24 refinement

Steps 1, 2, and 4 in Section 2.1 are reused directly. GKL use the same max-entropy spanning-tree algorithm and parity correction from Step 1, the same hierarchy and polygon structure from Step 2, and the same repair vectors and parameter balancing from Step 4. Within Step 3 they also retain the definitions of good bundles and happiness events, as well as the KKO matching and payment arguments. Thus GKL do not change the sampled tree, the combinatorial architecture, or the final O -join construction.

Only the probability certification in Step 3 is modified. A maximum-entropy spanning-tree distribution is strongly Rayleigh. Therefore its multivariate generating polynomial is real stable. If $A_1, \dots, A_m$ count tree edges in disjoint edge sets, then the probability of an exact pattern $A_i = \kappa_i$ is a coefficient of this polynomial, while the expectations of the A_i are entries of its gradient at the all-ones vector. GKL prove new capacity bounds that convert control of these expectations, including the expectations of partial sums, into a lower bound for the desired coefficient [GKL24, Theorem 4.1 and Corollary 4.2]. This replaces several case-specific Bernoulli-sum estimates in KKO by one multivariate stable-polynomial inequality. For the harder configurations, GKL first use capacity to reduce to a structured special case and then combine it with the original log-concavity and stochastic-dominance arguments.

How the modified probability argument works. The contrast with the KKO probability lemma is especially useful here. That lemma turns lower-tail and upper-tail estimates for every partial sum into a joint exact-count estimate, but its dependence on the tail parameter is doubly exponential in the number of counted sets. KKO observed that suitable expectation bounds imply the required tail estimates, yet the resulting generic constant was too small for their final analysis, so they proved the important happiness estimates by ad hoc arguments. GKL instead show, under the slightly stronger condition that every partial-sum expectation stays a fixed distance from the neighboring integer, that the exact-count probability has a simply exponential lower bound, with the correct dependence on that distance [GKL24, Theorems 2.5 and 2.6]. Their proof productizes a stable polynomial into affine factors, reduces the associated matrices to extreme points supported on bipartitioned forests, and inducts on forest leaves [GKL24, Section 3]. Removing the dependence on the total degree is what makes this capacity bound effective for the TSP events.

Quantitative effect of the modification. For events 2–5, the KKO coefficients of $\epsilon_{1/2}^2$ are 0.005, 0.006, 0.005, and 0.020, respectively. GKL replace them by 0.039, 0.038, 0.0498, and 0.0485

[GKL24, Section 4.2]; the first event is unchanged. At $\epsilon_{1/2} = 0.0002$, the KKO and GKL lower bounds for the first five events are, respectively,

	1	2	3	4	5	
KKO	1.50	0.20	0.24	0.20	0.80	times 10^{-9} .
GKL	1.50	1.56	1.52	≥ 1.90	≥ 1.90	

The sixth event is far from the bottleneck; GKL note that its threshold changes slightly when p is raised and record that it remains above the new common value. Thus the minimum rises from $p_{\text{KKO}} = 2 \cdot 10^{-10}$ to $p_{\text{GKL}} = 1.5 \cdot 10^{-9}$, an improvement by a factor of 7.5 [GKL24, Section 4.2].

How the modification improves ϵ . Nothing after this substitution requires a new TSP construction. The KKO payment theorem still makes ϵ_P linear in p , and the final choice of η still makes the approximation saving quadratic in p . Consequently, the 7.5-fold improvement in p becomes a factor of $7.5^2 = 56.25$: the same parameterized formula changes from $3.88 \cdot 10^{-36}$ to $2.1825 \cdot 10^{-34}$. GKL package this dependence as $9.7p^2 \cdot 10^{-17}$ and report $2.18 \cdot 10^{-34}$ [GKL24, Lemma 4.5 and Corollary 4.6]. Comparing the displayed theorem guarantees $1.00 \cdot 10^{-36}$ and $2.18 \cdot 10^{-34}$ gives a factor of $218 > 10^2$ [KKO21, Theorem 1.1]. This ratio is larger than 56.25 because the KKO theorem states its final constant conservatively: against the sharper pre-GKL bound $4.11 \cdot 10^{-36}$ quoted by GKL, the ratio is about 53 [GKL24, Sections 2 and 4.2]. In short, KKO supply the combinatorial mechanism that turns local happy events into a cheap O -join, whereas GKL supply a stronger real-stable-polynomial method for proving that those same events occur often enough.

2.3 Summary of our approach

Steps 1 and 2 retain the structural framework of Section 2.1. We use the same subtour-LP reduction, max-entropy spanning-tree algorithm, hierarchy and polygon representation of near-minimum cuts, and conversion of a feasible slack vector into a cheap tour. The new work strengthens the quantitative slack certificate under the exact tree law. The finite-precision comparison is applied only after all layers have been combined, to the actual tree-plus-minimum-join cost.

Section 4 uses four parameter choices. The probability estimates, localized matching, and conditional parity bound feed the same ancestor and payment argument for each row. The separated repairs and aggregate-credit lemma then give a centered slack vector, and one finite-layer lemma converts its threshold-dependent guarantees into all-cut feasibility. The hierarchy domain and all scalar conditions are checked for each complete tuple.

Step 3 strengthens the local probability and parity estimates. For a fixed row, a half bundle is good at conditional probability $\gamma_{\text{good}}h$, with that row's parameters. This class is not assumed to be contained in the source good-half class. An electrical-energy argument bounds the smaller remote-vertex mean increase by 0.22 throughout $h \leq 0.0055$. The competing-half calculation then gives probability greater than 0.01330 for one of two incident half bundles, so bad half bundles form a matching. Two Newton inequalities and a concave rank-balance expression show that upward mass at least $1/2 + k_{\text{good}}h$ forces goodness (Lemmas 4.10 and 4.17).

For the window event, we condition on the bundle being selected before estimating its residual lower tail. A pointwise inequality in the binary covering coupling gives

$$L_* := \frac{(\gamma_{\text{good}} - 2)h - 12d_0}{3/2 - h - 3d_0} > 0.00261653801701028.$$

Factoring the parallel-edge choice and applying full-rank monotonicity to deletion preserve this total-rank lower tail. The distinct window-premise subcount still retains its additive deletion

allowance. The conditional rank-three atom exceeds $c_3 = 0.248$, giving central mass $m_* = L_*/(1+L_*/(2c_3))$. Three rank ranges, followed by deletion of one degree block and a quadratic split, prove the oriented window estimate. The direct mixed event retains its actual rank-two probability in $q\psi(b_D - q^2/(2c_3))$, rather than using a fixed split throughout, and restores the independent endpoint-selection factor 1.96ω . The low-mean CDF domain extends through 1.512.

For the non-half events, full projected-rank conditioning changes each subunion mean by its rank-excess budget. In the small case, a sharper excess-tail estimate and closed-interval concavity bounds handle small rank-three probability; the complementary range uses up/down tails. In the large case, proper-cut support and Newton bounds control the rank-three means, and a homogeneous cubic split gives the stronger event probability. All dimension reductions and shared interval endpoints are explicit. The paired proof keeps separate lower and upper excess moments. Nested-event mean comparisons and an asymmetric binary separator sharpen both branches. For upper subset means, deleting the residual block before the final binary selection gives the same final law with stronger bounds. Both complete subset tables are retained. Proper-cut exclusions give coefficient factors $e^{-2}/2$ and $e^{-1}/2$ in the low and high branches, respectively.

For polygons, a small-excess joint-atom estimate and a complete max-flow cut proof preserve both separate marginal budgets. Each row's top and polygon rates are p_j and $P_j = s_j p_j$. Independent thinning preserves the conditional laws; paired reductions share one coin. The crossing cutoff ω_j is separate from the half-bandwidth h_j . Event classification is repeated for each complete parameter tuple in Definition 4.2.

At a bad endpoint, both bundle states contribute to the same badness budget. An outward-rounded bootstrap and closed-interval certificate bound its covering mean difference by the row's $\Delta_{*,j}$. For a polygon child, its boundary count reduces outside an explicit exceptional event to a Bernoulli sum of degree at most two, independent of the internal child-tree count. The overlap thresholds 0.0025 and 0.25 give stronger bottom savings. Neither argument divides by the rare polygon-event probability.

The ancestor proof retains actual bottom savings, including terminal root-bad edges and all large-, small-, and fractional-mass cases. Only a fractional atom incident to an internal bad bundle receives the matching discount. The top and bottom credits are $a_j, a_{\text{bot},j}$. The two reduction rates remain separate. One-sided and two-sided repairs retain coefficients 21 and 5, and the one-sided repair is supported on bottom edges.

Step 4 mixes threshold-dependent cutwise credit. Fix a hierarchy at threshold u before choosing a parameter row. In row j , put $W_j = s_j^{\text{pay}} + r_j^{(1)}$ and retain the edgewise credit $m_j(u) = \min\{a_j, a_{\text{bot},j} - R_1(u)u\} > 0$. At a child incident to a bad bundle, the horizontal credit is the sum of a baseline term $a_j^{(4)}I$ and a nonnegative term proportional to the upward bad mass; neither term is discarded. Together with the upward good-edge credit, this gives the degree-parent bound $A_j(u)$ in Lemma 4.48. The remaining one-sided cuts retain bottom credit $B_j(u) = (1 - d_0)(a_{\text{bot},j} - R_1(u)u)$.

Mix the row witnesses with weights w_j . Their good sets may differ, but the two cut classes agree because the hierarchy is fixed. Hence the guaranteed cut credit is

$$M_w := \min\left\{\sum_j w_j A_j(u), \sum_j w_j B_j(u)\right\}, \quad \bar{m} := \sum_j w_j m_j(u).$$

The minimum is taken after mixing. Define

$$C_w := \frac{M_w}{(1 - \bar{m})(2 + u) + M_w}, \quad \lambda := \frac{2 + u}{(1 - \bar{m})(2 + u) + M_w}.$$

For $W = \sum_j w_j W_j$, the centered vector $\lambda(W - \mathbb{E}[W]) - C_w \beta x$ preserves the coordinate and one-sided cut bounds: $\lambda(1 - \bar{m}) + C_w = 1$ and $\lambda M_w = C_w(2 + u)$. Adding one two-sided repair leaves

saving $C_w - R_2(u)u$.

Use 50000 equally spaced thresholds up to $H = 10^{-9}$, all on the same sampled tree. An explicit five-interval schedule uses either one row or two rows whose mixed cut credits agree. Coordinate losses telescope, and cut excess pays for layers below that cut's threshold. The rational ledger and finite-layer argument give

$$B_* > 1.6330510103142689892774 \cdot 10^{-18}.$$

The definition and finite-sum proof are in Lemma 4.51. A one-time implementation loss 10^{-28} still leaves $\varepsilon > 1.63305 \cdot 10^{-18}$. The centered expectations and auxiliary layers are analysis witnesses, not computations required of the algorithm.

3 Preliminaries

Section 3.1 introduces the notation for the subtour-LP solution, cuts, the KKO hierarchy, and the exact and approximate max-entropy distributions. Section 3.2 records the standard reduction to the spanning-tree polytope and the finite-precision max-entropy guarantee. Finally, Section 3.3 bounds the tour-cost loss when passing to the approximate tree law.

3.1 Notation

Throughout the paper, references to numbered statements in [KKO21], [KKO22], and [GKL24] use, respectively, the full versions arXiv:2007.01409v6, arXiv:2105.10043v3, and arXiv:2311.09072v2.

Let x^0 denote a feasible subtour-LP solution after adjoining the zero-cost edge $e_0 = \{u_0, v_0\}$, and set $E := \text{supp}(x^0) \setminus \{e_0\}$. Let x be the restriction of x^0 to E , let $G = (V, E)$, and write $n := |V|$. We use OPT_{LP} for the optimum subtour-LP value.

For a vector $z \in \mathbb{R}^E$ and an edge set $F \subseteq E$, write $z(F) := \sum_{e \in F} z_e$ and $\text{supp}(z) := \{e \in E : z_e > 0\}$. For a nonnegative edge-cost vector c , write $c(z) := \sum_{e \in E} c_e z_e$.

For a vertex set $S \subseteq V$, let $E(S) := \{\{u, v\} \in E : u, v \in S\}$ and $\delta(S) := \{\{u, v\} \in E : |\{u, v\} \cap S| = 1\}$. The notation $e_0 \in \delta(S)$ is shorthand for $|\{u_0, v_0\} \cap S| = 1$ in the augmented graph; sums over $\delta(S)$ remain over E unless the e_0 coordinate is explicitly included. If $T \subseteq E$ is a spanning tree and $F \subseteq E$, write $F_T := |F \cap T|$. In particular, $\delta(S)_T$ is the number of tree edges crossing the cut S .

When invoking the KKO hierarchy and polygon classification, we identify u_0 and v_0 . Thus every classified cut satisfies $e_0 \notin \delta(S)$; cuts with $e_0 \in \delta(S)$ are handled separately by setting $y_{e_0} = 1$. A cut S with $e_0 \notin \delta(S)$ is η -near-minimum if $x(\delta(S)) < 2 + \eta$.

We use the hierarchy and polygon terminology of [KKO22, Appendix B]. The construction in the proof of [KKO22, Theorem B.3], together with [KKO22, Theorem 6.2 and Fact B.4], produces for every x and $\eta \leq 10^{-12}$ a valid hierarchy whose root is $V \setminus \{u_0, v_0\}$, whose one-sided and two-sided classes exhaust the relevant η -near-minimum cuts, and whose added hierarchy cuts are 7η -near-minimum. In the payment estimates we use the conservative uniform envelope $d := 14\eta$, consistent with [KKO21, Fact 4.34]; the one-sided repair estimate retains the sharper 7η bound. For a non-root hierarchy cut S , write $\mathbf{p}(S)$ for its parent, $\delta^\uparrow(S) := \delta(S) \cap \delta(\mathbf{p}(S))$, and $\delta^\rightarrow(S) := \delta(S) \setminus \delta(\mathbf{p}(S))$. The children of a hierarchy cut are called its atoms. For disjoint atoms u, v , the set $E(u, v)$ is the edge bundle with one endpoint in each. Hierarchy cuts are classified as degree cuts or near-cycle cuts. Following [KKO22, Definition B.1], a *triangle cut* is a hierarchy cut with exactly two children. Separately, the three-atom degree configuration in [KKO21, Case 1 in the proof of Lemma 7.6] will always be called a *three-atom degree configuration*. The KKO polygon classification partitions the

relevant near-minimum cuts into those crossed on both sides and those crossed on at most one side. When applying the KKO payment theorem, we follow [KKO21, Definition 4.31] and regard a triangle cut as a degenerate polygon cut with $C = \emptyset$.

Whenever a KKO hierarchy is used, write d for the uniform error envelope just described. The one-sided repair theorem retains its sharper error 7η . For the final parameter choice, Lemma 4.42 extends this hierarchy interface to $\eta \leq 1.4 \cdot 10^{-9}$ directly from the same construction and its unrounded geometric bounds. As in [KKO21, Definition 5.13], edges incident to u_0 or v_0 are declared bad and carry zero payment; they are neither top nor bottom edges.

We use μ for the exact max-entropy distribution on spanning trees with marginals x , and μ_λ for a computable λ -uniform approximation. Unless a subscript is displayed, probabilities and expectations are taken over the current random spanning tree and any auxiliary randomness in the slack vectors.

3.2 Setup and max-entropy approximation

We first record the standard reduction that places the subtour-LP solution in the spanning-tree polytope, together with the finite-precision max-entropy guarantee.

Lemma 3.1 (Subtour-LP reduction and max-entropy approximation). *An optimal extreme point x^0 may be written in the form above so that x belongs to the spanning-tree polytope and*

$$c(x) = \text{OPT}_{\text{LP}}.$$

Every positive coordinate of x^0 is at least $1/n!$. For every rational $\delta > 0$, one can compute, in time polynomial in n and $\log(1/\delta)$, a λ -uniform spanning-tree distribution μ_λ satisfying

$$\Pr_{T \sim \mu_\lambda} [e \in T] \leq (1 + \delta)x_e \quad \text{for every } e \in E.$$

Moreover, there is an explicitly computable universal constant $C_{\text{stab}} > 0$, independent of the instance, such that, for $0 < \delta \leq 1$, the exact and approximate distributions satisfy

$$q := \|\mu - \mu_\lambda\|_1 \leq C_{\text{stab}} n^3 \sqrt{\delta(1 + \log(n/\delta))}.$$

Proof. The reduction and the spanning-tree-polytope assertion are from [KKO22, Section 2.1 and Fact 2.2]. The approximation algorithm is [KKO22, Theorem 2.1], and the coordinate bound is recorded in [KKO22, Section 6.2]. Let x^λ be the marginal vector of μ_λ , and define $E_+ := \{e : \lambda_e > 0\}$. Since $x_e^\lambda = 0$ for $e \notin E_+$, every spanning-tree law with marginals x^λ is supported on trees contained in E_+ . On this support,

$$\log \mu_\lambda(T) = \sum_{e \in T} \log \lambda_e - \log Z_\lambda.$$

Thus, for every spanning-tree law ν with marginals x^λ ,

$$0 \leq D(\nu \| \mu_\lambda) = H(\mu_\lambda) - H(\nu),$$

because ν and μ_λ have the same expectation of the preceding display. Hence μ_λ is the maximum-entropy law with marginals x^λ , also when some weights vanish. Since $x_e^\lambda \leq (1 + \delta)x_e$ and both marginal vectors have coordinate sum $n - 1$,

$$\|x - x^\lambda\|_1 \leq 2(n - 1)\delta.$$

Apply the quantitative form [SV19, Theorem 10] to the spanning-tree polytope. Its ambient dimension is at most n^2 , its unary facet complexity is one, the diameter of its vertices is smaller than $\sqrt{2n}$, and the number of spanning trees is at most n^{n-2} . Substitution in that theorem gives

$$\|\mu - \mu_\lambda\|_1 \leq C_{\text{stab}} n^3 \sqrt{\delta(1 + \log(n/\delta))}$$

after increasing one universal constant. The proof of the cited theorem is effective, so we fix any computable value supplied by that proof; no optimization of this value is needed here. $\square$

3.3 Change of tree law

We record the output-cost comparison used after the slack construction.

Lemma 3.2 (Change of tree law). *Let $J(T)$ denote the cost of a minimum join for the odd vertices of T . For the law μ_λ in Lemma 3.1,*

$$\mathbb{E}_{\mu_\lambda}[c(T) + J(T)] \leq \mathbb{E}_\mu[c(T) + J(T)] + (\delta + \frac{q}{2})c(x).$$

Proof. For every spanning tree T , the vector that equals $x/2$ on E and one on the zero-cost edge e_0 is a feasible fractional join for the odd vertices of T . Therefore

$$0 \leq J(T) \leq c(x)/2.$$

Moreover,

$$\mathbb{E}_{\mu_\lambda}[c(T)] = c(x^\lambda) \leq (1 + \delta)c(x), \quad \mathbb{E}_\mu[c(T)] = c(x).$$

Using $q = \|\mu - \mu_\lambda\|_1$ gives

$$\mathbb{E}_{\mu_\lambda}[J(T)] - \mathbb{E}_\mu[J(T)] \leq \sum_T |\mu_\lambda(T) - \mu(T)| J(T) \leq \frac{q}{2}c(x).$$

Adding the tree- and join-cost comparisons proves the claim. $\square$

4 Proof of upper bound

We mix four parameter choices on a finite threshold grid. Section 4.1 derives the main theorem from the layered slack vector. Sections 4.2–4.4 establish the probability estimates and localized matching. Section 4.5 proves the conditional parity, ancestor, and payment bounds, and Section 4.6 constructs the separated repairs. Section 4.7 combines them using aggregate expected credit. Finally, Section 4.8 proves the finite-layer lemma and evaluates the resulting certificate.

4.1 Proof of the main theorem

The layered slack-vector construction supplies the improvement under the exact max-entropy law. The finite-precision comparison then transfers this bound to the polynomial-time implementation and yields the main result.

Theorem 4.1 (Formal version of Theorem 1.1). *There exists an explicitly computable constant $\varepsilon_\star > 1.63305 \cdot 10^{-18}$ such that, for every fixed $0 < \varepsilon < \varepsilon_\star$ and every metric TSP instance, there is a randomized polynomial-time implementation of the max-entropy algorithm whose expected tour cost is at most $(\frac{3}{2} - \varepsilon) \text{OPT}_{\text{LP}}$. Consequently, the integrality gap of the subtour-elimination LP is at most $3/2 - \varepsilon_\star$.*

Proof. Solve the subtour LP and choose an optimal extreme point x^0 . Apply Lemma 3.1, and let x be the resulting vector in the spanning-tree polytope. Then

$$c(x) = \text{OPT}_{\text{LP}}.$$

Apply Lemma 4.51 and define $\varepsilon_\star := B_\star > 1.63305 \cdot 10^{-18}$, where $N = 50000$. The construction uses four certified parameter choices and couples all finitely many analysis layers to the same sampled tree. Each layer has a uniform edgewise certificate; the rows mixed within it share one geometric hierarchy, but their good-edge sets may differ. Different layers need not use the same hierarchy. For the sampled tree T , define $y_e := x_e/2 + Z_e$ for $e \in E$ and $y_{e_0} := 1$. Since $\beta(H) < 1/2$, every coordinate of y is nonnegative. A cut crossed by e_0 has y -value at least one, and every other cut containing an odd number of tree edges has y -value at least one by the same lemma. Thus y is a feasible fractional join for the odd vertices of T . The minimum join is no more expensive than y , so under the exact max-entropy law the expected tour cost is at most

$$c(x) + \frac{c(x)}{2} + \mathbb{E}[c(Z)] \leq \left(\frac{3}{2} - \varepsilon_\star\right)c(x).$$

It remains to control the computable approximation. Fix $0 < \varepsilon < \varepsilon_\star$ and define

$$\rho := \frac{\varepsilon_\star - \varepsilon}{2}.$$

Define

$$B := \min\left\{\frac{1}{n^2}, \frac{\rho}{2}, \left(\frac{\rho}{\sqrt{2}C_{\text{stab}}n^3}\right)^4\right\}$$

and choose the largest number of the form 2^{-m} , $m \in \mathbb{N}$, satisfying $\delta \leq B$. Then $B/2 < \delta \leq B$. Since $\delta \leq n^{-2}$ and $1 + \log n \leq n$,

$$1 + \log(n/\delta) \leq n + \log(1/\delta) \leq 2\delta^{-1/2}.$$

Lemma 3.1 therefore gives

$$q \leq \sqrt{2}C_{\text{stab}}n^3\delta^{1/4} \leq \rho, \quad q/2 + \delta \leq \rho.$$

The running time is polynomial because

$$\log(1/\delta) \leq \log(2/B) = O(\log n + \log(1/\rho)).$$

The comparison in Lemma 3.2 is applied once to the output-cost function $T \mapsto c(T) + J(T)$; the layers incur no separate change-of-law loss. Combining that lemma with the exact-law estimate above gives

$$\mathbb{E}_{\mu_\lambda}[c(T) + J(T)] \leq \left(\frac{3}{2} - \varepsilon_\star + \delta + \frac{q}{2}\right)c(x) \leq \left(\frac{3}{2} - \varepsilon_\star + \rho\right)c(x) < \left(\frac{3}{2} - \varepsilon\right)\text{OPT}_{\text{LP}}.$$

The integrality-gap conclusion follows because the expectation is an average over tours. Taking $\varepsilon \uparrow \varepsilon_\star$ proves the final claim. $\square$

4.2 Parameters and probability estimates

Definition 4.2 (Parameters for the final bound). Set $d_0 = 2 \cdot 10^{-8}$, $\epsilon_F = 0.1$, $t = 1$, $H = 10^{-9}$, and $N = 50000$. At each threshold $0 < u \leq H$, fix the geometric hierarchy before choosing any of the four complete parameter tuples below. In all local estimates fix one row and suppress its index. The final construction mixes the resulting payment vectors on this common hierarchy.

j	h_j	r_j	ω_j
1	0.00440137306	0.00193221105	0.00898297213
2	0.00433055899	0.00170123668	0.00926069877
3	0.00426797639	0.00149437120	0.00950875668
4	0.00423916601	0.00139842279	0.00962360603

j	$\gamma_{\text{good},j}$	$k_{\text{good},j}$	K_j	p_j
1	2.88916128109	5.80807677086	4.10703749964	0.00000832591
2	2.94495206496	5.92048255827	4.17148289388	0.00000897919
3	2.99580065381	6.02293349730	4.23004098925	0.00000957931
4	3.01971304430	6.07111352668	4.25752872334	0.00000986265

j	s_j	$\epsilon_{M,j}$	$s_{0,j}$	$\Delta_{*,j}$
1	1.75752770887	0.00304011064	0.00194924275	0.06276788700
2	1.75791072163	0.00315753889	0.00171703825	0.06291283589
3	1.75825415794	0.00326173286	0.00150888781	0.06304102775
4	1.75841365724	0.00330979901	0.00141228468	0.06310004120

Define $P = sp$ and retain

$$\begin{aligned}
\epsilon_B &= \frac{2((k_{\text{good}} + 1)h + 2d_0h)}{1 + 2k_{\text{good}}h} + 10^{-11}, & \vartheta &= \frac{(1-r)(1-2r-2d_0)}{1+d_0} - 10^{-9}, \\
\chi &= \vartheta r, & \sigma &= \chi + 2\chi^2 + d_0, \\
\rho_{\text{sm}} &= 1 - \frac{\sigma + 5d_0 + 2d_0\sigma + 6d_0^2}{1 - \sigma - 2d_0}, & \zeta &= \vartheta\rho_{\text{sm}} - 2d_0/r - 10^{-9}, \\
a &= \zeta rp, & a_{\text{bot}} &= p(s - (1+d_0)(7/4 + h/2 + 3d_0/2) - 2d_0s).
\end{aligned}$$

Also set

$$T_{18} = \sum_{i=0}^{18} \frac{(2(1 - \epsilon_M/2 - 3d_0))^i}{i!}, \quad q_0 = (1 + 1/T_{18})/2 + 10^{-12}, \quad \bar{q} = \epsilon_M/2, \quad q_{\text{unh}} = q_0 + 4d_0/(1 - \bar{q}).$$

Each half-bundle class is defined directly by conditional 2–2 probability at least $\gamma_{\text{good},j}h_j$. The first three $\gamma_{\text{good},j}$ are below three and the fourth is above three. No inclusion in a source good-half class is used. The crossing cutoff is ω , not h ; it does not replace h in the half-bandwidth or the full-pair saving. All events are reclassified at the row's own p . Put $w_0 := s_0(1 - s_0)$. All terminating decimals in the parameter tables are exact rationals. The local error satisfies $0 \leq d \leq d_0$, and the crossing cutoff has the separate domain $h \leq \omega \leq 0.01$.

All probability statements refer to the exact max-entropy tree law. The current hierarchy has error $0 \leq d \leq d_0$. For an arbitrary payment amplitude $\beta > 0$, put $\tau := t\beta = \beta$. Top events are thinned to probability p and polygon events to P . Whenever $x(\delta^\uparrow(S)) = 0$, set the degree-cut

increase $I_{e,S} := 0$. Matching conservation then forces the corresponding $m_{e,S} = 0$, so no quotient by zero occurs.

Scalar comparisons are verified over the rationals. For $x \geq 0$ we use

$$\sum_{j=0}^n \frac{x^j}{j!} \leq e^x \leq \sum_{j=0}^n \frac{x^j}{j!} + \frac{x^{n+1}}{(n+1)!} \frac{1}{1 - x/(n+2)}, \quad x < n+2,$$

with $n = 40$, or the shorter Taylor certificates specified below. The arithmetic rules and finite evaluations are detailed in Section 4.9.

Lemma 4.3 (Bernoulli atom and tail estimates). *For a Poisson-binomial variable Z , the following bounds hold in every finite dimension:*

<i>hypothesis</i>	<i>conclusion</i>
$\mathbb{E}[Z] \leq 1.01$	$\Pr[Z \leq 1] > 0.70$
$\mathbb{E}[Z] \leq 1.5$	$\Pr[Z \leq 1] \geq 7/16$
$\mathbb{E}[Z] \leq 1.504$	$\Pr[Z \leq 1] > 0.434$
$\mathbb{E}[Z] \leq 1.512$	$\Pr[Z \leq 1] > 0.4284$
$\mathbb{E}[Z] \leq 1.53$	$\Pr[Z \leq 1] > 0.4147$
$\mathbb{E}[Z] \leq 2.01$	$\Pr[Z \leq 2] > 0.66$
$\mathbb{E}[Z] \leq 2.5$	$\Pr[Z \leq 2] > 0.42$
$0.99 \leq \mathbb{E}[Z] \leq 1.512$	$\Pr[Z = 1] > 0.33$
$1 \leq \mathbb{E}[Z] \leq 1.2$	$\Pr[Z = 1] > 0.36$
$1.2 \leq \mathbb{E}[Z] \leq 2.5$	$\Pr[Z = 2] > 0.1637$
$1.49 \leq \mathbb{E}[Z] \leq 2.02$	$\Pr[Z = 2] > 1/4$
$1.47 \leq \mathbb{E}[Z] \leq 2.02$	$\Pr[Z = 2] > c_3 := 0.248.$

For $0 < \delta \leq 0.02$, one also has

$$0.99 \leq \mathbb{E}[Z] \leq 2 - \delta \implies \Pr[Z = 1] \geq \delta(1 - \delta/2), \quad (1)$$

$$\mathbb{E}[Z] \leq 3 - \delta \implies \Pr[Z \leq 2] \geq \delta(1 - \delta/3). \quad (2)$$

Moreover, $\Pr[Z = 1] \geq \mu e^{-\mu}$ for $0 \leq \mu := \mathbb{E}[Z] \leq 1$. For the same δ , if $\mathbb{E}[Z] \geq 1 + \delta$, then

$$\Pr[Z \geq 2] \geq \delta e^{-\delta} \geq \delta c(\delta), \quad c(x) := 1 - x + x^2/2 - x^3/6.$$

Proof. At fixed mean and dimension, choose a minimizer of the desired atom or tail with the fewest fractional Bernoulli parameters. For two parameters of fixed sum, the objective is affine in their product. A positive coefficient forces an endpoint, a zero coefficient permits an endpoint, and a negative coefficient forces equality. Thus all remaining fractional parameters are equal: a minimizer is a forced integer plus a binomial, as in [GKL24, Theorem A.9]. We use the inequality

$$\log(1 - z) \geq -z - \frac{z^2}{2(1 - z)}, \quad 0 \leq z < 1.$$

CDFs at one. Monotonicity permits the upper mean endpoint M . One forced success gives $2 - M$; an unforced binomial with $n \geq 2$ gives

$$F_n(M) = (1 - M/n)^{n-1} (1 + M - M/n).$$

The logarithmic inequality gives $(n-1)\log(1-M/n) \geq -M$ when $M \leq 2n/(n+1)$. The following finite checks and dimension-uniform lower bounds suffice:

M	dimensions checked exactly	bound in all larger dimensions
1.01	$2 \leq n \leq 20$	$e^{-M}(1 + 20M/21) > 0.70$
1.5	$n = 2, 3$	$e^{-M}(1 + 3M/4) > 7/16$
1.504	$n = 2, 3$	$e^{-M}(1 + 3M/4) > 0.434$
1.512	$n = 2, 3$	$e^{-M}(1 + 3M/4) > 0.4284$
1.53	$n = 2, 3$	$e^{-M}(1 + 3M/4) > 0.4147$

Dimensions at most one are immediate; equality $7/16$ occurs for $M = 1.5, n = 2$.

CDFs at two. Two forced successes give $3-M$, and one forced success reduces to the preceding CDF at mean $M-1$. In the unforced case, for $M \leq 4n/(n+2)$ the logarithmic inequality gives

$$\Pr[\text{Bin}(n, M/n) \leq 2] \geq e^{-M}(1 + M(1 - 2/n) + M^2(1/2 - 3/(2n) + 1/n^2)).$$

The parenthesized expression increases with n on the required ranges. At $M = 2.01$, check $3 \leq n \leq 99$ and use the displayed bound at $n = 100$ thereafter. At $M = 2.5$, check $3 \leq n \leq 9$ and use the bound at $n = 10$ thereafter. These exceed 0.66 and 0.42 , respectively. Dimensions at most two have CDF one.

Atoms at one. A forced success gives $2 - \mu$. Without one, the logarithmic bound gives

$$\Pr[\text{Bin}(n, \mu/n) = 1] \geq \mu e^{-D(n, \mu)}, \quad D(n, \mu) := \frac{\mu(n-1)(2n-\mu)}{2n(n-\mu)}.$$

For $\mu \in [0.99, 1.512]$, check $n = 2, 3$ at the mean endpoints. For every $n \geq 4$, $\mu < 2n/(n+1)$ gives $D(n, \mu) \leq \mu$; both $0.99e^{-0.99}$ and $1.512e^{-1.512}$ exceed 0.33 . On $[1, 1.2]$ this logarithmic condition holds already for $n \geq 2$, and $1.2e^{-1.2} > 0.36$. The same argument gives $\Pr[Z = 1] \geq \mu e^{-\mu}$ on $[0, 1]$. The one-variable and integer endpoints follow directly.

Atoms at two. For $\mu \in [1.2, 2.5]$, two forced successes leave a zero atom at least $1/2$. One forced success leaves an atom at one with residual mean in $[0.2, 1.5]$. Check one and two fractional variables directly; for $n \geq 3$, the logarithmic bound gives

$$\min\{0.2e^{-0.2}, 1.5e^{-1.5}\} > 0.1637.$$

Without a forced success, check $n = 2, 3, 4$ at the admissible mean endpoints. The atom is unimodal in its mean. For $n \geq 5$, $\mu < 4n/(n+2)$ gives

$$\Pr[Z = 2] \geq 0.4\mu^2 e^{-\mu} > 0.1637.$$

For the sharper target-two bounds below mean two, the zero- and one-forced-success branches of [KKO21, Lemma 2.21] are at least

$$1.49^2 e^{-1.49}/2 > 1/4, \quad 0.49e^{-0.49} > 1/4.$$

Above mean two, the correction exponent is at most 0.02 . For zero or one forced success its base is at least $1/4$. Since $0.97^{50} < 1/4$, the correction exceeds 0.97 . The two branches are consequently at least

$$0.97(2.02)^2 e^{-2.02}/2 > 1/4, \quad 0.97(1.02)e^{-1.02} > 1/4.$$

The Poisson factors decrease above their target. With two forced successes the bound is $0.98e^{-0.02} > 1/4$. On $[1.47, 2.02]$, only the lower endpoints change: $1.47^2 e^{-1.47}/2 > 0.248$ and $0.47e^{-0.47} > 0.248$.

Near-integer bounds. For Eq. (1), a forced success gives at least δ . With two unforced variables, concavity of $\mu(1 - \mu/2)$ gives $\delta - \delta^2/2$ at the upper endpoint; the other endpoint is larger. For $n \geq 3$,

$$D(n, \mu) \leq D(n, 2) = 2 + \frac{2}{n(n-2)} \leq 8/3,$$

so $0.99e^{-8/3} > 0.02$ suffices. One variable is immediate.

For Eq. (2), monotonicity permits mean $3 - \delta$ whenever attainable. Two forced successes give at least δ . One forced success leaves a CDF at one of mean $2 - \delta$. With two fractional variables this is $\delta - \delta^2/4$; with $n \geq 3$, its atom at one is at least $1.98e^{-8/3} > 0.02$. With no forced success, $n = 3$ gives $\delta - \delta^2/3 + \delta^3/27$. Check $n = 4, 5$ at mean three. For $n \geq 6$, the logarithmic condition $3 \leq 4n/(n+2)$ bounds the atom at two below by $(5/12)(2.98)^2e^{-3} > 0.02$. Smaller dimensions only improve the CDF.

Finally, decreasing Bernoulli parameters to total mean $1 + \delta$ can only decrease the event of at least two successes. At that mean, [KKO21, Lemma 2.21] has zero correction exponent for target two. Its one-forced branch is $\delta e^{-\delta}$, and its zero-forced branch is at least $e^{-1.2}/2 > \delta$. The cubic Taylor lower bound proves the last assertion. All finite comparisons use the rational exponential enclosures in Section 4.9; the logarithmic arguments cover every larger dimension. $\square$

Lemma 4.4 (Quadratic mean and tail bounds). *Let $J \in \{0, 1, 2\}$ have a real-rooted probability generating polynomial, and put $m = \Pr[J = 1]$ and $\psi(v) = v(1 - v/2)$. If $v \leq \mathbb{E}[J] \leq 2 - v$ for $0 \leq v \leq 1$, then $m \geq \psi(v)$.*

If both tails at one have probability at least $\alpha \in (0, 1)$, then

$$m \geq \phi(\alpha) := 2\sqrt{1 - \alpha} - 2(1 - \alpha) > \alpha(1 - \alpha).$$

In particular, for $0 < b < \alpha$, the rational comparison

$$4(\alpha - b)(1 - \alpha) - b^2 > 0$$

implies $m > b$.

Proof. Factor the count law into two Bernoulli factors, padding by a zero parameter if necessary. If their sum is μ , their product is at most $\mu^2/4$, so $m \geq \mu - \mu^2/2 \geq \psi(v)$ on $[v, 2 - v]$.

For the tail assertion, write the other probabilities as p_0, p_2 . If $m < \alpha$, then $p_0, p_2 \geq \alpha - m$ and $p_0 + p_2 = 1 - m$. Hence real-rootedness gives

$$m^2 \geq 4p_0p_2 \geq 4(\alpha - m)(1 - \alpha).$$

Solving this quadratic gives $m \geq \phi(\alpha)$. If $m \geq \alpha$, the same conclusion follows from $\phi(\alpha) \leq \alpha$. For $t = \sqrt{1 - \alpha} \in (0, 1)$, $\phi(\alpha) - \alpha(1 - \alpha) = t(1 - t)^2(t + 2) > 0$. Finally $4(\alpha - m)(1 - \alpha) - m^2$ decreases with $m \geq 0$, which proves the rational test. The factorization describes the count distribution, not independence of the actual sampled edges. $\square$

Lemma 4.5 (Adjacent-rank balance). *Let a bivariate stable probability generating polynomial have degree-one and degree-two parts*

$$ax + by, \quad cx^2 + mxy + ey^2,$$

where $a, b > 0$. Put $P := a + b$ and $q := c + m + e > 0$. If both degree-at-most-two contributions to its coordinate means are at least L , and $W := P + 2q - 2L$, then

$$m \geq \frac{q}{2} - \frac{W^2}{8(P + q)}.$$

Proof. The adjacent-rank homogenization in [BBL09, Lemma 4.16], with a positive rescaling of the homogenizing variable, shows that

$$Q(x, y, z) := (ax + by)z + cx^2 + mxy + ey^2$$

is stable. In particular,

$$mab \geq cb^2 + ea^2.$$

Indeed, if $N := cb^2 - mab + ea^2 > 0$, take $x = b + i\tau$ and $y = -a + i\tau$. Solving $Q = 0$ for z gives

$$\operatorname{Im} z = \frac{N - \tau^2 q}{\tau(a + b)} > 0$$

for sufficiently small $\tau > 0$, contradicting stability.

Set $f := a - b$, $g := c - e$, and $s := f + 2g$. The two truncated means have sum $P + 2q$ and difference s , so $|s| \leq W$. Rewriting the preceding coefficient inequality gives

$$m \geq \frac{q}{2} + \frac{(P + q)f^2}{2P^2} - \frac{fs}{2P}.$$

Completing the square in f bounds the last two terms below by $-s^2/(8(P + q)) \geq -W^2/(8(P + q))$. $\square$

Lemma 4.6 (Newton tail and excess bounds). *Let (p_j) be a Poisson-binomial rank distribution with $p_1 \geq c > 0$ and $p_2 = q < 2c$. Put $z = q/(2c)$ and*

$$T_c(q) := \frac{2q^2}{3c} \frac{3 - 2z}{(1 - z)^2}, \quad T_c^{\text{ex}}(q) := \frac{2q^2}{3c} (1 - z)^{-2}.$$

Then

$$\sum_{j \geq 3} jp_j \leq T_c(q), \quad \sum_{j \geq 3} (j - 2)p_j \leq T_c^{\text{ex}}(q).$$

Proof. Newton's inequalities, after weakening their dimension factors, give

$$p_3 \leq \frac{2q^2}{3c}, \quad \frac{p_4}{p_3} \leq \frac{3}{4} \frac{p_3}{p_2} \leq z.$$

Log-concavity therefore gives $p_{3+i} \leq (2q^2/(3c))z^i$ for every $i \geq 0$. Sum $(3 + i)z^i$ and $(1 + i)z^i$ to obtain the two bounds. If $q = 0$ or $p_3 = 0$, gapless rank support makes the tail zero, so no division by zero is needed. $\square$

Lemma 4.7 (Concave adjacent-rank certificate). *Let X, Y have a stable joint probability generating polynomial, and put $Z := X + Y$. Suppose*

$$\mathbb{E}[X], \mathbb{E}[Y] \geq L_0, \quad \mathbb{E}[Z] \geq 1, \quad \Pr[Z = 1] \geq c > 0, \quad \Pr[Z = 2] = q < 2c.$$

Define

$$\Phi(q) := 1 - q - T_c(q), \quad W(q) := q + 1 - 2L_0 + 2T_c(q).$$

If $\Phi(q) > 0$, $W(q) \geq 0$, and $L_0 - 2q - T_c(q) > 0$, then

$$\Pr[X = Y = 1] \geq F(q) := q/2 - W(q)^2/(8\Phi(q)).$$

On every interval where these conditions hold, F is concave.

Proof. Write $P := \Pr[Z = 1]$. Lemma 4.6 bounds the mean contribution from ranks at least three by $T_c(q)$. Both degree-one coefficients are at least $L_0 - 2q - T_c(q) > 0$, and both truncated means are at least $L_0 - T_c(q)$. Also

$$P + q \geq 1 - q - T_c(q) = \Phi(q), \quad P \leq 1 - q.$$

Lemma 4.5 therefore gives the displayed bound. This applies after projection to the counted union; no outside coordinates are retained in a non-extremal rank condition. Compulsory monomial factors can be removed and the quotient polarized, since these operations preserve stability.

The power series of T_c has nonnegative coefficients, so $T'_c, T''_c \geq 0$, $W'' \geq 0$, and $\Phi'' \leq 0$. Thus

$$(W^2/\Phi)'' = \frac{2(W'\Phi - W\Phi')^2}{\Phi^3} + \frac{2WW''}{\Phi} - \frac{W^2\Phi''}{\Phi^2} \geq 0.$$

Hence $F'' \leq 0$, and its minimum on a compact interval is attained at an endpoint. $\square$

Lemma 4.8 (Binary covering and deleted mass). *Let a homogeneous strongly Rayleigh law have a binary coordinate E , with $0 < q := \Pr[E = 1] < 1$, and let F be a disjoint coordinate set. Then*

$$0 \leq \mathbb{E}[F_T] - \mathbb{E}[F_T \mid E = 1] \leq 1 - q, \quad 0 \leq \mathbb{E}[F_T \mid E = 0] - \mathbb{E}[F_T] \leq q.$$

Deleting a coordinate set D with positive deletion probability increases the mean of any disjoint coordinate set by at most $x(D)$.

Proof. Project away E . Its two conditional laws have adjacent total ranks. The coupling in [KKO21, Theorem 2.10] adds exactly one remaining coordinate when passing from $E = 1$ to $E = 0$. Their F -mean difference therefore lies in $[0, 1]$. Averaging with weights $q, 1 - q$ proves both inequalities. For deletion of D , negative association makes all surviving-coordinate changes nonnegative. Homogeneity makes their total change exactly $x(D)$, which proves the last assertion. We use the inclusion loss $1 - q$ in this form, rather than the q printed in the last inequality of [GKL24, Lemma A.1]. $\square$

Lemma 4.9 (Compulsory-cut coefficient bound). *For a polynomial with nonnegative coefficients, write*

$$\text{Cap}_\kappa(f) := \inf_{\mathbf{x} > 0} \frac{f(\mathbf{x})}{\prod_i x_i^{\kappa_i}}.$$

If $q(x, y)$ is real stable with nonnegative coefficients and $q(0, 0) = 0$, then

$$[xy]q \geq \frac{e^{-1}}{2} \text{Cap}_{(1,1)}(q).$$

Consequently, if $f(x, y, z)$ is real stable with nonnegative coefficients and $f(0, 0, z) = 0$ identically, then

$$[xyz]f \geq \frac{e^{-2}}{2} \text{Cap}_{(1,1,1)}(f).$$

Proof. We first prove the needed univariate contraction. Let $R(t)$ have nonnegative coefficients, only nonpositive real roots, and degree at most $D \geq 2$. Then

$$[t]R \geq \left(\frac{D-1}{D}\right)^{D-1} \text{Cap}_1(R) \geq e^{-1} \text{Cap}_1(R).$$

If $R(0) = 0$, letting t decrease to zero in $R(t)/t$ proves the stronger factor one. Otherwise write

$$R(t) = R(0) \prod_{i=1}^D (1 + a_i t), \quad a_i \geq 0,$$

padding with zero parameters if necessary, and set $A := \sum_i a_i$. For $A = 0$ the capacity is zero. For $A > 0$, the arithmetic-geometric mean inequality gives

$$R(t) \leq R(0)(1 + At/D)^D.$$

The infimum of the right side divided by t is attained at $t = D/(A(D-1))$. Hence

$$\text{Cap}_1(R) \leq R(0)A \left(\frac{D}{D-1}\right)^{D-1} = [t]R \left(\frac{D}{D-1}\right)^{D-1}.$$

The bound $(1 + 1/(D-1))^{D-1} \leq e$ proves the assertion. For degree one the capacity is exactly $[t]R$, and for degree zero it is zero, so the e^{-1} contraction holds in every degree.

Now let $N := \deg q$. The zero polynomial is immediate. If $N < 2$, scaling $x = y$ to infinity gives $\text{Cap}_{(1,1)}(q) = 0$. For $N \geq 2$, [BBL09, Theorem 4.5] shows that the homogenization

$$Q(x, y, w) := w^N q(x/w, y/w)$$

is stable. This closure theorem applies to all nonnegative stable polynomials, without a multiaffinity assumption. Since $q(0, 0) = 0$, $\deg_w Q \leq N - 1$, and homogeneity gives

$$\text{Cap}_{(1,1,N-2)}(Q) = \text{Cap}_{(1,1)}(q).$$

Let $H(x, y) := [w^{N-2}]Q$, the degree-two part of q . Differentiation followed by real specialization shows that H is stable or zero.

For $N \geq 3$, fix $x, y > 0$, put $D := N - 1$, and reverse the polynomial in w :

$$R(t) := t^D Q(x, y, 1/t).$$

It has nonnegative coefficients and only nonpositive real roots, with

$$[t]R = H(x, y), \quad \text{Cap}_1(R) = \inf_{w>0} \frac{Q(x, y, w)}{w^{N-2}}.$$

The univariate contraction, followed by division by xy and infimization over x, y , yields

$$\text{Cap}_{(1,1)}(H) \geq e^{-1} \text{Cap}_{(1,1)}(q).$$

For $N = 2$ this holds with factor one, since taking w down to zero in the capacity of Q leaves exactly H . Write $H = ax^2 + bxy + cy^2$. Stability gives $b^2 \geq 4ac$, and therefore

$$\text{Cap}_{(1,1)}(H) = b + 2\sqrt{ac} \leq 2b.$$

This proves the bivariate assertion, including degenerate quadratics.

Finally set $q(x, y) := [z]f(x, y, z)$. It is stable or zero and has $q(0, 0) = 0$. Apply the univariate contraction in z for each $x, y > 0$ and take the infimum after division by xy :

$$\text{Cap}_{(1,1)}(q) \geq e^{-1} \text{Cap}_{(1,1,1)}(f).$$

The bivariate assertion gives the claimed $e^{-2}/2$ factor. If q is zero, the displayed inequality forces the latter capacity to be zero, so the conclusion still holds. □

Lemma 4.10 (Electrical remote influence). *In the setting of [KKO21, Lemma 2.27], let u, v, w be distinct vertices and let $f = uv$, $g = vw$ have marginals in $[1/2 - e, 1/2 + e]$, where $0 \leq e < 1/6$. The smaller increase in the mean of $\delta(w) \setminus \{g\}$ on deleting f , or of $\delta(u) \setminus \{f\}$ on deleting g , is at most*

$$\frac{1/2 + e}{1/2 - e} \max_{0 \leq z \leq 1} \{z/(1+z) - (1/2 - e)z^2\} < 0.22 \quad \text{if } 0 \leq e \leq 0.00550006.$$

Consequently the bound 0.22 applies after endpoint-tree conditioning whenever $h \leq 0.0055$ and $d_0 \leq 2 \cdot 10^{-8}$.

Proof. The exact max-entropy law may lie on a face of the marginal polytope. To justify the weighted-tree calculation there, let $\bar{\mu}$ be uniform on all spanning trees of the support graph, with marginals $\bar{x}$, and let μ_t maximize entropy at $x_t := (1-t)x + t\bar{x}$, $0 < t < 1$. This is a relative-interior point, so μ_t is a positive weighted-tree law. Every subsequential limit ν has marginals x , and

$$H(\mu_t) \geq H((1-t)\mu + t\bar{\mu}) \longrightarrow H(\mu).$$

Continuity of entropy on the finite probability simplex and uniqueness of the maximizer imply $\nu = \mu$. Hence $\mu_t \rightarrow \mu$. Stability and the weighted-tree factorization identities pass to this limit. Conditional identities pass whenever their limiting conditioning probabilities are positive, as they are at every use below.

Let $f = uv$ and $g = vw$ be the two distinct incident edges, with conductances c_f, c_g and marginals a, b . Put $U := \delta(u) \setminus \{f\}$ and $W := \delta(w) \setminus \{g\}$. For a set F disjoint from f , write

$$I_f(F) := \mathbb{E}[F_T \mid f \notin T] - \mathbb{E}[F_T].$$

Let L^+ be the graph Laplacian pseudoinverse, and let b_j be an oriented incidence vector for edge j . Differentiating the reduced Laplacian determinant gives

$$\Pr[j \in T] = c_j b_j^T L^+ b_j, \quad \text{Cov}(\mathbf{1}_f, \mathbf{1}_j) = -c_f c_j (b_f^T L^+ b_j)^2 \quad (j \neq f).$$

Thus, for the unit-current voltage $V := L^+ b_f$,

$$I_f(F) = \frac{c_f}{1-a} \sum_{j \in F} c_j (b_j^T V)^2.$$

Eliminate all vertices other than u, v, w by harmonic extension. Write the reduced triangle conductances as A on uv , B on vw , and C on uw , and set $D := AB + AC + BC > 0$. Then $A \geq c_f$ and $B \geq c_g$. For an interior vertex j , let H_{jt} be its harmonic measure on the boundary. These numbers are nonnegative and sum to one, so

$$(V_w - V_j)^2 \leq \sum_{t \in \{u, v, w\}} H_{jt} (V_w - V_t)^2.$$

Multiplying by c_{wj} and summing, including direct boundary edges, gives the boundary-star inequality

$$\sum_{j \in \delta(w)} c_j (b_j^T V)^2 \leq C(V_w - V_u)^2 + B(V_w - V_v)^2.$$

Indeed, the coefficient at boundary vertex t is $c_{wt} + \sum_j c_{wj} H_{jt}$, exactly the corresponding Schur-complement conductance. In particular, reduction bounds the original star energy; it does not identify that energy with the total energy of the reduced network.

Take $V_v = 0$. The unit-current equations and the two marginal identities give

$$V_u = (B + C)/D, \quad V_w = C/D, \quad c_f = aD/(B + C), \quad c_g = bD/(A + C).$$

Subtract the actual g -energy from the star bound. Substitution and the symmetric calculation yield, with $z := C/\sqrt{(A + C)(B + C)} \in [0, 1)$,

$$I_f(W) \leq \frac{a}{1-a}(BC/D - bz^2), \quad I_g(U) \leq \frac{b}{1-b}(AC/D - az^2).$$

Both parentheses are nonnegative: before simplification their energy coefficients are $C, B - c_g$ and $C, A - c_f$, respectively. Also

$$\min\{AC/D, BC/D\} \leq z/(1 + z).$$

For example, if $A \leq B$, put $S := \sqrt{(A + C)(B + C)}$. Then $A \leq S - C$, so $A(S + C) \leq D$, proving the assertion. It follows that

$$\min\{I_f(W), I_g(U)\} \leq \frac{1/2 + e}{1/2 - e} \max_{0 \leq z \leq 1} \{z/(1 + z) - (1/2 - e)z^2\}.$$

For $f_e(z) := z/(1 + z) - (1/2 - e)z^2$, we have $f_e''(z) \leq -(1 - 2e)$. The quadratic tangent bound at $z_0 := 233/500$ gives

$$\max_{0 \leq z \leq 1} \{f_e(z)\} \leq f_e(z_0) + \frac{f_e'(z_0)^2}{2(1 - 2e)}.$$

The maximum and the prefactor increase with e . Exact rational substitution at $e = 0.00550006$ gives

$$\frac{1/2 + e}{1/2 - e} (f_e(z_0) + \frac{f_e'(z_0)^2}{2(1 - 2e)}) < 0.22.$$

Vanishing reduced conductances follow by continuity; the interior Dirichlet matrix is invertible because every interior component touches the boundary.

Finally, conditioning disjoint endpoint atoms to be trees and contracting them preserves the exterior weighted-tree law. Parallel edges aggregate with their summed conductance and preserve the bundle count. The endpoint-tree mean-change budget gives conditional half-width at most $h + 3d_0$, so the same bound applies to the half bundles. $\square$

Lemma 4.11 (Conditional rank-two split). *Let X, Y count disjoint coordinate blocks of a strongly Rayleigh law, let $Z = X + Y \geq 1$, and write $p_i := \Pr[Z = i]$ and $m := \mathbb{E}[Z]$. Suppose $p_2 > 0$ and $\mathbb{E}[X], \mathbb{E}[Y] \leq 2 - b$. If $0 < b - p_1 \leq 1$, then*

$$\Pr[X = Y = 1 \mid Z = 2] \geq \psi(b - p_1), \quad \psi(v) := v(1 - v/2).$$

If $m \geq 1 + \nu$ with $\nu > 0$, then

$$p_1 \leq e^{-\nu}, \quad p_1 \leq p_2/\nu, \quad p_2^2 \geq 2p_1p_3.$$

If also $m \leq 3$, then

$$p_3 \geq 1 - 3p_1 - 2p_2, \quad 3 - m \leq 2p_1 + p_2.$$

Proof. Project onto the union of the two coordinate blocks. The consecutive rank laws admit the one-coordinate monotone coupling of [KKO21, Theorem 2.10]. Thus, writing $\mu_X(j) := \mathbb{E}[X \mid Z = j]$,

$$\mu_X(j) \leq \mu_X(2) + \max\{j - 2, 0\}.$$

The rank support has no gaps. Averaging gives

$$\mu_X(2) \geq \mathbb{E}[X] - (m - 2 + p_1) = 2 - \mathbb{E}[Y] - p_1 \geq b - p_1.$$

The same holds for Y . The rank-two law is stable, and the two mean bounds give $\mathbb{E}[X \mid Z = 2] \in [b - p_1, 2 - b + p_1]$. Lemma 4.4 proves the split bound.

The real-rooted generating polynomial of Z has a compulsory factor of its variable. Hence $Z - 1$ is Poisson-binomial. Its zero atom is at most $e^{-\nu}$. When $p_1 > 0$, let t_j be the odds of its Bernoulli factors. Then

$$\frac{p_2}{p_1} = \sum_j t_j \geq m - 1 \geq \nu, \quad p_2^2 - 2p_1p_3 = p_1^2 \sum_j t_j^2 \geq 0.$$

The case $p_1 = 0$ is immediate. Finally, assigning rank at least four to every term beyond rank three gives $p_3 \geq 4 - m - 3p_1 - 2p_2$; assigning rank at least three beyond rank two gives $3 - m \leq 2p_1 + p_2$. $\square$

Lemma 4.12 (Projected full-rank mean changes). *Project a strongly Rayleigh law to a counted union with total rank R . If $\Pr[R = k] > 0$ and F is any subunion of the counted blocks, then*

$$\mathbb{E}[F] - \mathbb{E}[(R - k)_+] \leq \mathbb{E}[F \mid R = k] \leq \mathbb{E}[F] + \mathbb{E}[(k - R)_+].$$

The errors apply once to the union F , not once per block.

Proof. Consecutive supported full ranks admit a coupling differing by one coordinate [KKO21, Theorem 2.10]. Thus the conditional F -mean at rank j differs from the rank- k mean by a number in $[0, j - k]$ when $j \geq k$, and in $[j - k, 0]$ when $j \leq k$. Average these two inequalities. $\square$

We use the up/down inequality only on disjoint complementary counted sets with positive conditioning probability: for $A + B = n_A + n_B$,

$$\Pr[A \geq n_A \mid A + B = n_A + n_B] \geq \Pr[A \geq n_A] \Pr[B \leq n_B],$$

The lower-tail statement is symmetric. See [GKL24, Lemma A.10] and [KKO21, Lemma 5.4]. A stable count polynomial is polarized when needed, and every rank condition below is a full rank after projection. Outside coordinates are not retained under arbitrary subset-rank conditioning.

Lemma 4.13 (Homogeneous cubic split). *Let $X + Y + Z = 3$ have a stable homogeneous probability generating polynomial with all coordinate means in $[\ell, U]$, where $0 < \ell \leq 0.01$. For either row below,*

$$\Pr[X = Y = Z = 1] > c_U \ell e^{-\ell}.$$

U	F_U	v_U	c_U
1.504	0.434	0.21	0.3728
1.53	0.4147	0.19	0.3594

Proof. Choose the coordinate of smallest mean, say X , and put $\mu = \mathbb{E}[X]$. Then $\ell \leq \mu \leq 1$, and each other mean is at least $(3 - U)/2$. Lemma 4.3 gives $\Pr[X = 1] \geq \mu e^{-\mu} > 0$. Project to Y, Z and condition their full rank to two, which is precisely the event $X = 1$.

If $\mu \geq 0.02$, both unconditioned nonzero tails of Y, Z exceed $1 - e^{-(3-U)/2} > 0.52$, and their CDFs at one exceed F_U . The up/down inequality gives both conditional tails at one at least $\alpha_U = 0.52F_U$. The rational test in Lemma 4.4 gives conditional central mass greater than v_U , since

$$4(\alpha_U - v_U)(1 - \alpha_U) - v_U^2 > 0$$

in both rows. Hence the joint atom exceeds

$$0.02e^{-0.02}v_U > 0.02(0.98)v_U > 0.01c_U \geq c_U\ell e^{-\ell}.$$

If $\mu \leq 0.02$, put $W = Y + Z = 3 - X$. Lemma 4.12 gives

$$\begin{aligned} \mathbb{E}[Y \mid W = 2] &\geq \mathbb{E}[Y] - \Pr[X = 0] \\ &= 2 - \mathbb{E}[Z] - \Pr[X = 2] - 2\Pr[X = 3] \\ &\geq 2 - U - \mu^2/2 \geq 2 - U - 0.0002. \end{aligned}$$

Here $\mathbb{E}[X(X-1)] \leq \mu^2$ follows from the Bernoulli factorization. The same lower bound holds for Z . Lemma 4.4 gives conditional central mass at least $\psi(2 - U - 0.0002) > c_U$. Multiply by $\Pr[X = 1] \geq \ell e^{-\ell}$. Both row comparisons are exact rationals. The coordinate choice is made from the law, and all rank conditioning is after projection. $\square$

Lemma 4.14 (Homogeneous cubic bound from two marginal tails). *Let $X+Y+Z = 3$ have a stable homogeneous probability generating polynomial. Suppose every coordinate has nonzero probability at least $\alpha = 0.2574$ and probability at most one at least β , where $0 < \beta \leq 0.01$. Put*

$$c_0 = 0.4911 = 3(0.1637), \quad b = \frac{\beta}{1 + \beta/c_0}, \quad \psi(v) = v(1 - v/2).$$

Then $\Pr[X = Y = Z = 1] \geq b\psi(\alpha - b^2/c_0)$.

Proof. Choose a coordinate of largest mean, say X , and put $\mu = \mathbb{E}[X]$ and $p_j = \Pr[X = j]$. This choice depends on the law, not on its realization. Then $1 \leq \mu \leq 3 - 2\alpha < 2.5$. Gapless rank support, $p_0 + p_1 \geq \beta > 0$, and $\mu \geq 1$ imply $p_1 > 0$. The Bernoulli factorization of X has degree at most three. Padding by zero parameters and applying arithmetic-geometric mean gives

$$p_0 \leq (1 - \mu/3)^3, \quad \mu \leq 3(1 - p_0^{1/3}).$$

Each other coordinate has mean at least α and at least $3 - 2\mu$. Project to the other pair and condition its full rank to two, an event of probability $p_1 > 0$. Lemma 4.12 loses at most $\mathbb{E}[(Y + Z - 2)_+] = p_0$. Thus both conditional means are at least

$$\max\{\alpha - p_0, 6p_0^{1/3} - 3 - p_0\}.$$

This quantity exceeds 0.095: for $p_0 \leq 0.16$ use $\alpha - 0.16 = 0.0974$; above 0.16 the second expression increases, and $0.5425^3 < 0.16$ with $6(0.5425) - 3 - 0.16 = 0.095$ suffices. Lemma 4.4 bounds the conditional central atom below by $\psi(0.095)$. If $p_1 \geq 0.1$, the homogeneous quadratic mean bound gives joint atom at least $0.1\psi(0.095)$.

Suppose $p_1 < 0.1$. The rank-one bound in Lemma 4.3 on $[1, 1.2]$ forces $\mu > 1.2$, and its rank-two bound on $[1.2, 2.5]$ gives $p_2 > 0.1637$. Newton's degree-three inequality gives

$$p_1^2 \geq 3p_0p_2, \quad p_0 \leq p_1^2/c_0.$$

The inequality $b + b^2/c_0 < \beta \leq p_0 + p_1$ implies $p_1 \geq b$. Both conditional pair means are at least $\alpha - p_1^2/c_0 > 0$. For $f(x) = x\psi(\alpha - x^2/c_0)$,

$$f'(x) = \psi(\alpha) - 3(1 - \alpha)x^2/c_0 - 5x^4/(2c_0^2).$$

This decreases with x^2 and is positive at $x = 0.1$, by exact substitution. Thus the joint atom is at least $f(p_1) \geq f(b)$. Finally $f(b) \leq 0.01\psi(\alpha) < 0.1\psi(0.095)$, so the first case also suffices. The full-rank moment comparison follows from the adjacent-rank coupling of [KKO21, Theorem 2.10], after projection. No independence of actual sampled edges is used. $\square$

4.3 Good bundles and common events

For non-half top bundles and bottom bundles, retain [KKO21, Definition 5.13]. A half top bundle is good when its conditional 2–2 probability is at least $\gamma_{\text{good}}h$. We use this definition directly, without an inclusion assumption for the source good-half class. The selected-state tail, competing-half estimate, and structural trichotomy below establish its required interfaces directly.

Lemma 4.15 (Deletion and the total-rank lower tail). *Let ν be a strongly Rayleigh law on a finite ground set F , let $R := |T \cap F|$, and let $D \subseteq F$. If $\Pr[D_T = 0] > 0$, then for every integer k ,*

$$\Pr[R \leq k \mid D_T = 0] \geq \Pr[R \leq k].$$

Proof. The adjacent-rank monotone coupling in [KKO21, Theorem 2.10] makes $a_j := \Pr[D_T = 0 \mid R = j]$ nonincreasing on the rank support. The function $j \mapsto \mathbf{1}_{\{j \leq k\}}$ is also nonincreasing. For an independent copy R' of R , the covariance identity gives

$$\begin{aligned} 2 \operatorname{Cov}(a_R, \mathbf{1}_{\{R \leq k\}}) &= \mathbb{E}[(a_R - a_{R'})(\mathbf{1}_{\{R \leq k\}} - \mathbf{1}_{\{R' \leq k\}})] \\ &\geq 0. \end{aligned}$$

Divide the resulting joint-probability inequality by $\Pr[D_T = 0]$. The coupling uses full ranks of the projected law on F . $\square$

Lemma 4.16 (Selected-state tail of a good half bundle). *Let $e = (u, v)$ be a good half bundle. Conditional on the endpoint trees and on $e_T = 1$, its residual union count R satisfies*

$$\Pr[R \leq 2] \geq L_* := \frac{(\gamma_{\text{good}} - 2)h - 4\rho_t}{3/2 - h - \rho_t}, \quad \rho_t := 3d_0.$$

After deleting the degree remainder C , the same lower bound holds for the projected total residual rank. At the fixed parameters,

$$L_* > 0.00261653801701028.$$

Proof. Work first under the endpoint-tree law, let R denote its residual union count before conditioning on $E := e_T$, and put $q := \Pr[E = 1]$. Sequential maximum-rank conditioning has total mean-change budget at most $3d/2$. Thus, with the conservative $\rho_t = 3d_0$,

$$q \in [1/2 - h - \rho_t, 1/2 + h + \rho_t], \quad \mathbb{E}[\delta(u)_T], \mathbb{E}[\delta(v)_T] \in [2 - \rho_t, 2 + \rho_t].$$

The bundle is binary. Lemma 4.8 and the adjacent-rank coupling give a coupling of the residual union counts R_1, R_0 under $E = 1, E = 0$ with $R_0 - R_1 \in \{0, 1\}$. The union is a proper cut, so $R_1 \geq 1$. When $E = 0$, the two endpoint cuts each contribute a compulsory success, so $R_0 \geq 2$.

For every such pair of integer counts,

$$q\mathbf{1}_{\{R_1=2\}} + (1-q)\mathbf{1}_{\{R_0=4\}} \leq R_1 + (1-q)(R_0 - R_1) - 3 + (1+q)\mathbf{1}_{\{R_1 \leq 2\}}.$$

For $R_1 = 1$, necessarily $R_0 = 2$, and equality holds. For $R_1 = 2$, the two possible right sides are q and 1. For $R_1 = 3$, equality holds for both increments. For $R_1 \geq 4$, the inequality follows directly, with right side at least $R_1 - 3$.

Write $A := \Pr[R_1 \leq 2]$ and let g be the conditional 2–2 probability of the bundle. Its present- and absent-edge events require $R_1 = 2$ and $R_0 = 4$, respectively. Averaging the preceding inequality gives

$$g \leq \mathbb{E}[R] - 3 + (1+q)A \leq 1 - 2q + 2\rho_t + (1+q)A.$$

Here $\mathbb{E}[R] = \mathbb{E}[\delta(u)_T] + \mathbb{E}[\delta(v)_T] - 2q$. For $0 \leq A \leq 1$, the final expression decreases with q . Substitute $q \geq 1/2 - h - \rho_t$ and $g \geq \gamma_{\text{good}}h$ to obtain the claim. The numerator and denominator defining L_* are strictly positive.

After endpoint and union tree conditioning, the unique parallel-bundle choice factors from the residual weighted-tree law, as in Lemma 4.21. Splitting C into its bundle and residual parts shows that deleting the bundle part only restricts the independent selected edge. Project to the residual union and apply Lemma 4.15 to deletion of its residual part. This cannot decrease the total-rank lower tail. The conditions commute and all have positive probability by the outer bound in Lemma 4.21. The argument does not apply to the different window-premise subcount. $\square$

Lemma 4.17 (Good-bundle threshold). *In the setting of [KKO21, Lemma 5.16], if $x(\delta^\dagger(u)) \geq 1/2 + k_{\text{good}}h$, then the half bundle incident to u is good.*

Proof. Put $k := k_{\text{good}}$. Follow the source conditioning on u, v, S, W being trees, where $W := S \setminus u$, and define

$$X := \delta^\dagger(u)_T, \quad Y := \delta(v)_T - 1, \quad Z := X + Y.$$

The unshifted coordinate sets $\delta^\dagger(u)$ and $\delta(v)$ are disjoint. Projection and diagonalization of the conditional spanning-tree generating polynomial therefore give a stable bivariate polynomial. Since $\delta(v)_T \geq 1$, it is divisible by the variable corresponding to $\delta(v)$. Dividing by that variable preserves stability and gives the joint generating polynomial of (X, Y) . Thus Lemma 4.5 applies to these shifted counts.

The exact expectation estimates in [KKO21, Eq. (23)] imply

$$\mathbb{E}[X], \mathbb{E}[Y] \geq C := \frac{1}{2} + kh - 4d_0, \quad \mathbb{E}[Z] \geq 1 + \alpha, \quad \alpha := 2kh - 5d_0.$$

Also $\mathbb{E}[X] \leq 1 + d_0$, $\mathbb{E}[Y] \leq 1.5 - kh + 3d_0$, and hence $\mathbb{E}[Z] \leq 2.5 - kh + 4d_0 < 2.5$. Both individual means lie in $[1/2, 3/2]$. Here and below replacing the hierarchy error by d_0 only weakens the estimates; the source's rounded sufficient condition $k \geq 9$ is not used.

Define

$$q_c := \left(\frac{2\gamma_{\text{good}}}{0.164} + 0.01\right)h, \quad q := \Pr[Z = 2].$$

Suppose first that $q \geq q_c$. The four tail probabilities in the first case of [KKO21, Lemma 5.16] give

$$\epsilon_0 := (1 - e^{-1/2})\frac{7}{16}.$$

The up/down truncation inequality in [GKL24, Lemma A.10] gives both tails of X at one, conditional on $Z = 2$, at least ϵ_0 . The conditional bivariate rank-two law is stable, so its univariate rank

polynomial is real-rooted. Lemma 4.4 therefore implies $\Pr[X = 1 \mid Z = 2] > 0.164$. For a rational certificate,

$$e^{1/2} > \sum_{j=0}^7 \frac{1}{2^j j!} > \frac{10^6}{606531},$$

so $e^{-1/2} < 0.606531$. Put $\alpha_0 := (1 - 0.606531)7/16 < \epsilon_0$. The rational test in Lemma 4.4 applies because $0.164 < \alpha_0$ and

$$4(\alpha_0 - 0.164)(1 - \alpha_0) - 0.164^2 = 0.000067933561109375 > 0.$$

Consequently

$$\Pr[X = Y = 1] > q_c(0.164) > 2\gamma_{\text{good}}h.$$

Suppose now that $q < q_c < 0.1637$. The target-two bound in Lemma 4.3 forces $\mathbb{E}[Z] < 1.2$. The same lemma gives $\Pr[Z = 1] > 0.36$. Define

$$\begin{aligned} T(q) &:= T_{0.36}(q), & \Phi(q) &:= 1 - q - T(q), \\ W(q) &:= q - \alpha + 3d_0 + 2T(q), & F(q) &:= q/2 - W(q)^2/(8\Phi(q)). \end{aligned}$$

These are the quantities in Lemma 4.7 with $L_0 = 1/2 + kh - 4d_0$. The target-two bound in [KKO21, Lemma 2.21] gives

$$q \geq \alpha e^{-\alpha} \geq q_* := \alpha(1 - \alpha + \alpha^2/2 - \alpha^3/6).$$

Indeed, on $[1 + \alpha, 1.2]$, the one-forced-success branch is at least $\alpha e^{-\alpha}$, and the zero-forced branch is at least $e^{-1.2}/2 > \alpha$. The correction exponent is zero. Exact substitution gives

$$1/2 + kh - 4d_0 - 2q_c - T(q_c) > 0, \quad \Phi(q_c) > 0, \quad W(q_*) > 0.$$

Since T, W increase and Φ decreases, all hypotheses hold throughout $[q_*, q_c]$. Concavity therefore gives

$$\Pr[X = Y = 1] \geq F(q) \geq \min\{F(q_*), F(q_c)\}.$$

It remains to restore the outer conditioning. Put $A := 1/2 + kh < 1$. The source estimates give

$$\Pr[W \text{ is a tree} \mid u, v, S \text{ are trees}] \geq A - 2d_0,$$

and

$$\Pr[S \text{ is a tree} \mid u, v \text{ are trees}] \geq 1 - \frac{d_0}{2(1 - d_0)} \geq 1 - d_0.$$

Their product is at least $(1 - d_0)(A - 2d_0) \geq A - 3d_0 > A - 4d_0 > 1/2$. This proves goodness in the large-rank branch. In the small-rank branch, the two restored endpoint comparisons are

$$\begin{aligned} (1/2 + kh - 4d_0)F(q_*) - \gamma_{\text{good}}h &> 0, \\ (1/2 + kh - 4d_0)F(q_c) - \gamma_{\text{good}}h &> 0. \end{aligned}$$

Also $0 < q_* < q_c$, $1/2 + kh < 1$, and $kh > 4d_0$, so the source outer and mean calculations apply. This proves the goodness threshold. □

Lemma 4.18 (Competing half bundles on the enlarged domain). *For $0 < h \leq 0.0055$ and $0 \leq d_0 \leq \min\{h^2, 2 \cdot 10^{-8}\}$, one of any two half bundles incident to the same atom in [KKO21, Lemma 5.17] has conditional 2–2 probability greater than 0.01330. In particular, the bad half bundles form a matching for Definition 4.2.*

Proof. Use the source notation $\mathbf{e} = (u, v)$, $\mathbf{f} = (v, w)$, and

$$U := \delta(u) \setminus \mathbf{e}, \quad V := \delta(v) \setminus (\mathbf{e} \cup \mathbf{f}), \quad W := \delta(w) \setminus \mathbf{f}.$$

Condition on the three endpoint atoms being trees. Orient the bundles so that Lemma 4.10 bounds the increase of the W -mean on deleting $\mathbf{e}$ by 0.22. Put

$$\tau_0 := 0.073, \quad o := 0.494, \quad h_{\max} := 0.0055, \quad d_{\max} := 2 \cdot 10^{-8}.$$

The endpoint-tree and edge-selection factors exceed o on this box, since $(1 - 3d_{\max})(1/2 - h_{\max} - 3d_{\max}) > o$.

Suppose first that deleting $\mathbf{e}$ raises the V -mean by at least τ_0 . The source conditional-mean identities give, after removing a compulsory success from each of U_T and $(V \cup \mathbf{f})_T$, disjoint projected counts X, Y with

$$\mathbb{E}[X], \mathbb{E}[Y] \geq 1/2 - h - 3d_0, \quad 1 + v \leq \mathbb{E}[X + Y] \leq 1.5 + 2h + 3d_0 < 1.512,$$

where $v := \tau_0 - 2h - 3d_0$. Removing compulsory monomial factors and polarizing preserve stability. Put $q := \Pr[X + Y = 2]$. Lemma 4.3 gives a rank-one atom greater than 0.33, and [KKO21, Lemma 2.21] gives $q \geq ve^{-v}$. If $q \geq 0.11$, the up/down inequality [GKL24, Lemma A.10] and the quadratic split, with tail product 0.39(0.70), give split greater than 0.245. The restored probability is then greater than 0.494(0.11)(0.245) = 0.0133133.

For $q < 0.11$, use Lemma 4.7 at the worst mean bounds $L_0 = 1/2 - h_{\max} - 3d_{\max}$ and $c = 0.33$. Write its lower bound as $F(q)$. Put

$$v_{\min} := 0.073 - 2h_{\max} - 3d_{\max}, \quad q_* := v_{\min}(1 - v_{\min} + v_{\min}^2/2 - v_{\min}^3/6).$$

The function $v(1 - v + v^2/2 - v^3/6)$ is increasing for $0 \leq v \leq 0.073$, so $q \geq q_*$. Both degree-one coefficients and $\Phi(q)$ stay positive through $q = 0.11$, and $W(q) \geq 0$. Concavity reduces the whole interval to the exact comparisons

$$oF(q_*) > 0.01330, \quad oF(0.11) > 0.01330.$$

This proves the first branch uniformly in the box.

In the other branch, select $\mathbf{f}$ and delete $\mathbf{e}$. After removing the compulsory success, the source mean interval is contained in $[0.975, M]$, where

$$M := 1.5 + 0.22 + 0.073 + 3(0.0055) + 9 \cdot 2 \cdot 10^{-8} = 1.80950018.$$

Its atom at one exceeds 0.172. To see this in every dimension, use the fixed-mean extremizer reduction of Lemma 4.3. A forced success gives $2 - M > 0.172$. One summand is immediate. For $2 \leq n \leq 9$, check the unforced binomial atom at the two mean endpoints. For $n \geq 10$, $M < 20/11 \leq 2n/(n+1)$ gives the lower bound

$$\min\{0.975e^{-0.975}, Me^{-M}\} > 0.172.$$

Return to the law with $\mathbf{f}$ selected but without the extra deletion. Put $X := (V \cup \mathbf{e})_T$, $Y := W_T$, and $Z := X + Y$. This is a proper union cut, so $Z \geq 1$, and the source bounds give

$$\mathbb{E}[X], \mathbb{E}[Y] \leq 1.5 + 2h + d_0, \quad \mathbb{E}[Z] \geq 2 - 4h - 6d_0 = 1 + \nu_*, \quad \nu_* := 1 - 4h - 6d_0.$$

Negative association gives probability at least o for the extra deletion, so $q := \Pr[Z = 2] \geq q_{\min} := o(0.172)$. If $q \geq 0.11$, the tail product $0.6275(0.4284)$ again gives split greater than 0.245, which suffices. Otherwise Lemma 4.11 gives split at least

$$\psi(b(q)), \quad b(q) := 1/2 - 2h - d_0 - q/\nu_*.$$

The derivative of $q\psi(b(q))$ on $[q_{\min}, 0.11]$ is at least

$$\psi(b(0.11)) - 0.11(1 - b(0.11))/\nu_* > 0.$$

Increasing h, d_0 decreases b ; at their largest values, $oq_{\min}\psi(b(q_{\min})) > 0.01330$ by exact substitution. This proves both branches. Finally, $\gamma_{\text{good}}h < 0.01330$, so two incident bad half bundles are impossible. $\square$

Lemma 4.19 (Same-side mixed-bundle lower tail). *In the setting of [KKO21, Lemma 5.23], let $\mathbf{e} = (v, u)$ and $\mathbf{f} = (v, w)$ be good half top bundles, and let A, B, C be the degree partition of $\delta(v)$ with $x_{\mathbf{e}}(B), x_{\mathbf{f}}(B) \leq \omega$. For the fixed parameters and $d \leq d_0$, one bundle satisfies*

$$\Pr[U_T + (A - \mathbf{e})_T \leq 1] \geq 0.49(1 - M_{\text{same}}/2) > Kh,$$

after interchanging the bundles if necessary, where

$$U := \delta(u) - \mathbf{e}, \quad M_{\text{same}} := 1.72 + \omega + 4d_0 + \frac{4\omega + r + d_0}{0.49}.$$

Proof. Orient the bundles using Lemma 4.10. Under $\mathcal{H} := \{\mathbf{f} \notin T, u, v, w \text{ trees}\}$,

$$\mathbb{E}[U_T \mid \mathcal{H}] \leq x(U) + 0.22 + 3d, \quad \Pr[\mathcal{H}] \geq 0.49.$$

Set $c_{\mathbf{e}} := x_{\mathbf{e}}(C)$ and $c_{\mathbf{f}} := x_{\mathbf{f}}(C)$. The degree-partition identities give $x(A) \leq 1 + r + d - x(C)$, $c_{\mathbf{e}} + c_{\mathbf{f}} \leq x(C)$, and

$$x_{\mathbf{e}}(A) + x_{\mathbf{f}}(A) \geq 1 - 2h - 2\omega - c_{\mathbf{e}} - c_{\mathbf{f}}.$$

Thus $x(A - \mathbf{e} - \mathbf{f}) \leq 2h + 2\omega + r + d \leq 4\omega + r + d$. Endpoint-tree conditioning does not increase marginals on this external set. Dividing its mean by the probability of $\mathcal{H}$ and using $x(U) \leq 3/2 + h + d$ gives

$$\mathbb{E}[U_T + (A - \mathbf{e})_T \mid \mathcal{H}] \leq 1.72 + \omega + 4d + \frac{4\omega + r + d}{0.49} \leq M_{\text{same}}.$$

Markov's inequality now proves the claim. The exact comparison at d_0 is positive. No replacement of the residual-mass fraction by $9h + 3d$ is used. $\square$

Lemma 4.20 (Rank-two monotonicity bounds). *Let X, Y, V count disjoint coordinate sets of a strongly Rayleigh law, and put $R := X + Y + V \geq 1$. Suppose $q := \Pr[R = 2] > 0$, and write $p_1 := \Pr[R = 1]$. If $\mathbb{E}[X] \leq U$ and $\Pr[X + V \leq 1] \geq \lambda$, then*

$$\mathbb{E}[X \mid R = 2] \leq \frac{U}{1 - p_1}, \quad \Pr[Y \geq 1 \mid R = 2] \geq \frac{\lambda - p_1}{1 - p_1}.$$

Proof. Project to the union of the three counted sets. Consecutive full-rank laws admit the monotone coupling in [KKO21, Theorem 2.10]. For every $j \geq 2$ in the rank support, it gives

$$\mathbb{E}[X \mid R = j] \geq \mathbb{E}[X \mid R = 2], \quad \Pr[X + V \leq 1 \mid R = j] \leq \Pr[X + V \leq 1 \mid R = 2].$$

Average the first inequality and use nonnegativity at rank one. For the second, the event is automatic at rank one, so

$$\lambda \leq p_1 + (1 - p_1) \Pr[X + V \leq 1 \mid R = 2].$$

At rank two, $X + V \leq 1$ is equivalent to $Y \geq 1$. Since $q > 0$, the denominator $1 - p_1$ is positive. $\square$

Lemma 4.21 (Coupled oriented GKL window estimate). *In the setting of [GKL24, Lemma A.5], use the parameters in Definition 4.2. Let A, B, C be the degree partition at u , and suppose $x_{\mathbf{e}}(B) \leq \omega$ for the good half bundle $\mathbf{e} = (u, v)$. Write $V := \delta(v) - \mathbf{e}$. If*

$$\Pr[(A - \mathbf{e})_T + V_T \leq 1] + x_{\mathbf{e}}(B) \geq Kh,$$

then the bundle is 2-1-1 happy with probability greater than p .

Proof. Condition on the endpoint trees, $C_T = 0$, and $(u \cup v)_T$ being a tree, and call the resulting law ν . The full conditioning has probability at least

$$(1 - d_0)(1/2 - h - 3d_0) - (2r + d_0) \geq o_* := 1/2 - h - 2r - 5d_0 > 0.49.$$

The unique selected bundle edge is independent of the residual tree law. Indeed, after the endpoint trees are contracted, the bundle consists of parallel edges; conditioning their union to be a tree selects one of them. The weighted-tree product factorization separates this choice from the contracted residual tree. Deleting C imposes separate zero conditions on these factors and preserves independence. The boundary-law closure in Lemma 4.10 extends the identity to the exact max-entropy law.

Set $X := (A - \mathbf{e})_T$, $Y := (B - \mathbf{e})_T$, and use V_T for the third count. Their union count is $R := X + Y + V_T = \delta(u \cup v)_T \geq 1$. Lemma 4.16 gives

$$\Pr_{\nu}[R \leq 2] \geq L_* > 0.$$

The unrounded conditional-marginal rows in [KKO21, proofs of Lemmas 5.22 and A.1] give

$$2.5 - 3h - 3r - 8d_0 \leq \mathbb{E}_{\nu}[R] \leq 3 + 2h + 2r + 8d_0,$$

which is contained in $[2.47, 3.02]$. For the lower endpoint, the original residual mean is at least $3 - 2h - 2r - d_0$; endpoint-tree conditioning and bundle selection lose at most $1 - x(\mathbf{e}) + 2d_0 \leq 1/2 + h + 2d_0$. For the upper endpoint, the tree conditions and deletion cost at most $2d_0$ and $2r + 3d_0$, and selection cannot increase a disjoint count. These estimates imply the displayed conservative interval.

Remove the compulsory success from R . Lemma 4.3 gives $\Pr_{\nu}[R = 3] > c_3 := 0.248$. With $p_j := \Pr_{\nu}[R = j]$, Newton's inequality gives $p_1 \leq p_2^2/(2c_3)$. Consequently

$$q := p_2 > m_* := \frac{L_*}{1 + L_*/(2c_3)},$$

because $m_* + m_*^2/(2c_3) < L_* \leq p_1 + p_2$. This calculation will also be used for the direct mixed event. The conditional mean interval need not lie above 2.49, so its rank-three bound is 0.248, not $1/4$.

Let θ be the probability that the selected bundle edge lies in B . The orientation and the outer probability imply

$$0 \leq \theta \leq \theta_B := \omega/o_*.$$

Put $b_e := x_e(B)$ and $D := 1/2 + h + 3r$. Before the union-tree condition, the endpoint-tree conditions lose at most $2d_0$ from the B -bundle marginal, and the entire bundle has marginal at most D . It is binary after the endpoint trees are contracted. The parallel-edge factorization therefore gives selected B probability at least $(b_e - 2d_0)/D$ whenever the numerator is positive. Deleting the bundle part of C only renormalizes this probability upwards, while deleting the residual part does not change it. Thus in all cases

$$b_e \leq D\theta + 2d_0.$$

In the original counts, $A_T = X + 1$ when this edge is in A , and $B_T = Y + 1$ when it is in B . The source conditional-marginal rows give

$$\begin{aligned}\mathbb{E}_\nu[X] &\leq 1/2 + h + 2r + 8d_0 + \theta, \\ \mathbb{E}_\nu[Y] &\geq 1/2 - h - r - 4d_0 - \theta_B, \\ \mathbb{E}_\nu[V_T] &\geq 1 - 3r - 4d_0, \\ \mathbb{E}_\nu[X + Y], \mathbb{E}_\nu[V_T] &\leq 3/2 + h + 2r + 8d_0 < 1.512.\end{aligned}$$

The original individual counts also have means at most 1.512. The two sum means $\mathbb{E}_\nu[A_T + V_T], \mathbb{E}_\nu[B_T + V_T]$ are at least $2 - r - 2d_0$. Lemma 4.3, the zero-probability product bound, and [KKO21, Lemma 2.22] yield

$$\begin{aligned}\Pr_\nu[Y \geq 1] &> 0.37, & \Pr_\nu[V_T \geq 1] &> 0.629, \\ \Pr_\nu[X + Y \leq 1], \Pr_\nu[A_T \leq 1] &> 0.4284, & \Pr_\nu[B_T + V_T \geq 2] &> 0.59.\end{aligned}$$

For example, the first bound follows from $1 - e^{-(1/2 - h - r - 4d_0 - \theta_B)} > 0.37$. The assumed residual lower tail loses at most $2r + 3d_0$ on imposing the tree and C conditions. Selecting the binary bundle coordinate cannot decrease it, by negative association. Thus

$$\Pr_\nu[X + V_T \leq 1] \geq Kh - b_e - 2r - 3d_0 \geq \lambda - D\theta, \quad \lambda := Kh - 2r - 5d_0.$$

Project to the residual union and condition its full rank to two; denote this strongly Rayleigh law by ν_2 . The up/down inequality [GKL24, Lemma A.10], applied to the two disjoint complementary blocks each time, gives

$$\Pr_{\nu_2}[Y \geq 1] \geq 0.37(\lambda - D\theta), \quad \Pr_{\nu_2}[V_T \geq 1] \geq c := (0.629)(0.4284).$$

Whenever $\Pr_{\nu_2}[X = 0] > 0$, delete the entire X block. Negative association increases both nonzero probabilities, and the remaining counts satisfy $Y + V_T = 2$. Write

$$r_V := \Pr_{\nu_2}[V_T = 1], \quad c_Y := \Pr_{\nu_2}[X = 0, Y = V_T = 1].$$

At rank two, the alternative $V_T = 1$ pattern is $X = 1, Y = 0$. Independence of the selected bundle edge therefore gives the exact conditional happiness probability

$$(1 - \theta)c_Y + \theta(r_V - c_Y) = \theta r_V + (1 - 2\theta)c_Y.$$

Both outcomes are retained; in particular $1 - 2\theta > 0$.

Put

$$Q := 0.04, \quad Q_1 := 0.1, \quad z(q) := q^2/(2c_3), \quad A_0 := 1/2 - h - 2r - 8d_0.$$

The two pooled means $\mathbb{E}_\nu[X + Y], \mathbb{E}_\nu[V_T]$ are at most $2 - A_0$. Lemma 4.11 consequently gives $r_V \geq \psi(A_0 - z(q))$ for $q \leq Q_1$. The same source marginal rows bound both pooled means below by $1 - 3r - 4d_0 > 0.994$. For all q , their nonzero tails exceed 0.629 and their CDFs at one exceed 0.4284. Put $a_V := (0.629)(0.4284)$. The up/down inequality gives both conditional tails of V_T at least a_V . Since $0.245 < a_V$ and $4(a_V - 0.245)(1 - a_V) - 0.245^2 > 0$, Lemma 4.4 gives $r_V > 0.245$. For $m_* \leq q \leq Q$, Lemma 4.20 and Markov's inequality give

$$\Pr_{\nu_2}[X = 0] \geq d_\theta := \frac{A_0 - \theta - z(q)}{1 - z(q)}, \quad \Pr_{\nu_2}[Y \geq 1] \geq b_\theta := \frac{\lambda - D\theta - z(q)}{1 - z(q)}.$$

The exact comparisons

$$A_0 - \theta_B - z(Q_1) > 0, \quad \lambda - D\theta_B - z(Q) > 0,$$

$$c > \lambda > 0, \quad \lambda < 1/2, \quad \theta_B < 1/2, \quad 0 < m_* < Q < Q_1, \quad 1 - z(Q_1) > 0$$

make every conditioning positive and give $0 < b_\theta \leq \lambda < c$. After deleting X , Lemma 4.4 gives $c_Y > d_\theta b_\theta(1 - b_\theta)$. For fixed q , the resulting lower bound for happiness is

$$H_q(\theta) := \theta\psi(A_0 - z(q)) + (1 - 2\theta)d_\theta b_\theta(1 - b_\theta).$$

Differentiating, and bounding each negative term separately, gives

$$H'_q(\theta) \geq \psi(A_0 - z(q)) - \frac{(2A_0 + 1)\lambda + A_0 D}{(1 - z(q))^2} \geq \psi(A_0 - z(Q)) - \frac{(2A_0 + 1)\lambda + A_0 D}{(1 - z(Q))^2} > 0.$$

The minimum is therefore at $\theta = 0$. The rank-weighted factor is

$$F(q) := \frac{q(1 - \lambda)(A_0 - z(q))(\lambda - z(q))}{(1 - z(q))^3}.$$

Its derivative has the sign of

$$\mathcal{G}(z) := A_0\lambda + (5A_0\lambda - 3A_0 - 3\lambda)z + (5 - 3A_0 - 3\lambda)z^2 + z^3.$$

On $z(m_*) \leq z \leq z(Q)$, exact substitution gives

$$\mathcal{G}'(z) \leq 5A_0\lambda - 3A_0 + 10z(Q) + 3z(Q)^2 < 0.$$

Thus F' changes sign at most once, from positive to negative, so the minimum is at an endpoint. Restoring the outer condition gives probability greater than

$$W_0 := o_* \min\{F(m_*), F(Q)\} > p.$$

This argument uses the derivative sign, not a concavity assertion.

For $Q \leq q \leq Q_1$, keep d_θ and use $b_\theta := \eta(\lambda - D\theta)$, where $\eta := 0.37$. Deletion gives the same happiness expression H_q , whose derivative is at least

$$\psi(A_0 - z(Q_1)) - \eta \frac{(2A_0 + 1)\lambda + A_0 D}{(1 - z(Q_1))^2} > 0.$$

Again minimize at $\theta = 0$ and put $b := \eta\lambda$. The derivative of $q(A_0 - z(q))/(1 - z(q))$ has the sign of $A_0 - (3 - A_0)z(q) + z(q)^2$, which is positive because $A_0 - 3z(Q_1) > 0$. The restored probability is consequently greater than

$$W_1 := o_* Q \frac{A_0 - z(Q)}{1 - z(Q)} b(1 - b) > p.$$

Finally suppose $q \geq Q_1$. The original total count $A_T + B_T + V_T = R + 1$. Applying the up/down inequality to A_T and $B_T + V_T$ at full rank three gives

$$\Pr_{\nu}[A_T \leq 1 \mid R = 2] \geq \alpha := (0.4284)(0.59).$$

The independent bundle choice implies

$$\Pr_{\nu}[A_T \leq 1 \mid R = 2] = (1 - \theta) \Pr_{\nu_2}[X = 0] + \theta \Pr_{\nu_2}[X \leq 1].$$

Hence $\Pr_{\nu_2}[X = 0] \geq \alpha - \theta > 0$. Using $r_V > 0.245$ and the same deletion, the happiness probability exceeds

$$0.245\theta + (1 - 2\theta)(\alpha - \theta)\eta(\lambda - D\theta)(1 - \eta(\lambda - D\theta)).$$

Its derivative is at least $0.245 - \eta((1 + 2\alpha)\lambda + \alpha D) > 0$. Minimizing at $\theta = 0$ and restoring the outer condition gives

$$W_2 := o_* Q_1 \alpha b(1 - b) > p.$$

All three comparisons and derivative margins are exact rational inequalities for each of the four parameter rows. Every rank condition is imposed after projection to the full counted union; all other conditions are extremal or coordinate deletions. $\square$

Lemma 4.22 (Small-excess joint atom). *Let a homogeneous strongly Rayleigh law have disjoint counts X, Y , with $X + Y \geq 1$. Suppose $0 < \alpha \leq A_0 := 1/150$, $0 \leq \varepsilon < \alpha/100$, and*

$$\begin{aligned} \mathbb{E}[X], \mathbb{E}[Y] &\in [\alpha - \varepsilon, 2 + 3\varepsilon - \alpha], \\ \mathbb{E}[X + Y] &\in [1 + \alpha, 2 + 2\varepsilon]. \end{aligned}$$

Then $\Pr[X = Y = 1] \geq 0.94\alpha^2$.

Proof. Write $a = \min\{\mathbb{E}[X], \mathbb{E}[Y]\}$, b for the other mean, and $m = a + b$; rename X so that $\mathbb{E}[X] = a$. If $a < 1$, then $\Pr[X = 1] \geq ae^{-a} > 0$. Project away the entire X block. If the original fixed rank is n , conditioning $X = 1$ is conditioning the full remaining rank to $n - 1$. Thus Lemma 4.12 gives, with $p_0 = \Pr[X = 0] \leq e^{-a}$,

$$b - p_0 \leq \mathbb{E}[Y \mid X = 1] \leq m - 1 + p_0. \quad (3)$$

This uses homogeneity and projection to the complement, not arbitrary subset-rank conditioning while retaining the selected coordinates.

First suppose $a \leq 1.3\alpha$. Then $a \geq 0.99\alpha$ and $a < 1$. Using $e^{-a} \leq 1 - a + a^2/2$, the conditional lower mean is at least $\alpha - a^2/2 > 0.963\alpha$. Its upper mean is at most $2 - d_*\alpha$, where

$$d_* := 0.97 - 0.99^2 A_0/2 > 0, \quad d_*\alpha < 0.02, \quad d_*(1 - d_* A_0/2) > 0.963.$$

For conditional mean at most one, use the bound $\mu e^{-\mu}$; for conditional mean at least one, use the near-two atom bound in Lemma 4.3 with $\delta = d_*\alpha$. Both give an atom at least $0.963\alpha e^{-0.963\alpha}$. Hence the joint atom is at least

$$0.99(0.963)\alpha^2 e^{-(0.99+0.963)\alpha} \geq 0.99(0.963)e^{-(0.99+0.963)/150}\alpha^2 > 0.94\alpha^2.$$

Next suppose $a \geq 1.3\alpha$ and $a^2 \leq \alpha/2$. Then $a < 0.06$. The conditional lower mean in (3) is at least 0.75α . Its deficit from two is at least

$$a - a^2/2 - 2\varepsilon \geq (0.97(1.3) - 0.02)\alpha > \alpha.$$

The same two scalar branches give conditional atom at least $0.75\alpha e^{-0.75\alpha}$. The joint atom is at least

$$1.3(0.75)e^{-(1.3+0.75)/150}\alpha^2 > 0.94\alpha^2.$$

Finally suppose $a^2 \geq \alpha/2$. Since $\alpha \leq 1/150 < 1/128$, we have $a > 8\alpha$. All subset means are strictly between their neighboring target integers for target $(1, 1)$. The asymmetric capacity distances satisfy

$$\delta_1 = a, \quad \delta_2 = m - 1 \geq \alpha, \quad \epsilon_1 = 2 - b \geq a - 2\epsilon, \quad \epsilon_2 = 3 - m \geq 1 - 2\epsilon.$$

Each of $\delta_1\delta_2$, $\epsilon_1\delta_1$ and $\epsilon_1\epsilon_2$ is at least $a\alpha$. For the last product use

$$a - 2\epsilon \geq (399/400)a, \quad (399/400)(1 - A_0/50) > A_0 \geq \alpha.$$

The capacity form of [GKL24, Theorem 4.1 and Corollary 5.13] gives $\text{Cap}_{(1,1)} \geq a\alpha$. Projection and $X + Y \geq 1$ give zero constant term. The compulsory bivariate coefficient bound Lemma 4.9 gives joint atom at least $(e^{-1}/2)a\alpha > 4e^{-1}\alpha^2 > 0.94\alpha^2$. The three closed cases cover all a ; the first and third need not both apply, and their separating comparison $1.3^2 A_0 < 1/2$ is certified. $\square$

Lemma 4.23 (Accepted-side variation). *If $x, v \geq 0$, $0 \leq q < 1$, $\sum v_e = 1$, $\sum x_e \in [1 - \epsilon, 1 + \epsilon]$, and $(1 - q)v_e \leq x_e$, then $\sum_e |v_e - x_e| \leq 2q + \epsilon$.*

Proof. Put $w = x - (1 - q)v \geq 0$. Then $\sum |v - x| = \sum |qv - w| \leq q + \sum w = 2q + \sum x - 1 \leq 2q + \epsilon$. $\square$

Lemma 4.24 (Marginal-preserving polygon event). *Let a homogeneous strongly Rayleigh law have disjoint counts A_T, B_T , means in $[1 - \epsilon, 1 + \epsilon]$, and $A_T + B_T \geq 1$. Suppose $0 \leq \epsilon$ and $330\epsilon < \zeta < 1/200$. Put*

$$q = (\zeta - \epsilon)/2, \quad b = q - \epsilon, \quad c = q - 5\epsilon, \quad v = \frac{27}{4}(0.94)c^2.$$

There is a pair-dependent accepted event on which $A_T = B_T = 1$, with probability at least $v(1 - q)$, both separate marginal budgets at most ζ , and accepted marginals at most $x_e/(1 - q)$.

Proof. For $A' \subseteq A, B' \subseteq B$ of total mean at least $1 + \alpha$, delete their complement in $A \cup B$. The deletion probability is at least $\alpha - 2\epsilon$. Homogeneous deletion domination gives individual means in $[\alpha - \epsilon, 2 + 3\epsilon - \alpha]$ and sum in $[1 + \alpha, 2 + 2\epsilon]$. The compulsory union survives. Put $A_0 := 1/150$. For $100\epsilon < \alpha \leq A_0$, Lemma 4.22 gives the unconditioned target probability at least

$$0.94(\alpha - 2\epsilon)\alpha^2.$$

For $\alpha \geq A_0$, use that lemma at A_0 , but retain the actual deletion probability. The lower bound is then $0.94(\alpha - 2\epsilon)A_0^2$.

Use the bipartite network of [KKO21, Proposition 5.6]. Its middle capacities are $y_{ef} = \Pr[e, f \in T \mid A_T = B_T = 1]$, and its outer capacities are $\beta x_e, \beta x_f$, where $\beta = v/p_{11}$. The full-side case above proves $p_{11} > 0$. For a network cut put $\gamma = x(S_A) - x(S_B)$. Exterior arcs contribute at least $\beta(1 - \epsilon - \gamma)$. If $\gamma \leq b$, this suffices. Otherwise set $\alpha = \gamma - \epsilon$ and $x = \gamma - b > 0$. The domain gives $q > 102\epsilon$, so $\alpha > 100\epsilon$. For $\alpha \leq A_0$, the middle crossing probability is at least

$$0.94(\alpha - 3\epsilon)^3 = 0.94(x + c)^3 \geq vx,$$

using $(x + c)^3 - (27/4)c^2x = (x - c/2)^2(x + 4c) \geq 0$.

For $\alpha \geq A_0$, note $\varepsilon < 1/66000$, $c < q < 1/400$ and $x = \alpha - q + 2\varepsilon \leq \alpha + 2\varepsilon$. The ratio $(\alpha - 2\varepsilon)/(\alpha + 2\varepsilon)$ increases with α , and the exact inequality

$$(A_0 - 2/66000)A_0^2 > \frac{27}{4}(1/400)^2(A_0 + 2/66000)$$

therefore gives $0.94(\alpha - 2\varepsilon)A_0^2 \geq vx$. This retains the actual deletion factor in the large-cut case. Every source–sink cut has capacity at least $\beta(1 - q)$.

Take a flow of value at least this number. Given a selected pair, accept with probability its flow divided by y_{ef} ; zero-capacity pairs never occur. Flow conservation gives accepted side mass one and coordinate domination $x_e/(1 - q)$. Lemma 4.23, separately on each side, gives $2q + \varepsilon = \zeta$. Acceptance uses only the pair and one independent coin. Independent thinning preserves its conditional law. $\square$

Lemma 4.25 (Common-event probabilities). *At the parameters in Definition 4.2, the polygon event has probability at least P , and the small-, large-, mixed-, two-sided mixed-, and paired-bundle happiness events have probability at least p . Each event can be thinned to its designated probability without changing its conditional law.*

Proof. For each polygon cut S , use the construction in [KKO21, Definition 5.8], replacing its max-flow event by Lemma 4.24 with $\zeta = \epsilon_M$ and $\epsilon = 2d$. The unchanged polygon conditioning gives a homogeneous strongly Rayleigh law with $\mathbb{E}[A_T], \mathbb{E}[B_T] \in [1 - 2d, 1 + 2d]$, and $A_T + B_T = \delta(S)_T \geq 1$. The cut is proper even for a hierarchy-root polygon, so every premise of the lemma holds. Repeating the proof of [KKO21, Corollary 5.9], rather than invoking its specialization $\epsilon_M = 1/4000$, multiplies this probability by the probability that $C_T = 0$ and the polygon cut is a tree. The latter probability is at least $1 - 3d/2$. Define

$$q_M := (\epsilon_M - 2d_0)/2.$$

The unconditional probability is at least

$$\frac{27}{4}(0.94)(q_M - 10d_0)^2(1 - q_M)(1 - 3d_0/2) > P.$$

The new flow lemma requires $330(2d_0) < \epsilon_M < 0.005$. Both inequalities hold, and we also have $d_0 < \epsilon_M^2$. The enlarged polygon-payment domain is proved directly in Lemma 4.39. The lower bound decreases with d : the function $(q - 5\epsilon)^2(1 - q)$ increases with q and decreases with ϵ on $0 \leq 5\epsilon < q < 0.0025$, while q decreases as d increases. The two separate boundary marginal errors remain at most ϵ_M . Acceptance depends only on the boundary pair and an independent coin, so the internal/external factorization in [KKO21, Corollary 5.9] is preserved. Thin independently to probability P and call this event E_S . Its conditional marginals and the conditional Bernoulli-sum conclusion of that corollary are unchanged by the thinning.

Small non-half bundles. Use the conditioning of [KKO21, proof of Lemma 5.21]. The endpoint trees remain trees, and both the bundle and degree remainder are deleted. Its probability is at least $o_s := 1/2 + h - 2r - 3d_0 > 0$. Put $X = A_T$, $Y = B_T$, $W = V_T - 1$, and $R = X + Y + W$. The proper-cut constraints give $V_T \geq 1$, $X + Y \geq 1$, and hence $R \geq 1$. Divide the joint stable polynomial by the compulsory V variable and polarize. The target event is now $X = Y = W = 1$.

Write $q_e := x(\mathbf{e}) \leq 1/2 - h$. Before the conditioning, $x(A), x(B) \in [1 - r, 1 + d]$, $x(\delta(v)) \in [2, 2 + d]$, and $x_e(A) + x_e(B) \leq q_e$. The conditional-marginal calculation in the cited proof decreases any displayed union by at most $2d$ and increases it by at most $q_e + 2r + 3d$. For example,

$$\mathbb{E}[B_T + V_T] \geq x(B) + x(\delta(v)) - q_e - x_e(B) - 2d \geq 2 + 2h - r - 2d.$$

Substitution gives the complete shifted table for $0 \leq d \leq d_0$:

union	lower mean	upper mean
X, Y	$1/2 + h - r - 2d$	$3/2 - h + 2r + 4d$
W	$1/2 + h - 3d$	$1 + 2r + 4d$
$X + Y$	$3/2 + h - 2r - 2d$	$2 + 4r + 6d$
$X + W, Y + W$	$1 + 2h - r - 2d$	$2 + 2r + 5d$
R	$2 + 2h - 2r - 2d$	$3 + 2r + 6d$

The double entries denote both distinct subsets, not omitted constraints. At the final parameters every coordinate mean is at most 1.5, every pair mean is at most 2.01, and all individual nonzero probabilities exceed 0.39. Put

$$\delta_2 = 2h - r - 2d_0, \quad \delta_3 = 2h - 2r - 2d_0, \quad L_0 = 1/2 + h - r - 3d_0, \quad q_s = \delta_3 c(\delta_3), \quad Q_s = 0.04.$$

Every pair mean is at least $1 + \delta_2$ and $\mathbb{E}[R] \in [2 + \delta_3, 3.01]$. The shifted Poisson-binomial law $R - 1$ and Lemma 2.21 of [KKO21] give $q := \Pr[R = 3] \geq q_s > 0$. For shifted means above 1.2, the retained 0.1637 target-two bound is stronger than q_s ; below 1.2 the correction exponent is zero and the one-forced branch gives $\delta_3 e^{-\delta_3}$.

Small rank-three probability. Suppose $q \leq Q_s < 0.1637$. Then $\mathbb{E}[R - 1] < 1.2$, so $\Pr[R = 2] > 0.36$. Lemma 4.6, applied to $R - 1$, bounds $\mathbb{E}[(R - 3)_+]$ by $T_{0.36}^{\text{ex}}(q)$.

Cover $[q_s, Q_s]$ by the closed intervals $[l_i, u_i]$, where

$$l_i = q_s + (Q_s - q_s)i/128, \quad u_i = q_s + (Q_s - q_s)(i + 1)/128 \quad (0 \leq i < 128).$$

For the interval containing the actual q , put

$$T_i = T_{0.36}^{\text{ex}}(u_i), \quad \ell_i = L_0 - T_i, \quad \delta_i = \delta_2 - T_i, \quad v_i = \delta_i c(\delta_i).$$

Under the full rank-three law, Lemma 4.12 gives individual means at least ℓ_i and pair means at least $1 + \delta_i$. Choose a coordinate of largest mean and call the other two Y, Z . Then both remaining means are at most 1.5, and their sum belongs to $[1 + \delta_i, 2]$. This choice is made from the law, not from the sampled tree.

If $q' = \Pr[Y + Z = 2]$ under this rank-three law is at least 0.1, the individual nonzero tails 0.38 and CDF tails 7/16 give quadratic split greater than 0.15. Thus the joint atom $Y = Z = 1$ exceeds 0.015. If $q' < 0.1$, the retained target-two exclusion forces the pair mean below 1.2, so its rank-one atom exceeds 0.36, while $q' \geq v_i$. Apply Lemma 4.7 with $c = 0.36, L_0 = \ell_i$. Writing its function as F_i , the exact checks give

$$\ell_i - 0.2 - T_{0.36}(0.1) > 0, \quad \Phi(0.1) > 0, \quad W_i(v_i) > 0, \quad 0 < v_i < 0.1.$$

Monotonicity of $T, W, -\Phi$ and concavity of F_i cover the entire inner interval $[v_i, 0.1]$. Therefore the unconditional event is at least

$$o_s l_i \min\{0.015, F_i(v_i), F_i(0.1)\} > p.$$

All 128 inequalities are exact and their minimum exceeds p in each row. The evaluation rules and positive margins are given in Section 4.9. The displayed endpoints satisfy $l_0 = q_s, u_{127} = Q_s$, and $u_i = l_{i+1}$ for $0 \leq i < 127$.

Large rank-three probability. Suppose $q \geq Q_s$. The up/down inequality, applied to each coordinate and its complement at rank three, gives for each coordinate D

$$\Pr[D \geq 1 \mid R = 3] \geq \alpha := 0.39(0.66), \quad \Pr[D \leq 1 \mid R = 3] \geq \beta := (7/16)\delta_2 c(\delta_2).$$

The first uses the CDF at two, and the second uses the pair upper tail from Lemma 4.3. Here $0 < \beta < 0.01$. Put $c_0 := 0.4911$ and $b_* := \beta/(1 + \beta/c_0)$. Lemma 4.14, under this full rank-three conditioning, gives the unconditional lower bound

$$o_s Q_s b_* \psi(\alpha - b_*^2/c_0) > p.$$

These two ranges exhaust q . The small-event support used here is only $R \geq 1$, not the unproved stronger condition $R \geq 2$.

Large non-half bundles. Use the endpoint and union tree conditions and remainder deletion of [KKO21, proof of Lemma 5.22]. The conditioning probability is at least $o_l = 1/2 + h - 2r - 3d_0 > 0$. For the unshifted counts $X = A_T, Y = B_T, Z = V_T$, the support constraints are $X + Y \geq 1$ and $R := X + Y + Z \geq 2$. Exactly one bundle edge is selected, and $V = \delta(v) \setminus \delta(u)$. Thus

$$A_T + B_T + V_T = 1 + \delta(u \cup v)_T \geq 2.$$

The union is proper even if it equals the hierarchy root, since it omits the distinguished vertices. The unrounded conditional-marginal estimates of the cited proof give the complete table

union	lower mean	upper mean
X, Y	$1/2 + h - r - 2d$	$3/2 - h + r + 3d$
Z	$1 - 3d$	$3/2 - h + 2r + 4d$
$X + Y$	$2 - 3r - 3d$	$5/2 - h + 3d$
$X + Z, Y + Z$	$2 - 3r - 4d$	$3 - (2h - r - 6d)$
R	$3 - 3r - 5d$	$4 - (2h - 6d),$

Put $\Delta_p := 2h - r - 6d_0$, $\Delta_t := 2h - 6d_0$, and $U_0 = 3/2 - h + 2r + 4d_0$. To justify the upper rows, under only endpoint trees put $q = y(e)$, $a_A = y(e(A))$, $c_e = y(e(C))$, and $D = q - c_e$. Then $q \geq 1/2 + h - 2d$, $D > 0$, $0 \leq a_A \leq D \leq 1$, and $y(A) + y(C) \leq 1 + r + d$. Selecting the bundle and deleting its C part makes its independent A probability a_A/D . Residual C deletion costs at most $y(C) - c_e$. Using $a_A(1/D - 1) \leq 1 - D$ gives respectively $2 + r + d - q$, $1 + y(\delta(u)) - q$, $4 + r + 2d - 2q$, and $1 + y(\delta(u)) + y(\delta(v)) - 2q$ for the relevant upper means. Substitution proves the upper rows. The lower rows follow from the original endpoint-tree and deletion estimates.

These lower endpoints imply $U_0 < 1.5$, pair means at most $3 - \Delta_p$, $\mathbb{E}[X + Y] \leq 2.5$, and pair upper tails at two greater than 0.59. For the latter, thin the Bernoulli parameters to mean $2 - 3r - 4d_0 \in (1, 2)$ and use the two branches of [KKO21, Lemma 2.22].

The shifted rank $R - 2$ has mean in $[0.99, 2 - \Delta_t]$. Lemma 4.3 gives

$$q := \Pr[R = 3] \geq q_l := \Delta_t(1 - \Delta_t/2) > 0.$$

If $q \leq 0.12$, its mean cannot lie in $[0.99, 1.49]$, where the rank-one atom exceeds 0.33. Hence $\mathbb{E}[R - 2] > 1.49$ and Lemma 4.3 gives $\Pr[R = 4] > 1/4$. Newton yields $\Pr[R = 2] \leq 2q^2$. Lemma 4.12 at rank three therefore gives

$$\mathbb{E}[D \mid R = 3] \leq U_0 + 2q^2, \quad \mathbb{E}[D \mid R = 3] \geq \Delta_p - 2q^2$$

for every coordinate D , using its complementary pair for the lower bound.

The range $q_l \leq q \leq Q_l = 0.025$. The conditional means lie below 1.504 and above $\ell(q) = \Delta_p - 2q^2 > 0$, with $\ell(q) < 0.01$. Lemma 4.13 gives full probability at least

$$o_l(0.3728)q\ell(q)e^{-\ell(q)}.$$

The derivative of $q\ell(q)e^{-\ell(q)}$ has the sign of

$$\Delta_p + (4\Delta_p - 6)q^2 - 8q^4.$$

This expression strictly decreases as a function of q^2 , so the original function has no interior minimum. Checking its two endpoints gives probability greater than p .

The range $Q_l \leq q \leq 0.12$. Every rank-three coordinate mean is below 1.53. The original nonzero tails for X, Y exceed 0.39; their complementary pairs have CDF at two at least $\Delta_p(1 - \Delta_p/3)$ by Lemma 4.3. The original Z nonzero tail exceeds 0.63, and $\Pr[X + Y \leq 2] > 0.42$. Thus every conditional nonzero probability, and hence every conditional mean, is at least

$$\alpha_l := 0.39\Delta_p(1 - \Delta_p/3) \in (0, 0.01).$$

Lemma 4.13 gives full probability at least

$$o_l Q_l(0.3594)\alpha_l e^{-\alpha_l} > p.$$

The range $q \geq 0.12$. The support $X + Y \geq 1$ gives $Z \leq 2$ under $R = 3$. Each original CDF at one is at least $7/16$, and each complementary pair upper tail at two exceeds 0.59. Hence the conditional CDF of Z at one is at least $\beta_l = (7/16)(0.59)$. Its conditional nonzero tail exceeds $0.63(0.42) > \beta_l$. The stable quadratic discriminant therefore gives $\Pr[Z = 1 \mid R = 3] > 0.238$. The rank-three nonzero and CDF tails of X, Y are at least α_l, β_l . Project to X, Y and condition the full rank to two; the conditioning is positive by the preceding bound. Its two nonzero tails are at least $\alpha_l \beta_l$, so Lemma 4.4 gives split greater than $\alpha_l \beta_l(1 - \alpha_l \beta_l)$. The full probability exceeds

$$o_l(0.12)(0.238)\alpha_l \beta_l(1 - \alpha_l \beta_l) > p.$$

The three closed ranges cover every possible q . All projections, support exclusions, and positive conditioning margins have been specified.

For the mixed-bundle event, Lemma 4.19 supplies the premise of Lemma 4.21: after orienting $\mathbf{e}$ from v to u , its residual set V is the set U used there, and its probability exceeds Kh .

For the two-sided mixed event in [GKL24, Lemmas A.7 and A.12], use their law ν conditioned on $C_T = 0$ and $u, v, u \cup v$ being trees. This conditioning has probability at least 0.49. Let $X := (A - \mathbf{e}) \dot{\cup} (B - \mathbf{e})$ and $Y := \delta(v) - \mathbf{e}$, which are disjoint. The source conditional-marginal bounds, with a conservative error allowance, give

$$\mathbb{E}_\nu[X_T], \mathbb{E}_\nu[Y_T] \in [1 - 3r, 3/2 + h + 2r + 8d_0] \subset [0.994, 1.512],$$

$$\mathbb{E}_\nu[X_T + Y_T] \in [5/2 - 3r - 3h - 8d_0, 3 + 2h + 2r + 8d_0] \subset [2.47, 3.02].$$

For the individual upper bounds, the original residual cut has mean at most $3/2 + h + d_0$; the endpoint trees and deletion contribute at most $2d_0$ and $2r + 3d_0$, and selecting the bundle cannot increase a disjoint residual count. The total-count bounds follow from the same calculation given in the window proof.

The projected ranks are Bernoulli sums. Their zero-probability product bound and Lemma 4.3 give

$$\Pr_\nu[X_T \geq 1], \Pr_\nu[Y_T \geq 1] > 0.629, \quad \Pr_\nu[X_T \leq 1], \Pr_\nu[Y_T \leq 1] > 0.4284.$$

Project to $X \cup Y$ and condition its full rank to two; this preserves stability. Lemma A.10 of [GKL24] gives both conditional tails of X_T at least $(0.629)(0.4284)$. The same quadratic test as in Lemma 4.21 yields

$$\Pr_\nu[X_T = Y_T = 1 \mid X_T + Y_T = 2] > 0.245.$$

Furthermore $X_T + Y_T = \delta(u \cup v)_T \geq 1$, and Lemma 4.16 gives its lower tail at least L_* . The forced-one central-rank calculation in the window proof applies throughout the displayed mean interval, and hence

$$\Pr_\nu[X_T + Y_T = 2] > m_* = \frac{L_*}{1 + L_*/(2c_3)}.$$

Put $q := \Pr_\nu[X_T + Y_T = 2]$ and

$$b_D = 1/2 - h - 2r - 8d_0, \quad v_D = b_D - (0.1)^2/(2c_3).$$

The same target-two bound gives $\Pr_\nu[X_T + Y_T = 3] > c_3$. Newton's inequality for the shifted rank therefore yields

$$\Pr_\nu[X_T + Y_T = 1] \leq q^2/(2c_3).$$

For $m_* \leq q \leq 0.1$, Lemma 4.11 gives conditional central split at least $\psi(b_D - q^2/(2c_3))$. Both mean arguments are in $(0, 1)$, and

$$\psi(v_D) - \frac{(0.1)^2}{c_3}(1 - v_D) > 0.$$

Therefore $q\psi(b_D - q^2/(2c_3))$ increases on this interval. For $q \geq 0.1$, use the preceding 0.245 split, which gives joint probability at least $0.0245 > m_*\psi(b_D - m_*^2/(2c_3))$. The unique parallel-edge choice is independent of the residual tree law. Its required endpoint probability is at least $(\omega - 2d_0)/(1/2 + h + 3r) > 1.96\omega$, and the outer event has probability at least 0.49. Hence the direct-mixed event has probability greater than

$$(1.96)(0.49)\omega m_*\psi(b_D - m_*^2/(2c_3)) > p.$$

This use requires mass at least ω in both degree parts. The oriented window estimate handles the complementary cases.

The sixth event is the paired event in Lemma 4.31, which uses Lemma 4.21 with the fixed K . Thus the top-bundle events have probability at least p , and the polygon event has probability at least P . Independent thinning gives their designated probabilities while preserving each conditional law. The remaining numerical premise in [KKO21, Theorem 5.28] holds as well: $p > 0.005h^2$. $\square$

Lemma 4.26 (Half-budget and small-remainder polygon bounds). *Let $E_{\mathcal{P}}$ be the accepted event of a polygon ancestor $\mathcal{P}$, including its independent thinning to probability P . For every proper descendant u in the hierarchy,*

$$\Pr[\delta(u)_T \text{ odd} \mid E_{\mathcal{P}}] < q_0.$$

If u is a polygon cut, its conditional left and right unhappiness probabilities are at most q_{unh} .

Proof. Write x' for the marginals after making the parent a tree and deleting its remainder, before accepting the boundary pair. On either accepted boundary side, the accepted marginal vector v satisfies

$$\sum_e v_e = 1, \quad \sum_e |v_e - x'_e| \leq \epsilon_M, \quad \left| \sum_e x'_e - 1 \right| \leq 2d.$$

Positive and negative variation therefore each have mass at most $\epsilon_M/2 + d$. For every subset F of one side,

$$|v(F) - x'(F)| \leq \epsilon_M/2 + d.$$

The internal parent tree and the contracted tree are independent. Making the parent a tree changes an internal mean by an amount in $[0, d/2]$. On surviving exterior coordinates, the tree and remainder

conditions change an edge-set mean by an amount in $[-d/2, d]$; a deleted exterior part has original mass at most d . Consequently every count contained in the parent and its boundary, and meeting at most one surviving boundary side, changes in mean by at most $\epsilon_M/2 + 3d$ under $E_{\mathcal{P}}$.

The internal count is Poisson-binomial and independent of the exterior count, which is zero or one. Their sum is therefore Poisson-binomial; this does not assert that the entire accepted pair law is strongly Rayleigh. By [KKO21, Observation 4.32], the exterior part of every descendant cut is contained in one parent side or in its deleted remainder. The cut count has a compulsory success. After removing it, the mean lies between $1 - \epsilon_M/2 - 3d_0$ and $1 + \epsilon_M/2 + 4d_0 < 1.2$. The parity bound in [KKO21, Corollary 2.17], followed by the Taylor bound defining q_0 , proves the first assertion.

Now let A, B, C be the descendant polygon partition. There is stronger control on the small remainder $F := C$, whose original mass is at most d . Flow conservation in Lemma 4.24 gives $v_e \leq x'_e/(1 - q(d))$, where

$$q(d) := (\epsilon_M - 2d)/2 \leq q(0) = \bar{q}.$$

The internal part gains at most $d/2$, and the surviving exterior part gains at most d before acceptance. Hence

$$\begin{aligned} \mathbb{E}[F_T \mid E_{\mathcal{P}}] &\leq x(F \cap E(\mathcal{P})) + d/2 + \frac{x(F \cap \delta(\mathcal{P})) + d}{1 - \bar{q}} \\ &\leq d/2 + \frac{2d}{1 - \bar{q}} \leq \frac{3d}{1 - \bar{q}}. \end{aligned}$$

Deleted exterior coordinates contribute zero. The uniform denominator uses $q(0) = \bar{q}$, not $q(d_0)$. The side count $A_T \mid E_{\mathcal{P}}$ is also Poisson-binomial, with mean at least $1 - \epsilon_M/2 - 4d_0$ and at most $1 + \epsilon_M/2 + 4d_0 < 1.2$. For $f(x) := (1 + e^{-2x})/2$, $|f'(x)| < 1$ for $x > 0$. Its even probability is thus at most $q_0 + d_0$. The union bound and Markov's inequality give

$$\begin{aligned} \Pr[u \text{ not left happy} \mid E_{\mathcal{P}}] &\leq q_0 + d_0 + \frac{3d_0}{1 - \bar{q}} \\ &\leq q_0 + \frac{4d_0}{1 - \bar{q}} = q_{\text{unh}}. \end{aligned}$$

The right side is symmetric. Independent thinning preserves every conditional distribution used in the proof. $\square$

4.4 Paired bundles and localized matching

Lemma 4.27 (Monotone marginal-shift bound). *Let ν_- be a strongly Rayleigh law on E , let $F \subseteq E$, and let ν_+ be the law obtained by a positive-probability conditioning on $F_T = 0$. Let $D \subseteq L \subseteq E \setminus F$, and suppose that for some j and nonnegative ℓ_-, ℓ_+, u_-, u_+ ,*

$$\mathbb{E}_{\nu_-}[L_T] \in [j - \ell_-, j + u_-], \quad \mathbb{E}_{\nu_+}[L_T] \in [j - \ell_+, j + u_+].$$

Then

$$0 \leq \mathbb{E}_{\nu_+}[D_T] - \mathbb{E}_{\nu_-}[D_T] \leq u_+ + \ell_-.$$

If instead $F_T \leq 1$ almost surely under ν_- and ν_+ is the law obtained by a positive-probability conditioning on $F_T = 1$, then

$$0 \leq \mathbb{E}_{\nu_-}[D_T] - \mathbb{E}_{\nu_+}[D_T] \leq u_- + \ell_+.$$

For the zero conditioning, the upper bound $u_+ + \ell_-$ remains valid when $D \subseteq L \subseteq E$ and $F \cap L \subseteq D$; in this extension no lower sign is asserted for the change of D_T .

Proof. By [KKO21, Fact 2.11], every coordinate outside F has nonnegative marginal change under deletion. Since $L \setminus D$ is disjoint from F , including in the overlap extension,

$$\mathbb{E}_{\nu_+}[D_T] - \mathbb{E}_{\nu_-}[D_T] \leq \mathbb{E}_{\nu_+}[L_T] - \mathbb{E}_{\nu_-}[L_T] \leq u_+ + \ell_-.$$

The change on D is nonnegative when D is disjoint from F . The upward-conditioning part of the same fact reverses the signs and gives the selection bound by the identical subtraction. Both conditionings preserve stability. Deletion is specialization at zero. For the binary selection, write the generating polynomial as $p = p_0 + p_1$ according to its degree in the F variables. Positive rescaling and the limit

$$t^{-1}p(tz_F, z_{E \setminus F}) \rightarrow p_1(z_F, z_{E \setminus F})$$

as $t \rightarrow \infty$ give stability of the nonzero conditioned polynomial. No arbitrary subset-rank conditioning is used. $\square$

Lemma 4.28 (Conditioning excess moments). *Let X be integer-valued, let j be an integer, and let $l, u \geq 0$. Suppose $\mathbb{E}[(j - X)_+] \leq l$ and $\mathbb{E}[(X - j)_+] \leq u$. For any event A with $\Pr[A] \geq P > 0$,*

$$j - l/P \leq \mathbb{E}[X \mid A] \leq j + u/P.$$

The assertion does not require independence or a central-atom bound.

Proof. Intersect each nonnegative excess variable with A , bound its expectation by its unconditional expectation, and divide by $\Pr[A]$. Subtract the resulting lower and upper excess bounds. $\square$

Lemma 4.29 (Nested conditioning of excess moments). *Let $A \subseteq H$ be positive-probability events, with actual probabilities $a = \Pr[A] \leq b = \Pr[H]$. Suppose*

$$\mathbb{E}[(j - X)_+] \leq l, \mathbb{E}[(X - j)_+] \leq u, \quad 0 \leq l \leq u.$$

Then

$$\begin{aligned} \mathbb{E}[X \mid A] - \mathbb{E}[X \mid H] &\leq u/a + (l - u)/b, \\ \mathbb{E}[X \mid H] - \mathbb{E}[X \mid A] &\leq l/a + (u - l)/b. \end{aligned}$$

If $a \geq A_ > 0$ and $b \geq H_* > 0$, the right sides may be replaced by $u/A_* - (u - l)$ and $l/A_* + (u - l)/H_*$, respectively.*

Proof. Writing $Z = X - j$ gives

$$\mathbb{E}[X \mid A] - \mathbb{E}[X \mid H] = (1/a - 1/b) \mathbb{E}[Z1_A] - \mathbb{E}[Z1_{H \setminus A}]/b.$$

The two coefficients are nonnegative. Bound the first positive part by u and the second negative part by l to obtain the first inequality. Reverse their roles for the second inequality. For the relaxed first bound use $b \leq 1$ and $l - u \leq 0$; for the second use $b \geq H_*$ and $u - l \geq 0$. $\square$

Lemma 4.30 (Asymmetric binary separator). *Let a homogeneous strongly Rayleigh law have a binary counted block E and a disjoint count D . Put $q = \Pr[E = 1]$, $S = D + 2E$. If*

$$\mathbb{E}[(k - S)_+] \leq L, \mathbb{E}[(S - k)_+] \leq U,$$

then

$$qL + (1 - q)U \geq q(1 - q).$$

Proof. For $q = 0, 1$ the conclusion is immediate. Otherwise collapse the binary block to one coordinate and project it away. Homogeneity makes the conditional complement ranks consecutive. The full-rank coupling in [KKO21, Theorem 2.10] gives $D_0 - D_1 \in \{0, 1\}$. Thus $S_0 = D_0$, $S_1 = D_1 + 2$ satisfy $S_1 - S_0 \geq 1$. Pointwise,

$$(k - S_0)_+ + (S_1 - k)_+ \geq S_1 - S_0 \geq 1.$$

Their expectations are at most $L/(1 - q)$ and U/q . Multiply by $q(1 - q)$. $\square$

Lemma 4.31 (Paired-bundle estimate). *In the setting of [KKO21, Lemma 5.27], use the parameters in Definition 4.2 and use p as the threshold for a good 2-1-1 event. Assume $x_e(B), x_f(A) \leq \omega$. If the two bundles are not 2-1-1 good, then their joint 2-2-2 happiness event has probability greater than p .*

Proof. By hypothesis, $x_e(B) \leq \omega$ and $x_f(A) \leq \omega$. Thus each bundle satisfies the orientation required by Lemma 4.21, exchanging A, B for the second bundle. Orient $\mathbf{e}$ from v to u , so the residual set V in Lemma 4.21 is the present set U . By that lemma, non-goodness implies

$$\Pr[U_T + (A - \mathbf{e})_T \leq 1] + x_e(B) < Kh.$$

For the present bound we retain the weaker consequence with the nonnegative bundle mass omitted, so all subsequent budgets are unchanged. Put $U := \delta(u) - \mathbf{e}$, $W := \delta(w) - \mathbf{f}$, and $Z := \delta(u) \cap \delta(w)$. We retain the mass at zero in the source first-moment calculation. Let $X := U_T + (A - \mathbf{e})_T$, $p_j := \Pr[X = j]$, and put

$$a_{\text{tail}} := Kh, \quad v_{\text{tail}} := 2h + \omega + r + 3d, \quad t_{\text{tail}} := p_0 + p_1 < a_{\text{tail}}, \quad q := \Pr[X \geq 3].$$

The two blocks defining X are disjoint. Its rank law under the original strongly Rayleigh distribution is therefore Poisson-binomial. The degree partition satisfies $x(A) + x(C) \leq 1 + r + d$, because $x(B) \geq 1 - r$ and $x(\delta(v)) \leq 2 + d$. Also $x_e(C) \leq x(C)$ and $x_e(B) \leq \omega$. The unrounded identity preceding [KKO21, Eq. (56)] therefore gives

$$\begin{aligned} \mathbb{E}[X] &= x(\delta(u)) + x(A) - 2x(\mathbf{e}) + x_e(B) + x_e(C) \\ &\leq 2 + 2h + \omega + r + 2d \leq 2 + v_{\text{tail}}. \end{aligned}$$

Since

$$\mathbb{E}[X] \geq 2(1 - t_{\text{tail}} - q) + 3q,$$

we have $q \leq 2t_{\text{tail}} + v_{\text{tail}}$ and

$$p_2 > 1 - 3a_{\text{tail}} - v_{\text{tail}} > 0.$$

Every Poisson-binomial law satisfies $p_1^2 \geq 2p_0p_2$. If $p_0 > 0$, write the Bernoulli odds as $w_i \geq 0$; then $p_1 = p_0 \sum_i w_i$ and $p_2 = p_0 \sum_{i < j} w_i w_j$, so the inequality follows by retaining the diagonal terms in $(\sum_i w_i)^2$. If a success is forced, $p_0 = 0$ and the inequality is immediate. Consequently,

$$p_0 < \frac{a_{\text{tail}}^2}{2(1 - 3a_{\text{tail}} - v_{\text{tail}})}.$$

For both raw central identities, put

$$r_0 := \frac{(Kh)^2}{2(1 - 3Kh - v_{\text{tail}})}, \quad \ell_{\text{ex}} := Kh + r_0, \quad u_{\text{ex}} := v_{\text{tail}} + \ell_{\text{ex}}.$$

The same calculation gives

$$\mathbb{E}[(2 - X)_+] = p_1 + 2p_0 < \ell_{\text{ex}}, \quad \mathbb{E}[(X - 2)_+] = \mathbb{E}[X] - 2 + \mathbb{E}[(2 - X)_+] < u_{\text{ex}}.$$

In the paired setting, the two raw central identities have sum $D_T + 2Z_T$, where D is disjoint from the binary block Z after the three endpoint-tree conditions. This is the set identity used in [KKO21, Claim A.2], not an assumption about independent counts.

The endpoint-tree event has probability at least $1 - 3d_0$. Define $q_Z := \Pr[Z_T = 1 \mid u, v, w \text{ trees}]$. Subadditivity of positive parts bounds the lower and upper excess of the sum by $2\ell_{\text{ex}}$ and $2u_{\text{ex}}$. Thus Lemma 4.30 applies with $k = 4$,

$$L_{\text{sep}} = \frac{2\ell_{\text{ex}}}{1 - 3d_0}, U_{\text{sep}} = \frac{2u_{\text{ex}}}{1 - 3d_0}.$$

Define $z_L^{(0)} = z_H^{(0)} = 1/2$ and, for $i = 0, 1, 2$,

$$z_L^{(i+1)} = \frac{U_{\text{sep}}}{1 - L_{\text{sep}}/(1 - z_L^{(i)})}, z_H^{(i+1)} = \frac{L_{\text{sep}}}{1 - U_{\text{sep}}/(1 - z_H^{(i)})}.$$

Set $z_L = z_L^{(3)}$, $z_H = z_H^{(3)}$. All denominators are positive and the sequences decrease. For $q_Z \leq 1/2$, the separator inequality implies $q_Z \leq U_{\text{sep}}/(1 - L_{\text{sep}}/(1 - q_Z))$; the right side increases with q_Z on this domain. Induction therefore gives $q_Z \leq z_L$. For $q_Z \geq 1/2$, interchange $L_{\text{sep}}, U_{\text{sep}}$ and $q_Z, 1 - q_Z$ to obtain $q_Z \geq 1 - z_H$.

The three iterations use only exact rational operations.

The six probabilities and sharper endpoints.

Put $A' := A - \mathbf{e}$, $B' := B - \mathbf{f}$, $Z := \delta(u) \cap \delta(w)$, $U_0 := U \setminus Z$, $W_0 := W \setminus Z$, and $U_1 := U' := U_0$, $W_1 := W' := W_0$. After conditioning on the three endpoint trees, $Z_T \leq 1$. The residual blocks A', U_0, W_0 are pairwise disjoint. In the low branch impose the three endpoint trees, $Z_T = 0$, $C_T = 0$, $\mathbf{e}(B)_T = 0$, $\mathbf{f}_T = 0$, and $\mathbf{e}(A)_T = 1$. In the high branch replace $Z_T = 0$ by $Z_T = 1$. Put $a_0 = 1/2 - h - \omega - 2r - 4d_0$ and

$$\begin{aligned} P_{0b} &= (1 - 3d_0)(1 - z_L - 2r - \omega - 4d_0), \\ P_{0p} &= (1 - 3d_0)(1/2 - z_L - 2r - d_0 - h - \omega), & P_0 &= P_{0p}a_0, \\ P_{1b} &= (1 - 3d_0)(1 - z_H)(1 - 2r - \omega - 4d_0), \\ P_{1p} &= (1 - 3d_0)(1 - z_H)(1/2 - 2r - h - \omega - 4d_0), & P_1 &= P_{1p} \frac{a_0 - z_H}{1 - z_H}. \end{aligned}$$

They bound probabilities before deleting $\mathbf{f}$, after deleting $\mathbf{f}$, and after the last selection. All are positive. For the low branch use union bounds under the endpoint-tree law; the deleted external means are at most $2r + d$, ω , and $1/2 + h + d$. The endpoint-tree $\mathbf{e}(A)$ marginal is at least a_0 , and disjoint deletions increase it. In the high branch, selecting binary Z decreases each disjoint deleted mean. If its actual success probability is $q_Z \geq 1 - z_H$, the final selection has conditional probability at least $(a_0 + q_Z - 1)/q_Z \geq (a_0 - z_H)/(1 - z_H)$. These give the six bounds above.

The overlap $\mathbf{f}(A)$ is charged before deletion. Define

$$a_{0p} = a_0 - \omega/P_{0p}, a_{1p} = a_0 - \omega/P_{1p} - z_H.$$

These are lower bounds for the residual $A' = A - \mathbf{e}$ mean before the final selection. Lemma 4.29 gives

$$\ell_0 = a_{0p} - v_{\text{tail}}/P_{0p} - \ell_{\text{ex}}/P_0, \quad u_0 = 1/2 + h + 2\omega + 4r + 6d_0 + z_L + u_{\text{ex}}/P_{0p} - v_{\text{tail}}, \quad (4)$$

$$\ell_1 = a_{1p} - v_{\text{tail}}/P_{1p} - \ell_{\text{ex}}/P_1, \quad u_1 = 1/2 + h + 2\omega + 4r + 6d_0 + u_{\text{ex}}/P_{1p} - v_{\text{tail}}. \quad (5)$$

Both residual $A', B' = B - \mathbf{f}$ means are at most u_i before and after the last selection, and the final A' mean is at least ℓ_i .

For completeness, before the $\mathbf{f}$ deletion the raw mass estimate gives base $1/2 + h + 2\omega + 4r + 6d_0$, with the additional z_L only in the deletion branch. Deleting $\mathbf{f}$ increases a containing mean by at most $u_{\text{ex}}/P_{ip} - (u_{\text{ex}} - \ell_{\text{ex}})$. The same upper estimate applies to the overlap by the bounded marginal-shift lemma. For the lower mean, delete the overlap at cost ω/P_{ib} and use nonnegative changes elsewhere. The final disjoint selection loses at most $\ell_{\text{ex}}/P_i + (u_{\text{ex}} - \ell_{\text{ex}})/P_{ip}$. The high branch's fixed Z success shifts the center from two to one but leaves these excess budgets unchanged. This proves Eqs. (4)–(5).

Reorder the last two conditions for upper means.

For the final paired law, the upper subset means may be bounded by first deleting B' , then selecting $\mathbf{e}(A)$, while lower bounds may use the original order. Put

$$\hat{\omega}_i = u_{\text{ex}}/P_{ip}, \hat{v}_i = u_{\text{ex}}/(P_{ip}(1 - u_i)), \quad j_i = 2 - i.$$

The upper endpoints in the two tables below follow by reordering. The full event has probability at least $P_i(1 - u_i)$.

Let H_i be the event through $\mathbf{f}_T = 0$, before the last selection. Under H_i , $\mathbb{E}[A'] \geq a_{ip}$ and $\mathbb{E}[A'], \mathbb{E}[B'] \leq u_i < 1$. The containing ranks $X = A' \cup U_i$, $Y = B' \cup W_i$ have centers j_i and upper excess u_{ex} inherited from their raw identities. Deleting B' first has total probability at least $P_{ip}(1 - u_i)$. Thus $\mathbb{E}[X]$ afterwards is at most $j_i + \hat{v}_i$. Its A' part still has mean at least a_{ip} , by negative association. The A' upper mean is at most $2u_i$, by homogeneous deleted-mass domination. After deletion, $Y = W_i$, whose mean is at most the pre-deletion Y mean $j_i + \hat{\omega}_i$. The means of $A' + W_i$, $U_i + W_i$ and the full union are at most their pre-deletion containing means $A' + Y$, $X + Y - A'$ and $X + Y$. These give every displayed endpoint.

The final binary selection $\mathbf{e}(A)_T = 1$ is disjoint from all remaining counted blocks. It can only decrease each of their union means. It has positive probability: preceding disjoint deletions increase its marginal, and the low/high branch lower bounds are a_0 and $(a_0 - z_H)/(1 - z_H)$. The two conditioning events commute and define exactly the same final law. The original order supplies probability $P_i(1 - u_i)$ and the lower endpoints. No independence of these two events is asserted.

To state both full tables, put

$$g_0 = \ell_{\text{ex}}/P_0, \psi_i = \ell_{\text{ex}}/(P_i(1 - u_i)).$$

The low-branch target is $(1, 1, 2)$:

Union	lower mean	upper mean
A'	ℓ_0	$2u_0$
U_0	$2 - g_0 - u_0$	$2 + \hat{v}_0 - a_{0p}$
W_0	$2 - \psi_0$	$2 + \hat{\omega}_0$
$A' \cup U_0$	$2 - \psi_0$	$2 + \hat{v}_0$
$A' \cup W_0$	$\ell_0 + 2 - \psi_0$	$2 + \hat{\omega}_0 + u_0$
$U_0 \cup W_0$	$4 - 2\psi_0 - 2u_0$	$4 + 2\hat{\omega}_0 - a_{0p}$
$A' \cup U_0 \cup W_0$	$4 - 2\psi_0$	$4 + 2\hat{\omega}_0$

The high-branch target is $(1, 0, 1)$:

Union	lower mean	upper mean
A'	ℓ_1	$2u_1$
U'	0	$1 + \widehat{v}_1 - a_{1p}$
W'	$1 - \psi_1$	$1 + \widehat{w}_1$
$A' \cup U'$	$1 - \psi_1$	$1 + \widehat{v}_1$
$A' \cup W'$	$\ell_1 + 1 - \psi_1$	$1 + \widehat{w}_1 + u_1$
$U' \cup W'$	$1 - \psi_1$	$2 + 2\widehat{w}_1 - a_{1p}$
$A' \cup U' \cup W'$	$2 - 2\psi_1$	$2 + 2\widehat{w}_1$

The lower columns are the original-order containing-mean calculation, using the new ℓ_i, u_i . Every row is ordered and strictly within its target's neighboring integers in all four tuples. For each branch let L_S, U_S denote its table endpoints and κ_S its target sum. Define

$$d_k := \min_{|S|=k} \{L_S - \kappa_S + 1\}, \quad e_k := \min_{|S|=k} \{\kappa_S + 1 - U_S\} \quad (k = 1, 2, 3).$$

The capacity is at least

$$\min\{d_1 d_2 d_3, e_1 d_1 d_2, e_1 e_2 d_1, e_1 e_2 e_3\}.$$

This is the capacity bound in [GKL24, proof of Theorem 4.1 and Corollary 5.13] with all subset hypotheses explicitly verified. The low branch retains $W_0 \geq 1$ and $A' + U_0 \geq 1$, both proper-cut counts. Divide out its compulsory success and use factor $e^{-2}/2$. In the high branch, the selected $\mathbf{e}, Z$ edges connect the proper three-atom union, whose boundary is $A' + U' + W' \geq 1$. At target zero in U' , capacity equals that of specialization $U' = 0$. Positive capacity excludes the zero polynomial; the remaining bivariate polynomial has zero constant term, giving $e^{-1}/2$. This last operation is a capacity specialization, not a new stochastic mean law. Writing $\mathcal{C}_i$ for the four-product minimum in branch i , the full probabilities are at least

$$P_0(1 - u_0)(e^{-2}/2)\mathcal{C}_0 > p, \quad P_1(1 - u_1)(e^{-1}/2)\mathcal{C}_1 > p.$$

Both strict comparisons hold in every row.

In the high branch $Z_T = 1$, and the target implies $U_T = 1, W_T = 2$; in the low branch these identities hold directly. With the fixed selected bundle edge, all three boundary degrees are two. Deletions and the binary extremal selections preserve stability at every step. $\square$

Lemma 4.32 (Localized matching). *For a degree cut S and its child u , put $U_u := x(\delta^\dagger(u))$, and let $b(u)$ be the indicator that u is incident to an internal bad bundle of S . Define*

$$F_u := 1 - \epsilon_B \mathbf{1}_{\{\epsilon_F \leq U_u \leq 1 - \epsilon_F\}} b(u),$$

$$Z_u := 1 + \mathbf{1}_{\{|\mathcal{A}(S)| \geq 4, U_u \leq \epsilon_F\}}, \quad \alpha := 2d_0.$$

For every $0 \leq d \leq d_0$, there are nonnegative matching values $m_{\mathbf{e},u}$ such that, for every good internal bundle $\mathbf{e} = (u, v)$,

$$m_{\mathbf{e},u} F_u + m_{\mathbf{e},v} F_v \leq (1 + \alpha) x_{\mathbf{e}}, \quad \sum_{\substack{\mathbf{e} \text{ good} \\ \mathbf{e} \ni u}} m_{\mathbf{e},u} = U_u Z_u.$$

Thus the fractional discount is required only at atoms incident to a bad bundle.

Proof. Lemma 4.17 replaces the published $k = 9$ endpoint threshold by k_{good} , while Lemma 4.18 supplies the source fact that an atom is incident to at most one bad bundle. We check every cut family in the max-flow proof of [KKO21, Lemma 6.2] with the localized F_u . The source network

gives each good bundle supply $(1 + \alpha)x_{\mathbf{e}}$ and each child sink demand $U_u F_u Z_u$. Saturating these demands and dividing the incoming flow at u by $F_u > 0$ gives the stated identities. The extra scalar comparisons needed below are

$$0 < \epsilon_B < 1, \quad 1/2 + k_{\text{good}}h < 1 - \epsilon_F, \quad 2\epsilon_F < (1/2 + k_{\text{good}}h)(1 - \epsilon_B).$$

For a three-atom degree cut, a hypothetical bad bundle forces the third atom to have degree at least $3 - (4k_{\text{good}} + 2)h$. Thus the contradiction in [KKO21, Lemma 6.4] remains valid because

$$3 - (4k_{\text{good}} + 2)h > 2 + d_0.$$

There are therefore no bad bundles in this case, and $F_u = Z_u = 1$. The corresponding source-only max-flow cut has capacity margin

$$(1 + \alpha)(2 - d_0/2) - (2 + d_0) > 0.$$

For the source cuts with at least five atoms, Eqs. (28)–(29) of the source give, with $n := |\mathcal{A}(S)|$,

$$M_{\geq 5}(n) := (1 + \alpha)(n - 1 - \frac{d_0}{2} - \frac{n}{2}(\frac{1}{2} + h)) - (2 + d_0 + \epsilon_F n).$$

Its coefficient in n is positive, so the worst case is $n = 5$, where $M_{\geq 5}(5) > 0$. These bounds use only $F_u \leq 1$ and $U_u Z_u \leq U_u + \epsilon_F$, both unchanged by localization. For a four-atom cut with at most one bad bundle, the corresponding margin is

$$M_{4, \leq 1} := (1 + \alpha)(\frac{5}{2} - \frac{d_0}{2} - h) - (2 + d_0 + 4\epsilon_F) > 0.$$

These are the two places where the source used only $h \leq 0.01$, $\epsilon_F \leq 0.1$, and $\alpha \geq 2d$. In the four-atom case with two bad bundles, the argument forcing every upward mass to be ϵ_F -fractional remains valid because

$$\frac{2 - \epsilon_F}{3} > 1/2 + k_{\text{good}}h.$$

The two bad bundles cover all four atoms. Thus every upward mass is fractional, every $b(u) = 1$, and all four sink factors are $1 - \epsilon_B$. The remaining capacity comparison is exactly

$$M_{4,2} := (1 + \alpha)(2 - 2h - d_0/2) - (2 + d_0)(1 - \epsilon_B) > 0.$$

It remains to check the nontrivial max-flow cut in [KKO21, Eq. (30)]. Let W be a nonempty proper subset of the atoms, put $w := |W|$, and let $k \leq w$ be the number of distinct bad bundles adjacent to W . Since bad bundles form a matching, choose one distinct endpoint in W for each such bundle. For a chosen endpoint with $U_u \leq \epsilon_F$, its demand is at most $2\epsilon_F$, and hence at most $(1/2 + k_{\text{good}}h)(1 - \epsilon_B)$. Otherwise the bad-endpoint bound gives $\epsilon_F < U_u \leq 1/2 + k_{\text{good}}h < 1 - \epsilon_F$; its factor is $F_u = 1 - \epsilon_B$ and the same demand bound holds. For every unchosen atom use $U_u F_u Z_u \leq 1 + d$. The total sink demand in W is therefore at most

$$k(1/2 + k_{\text{good}}h)(1 - \epsilon_B) + (w - k)(1 + d).$$

By [KKO21, Lemma 6.3], the internal-bundle incidence at W is at least $w - d/2$. Removing the k bad bundles leaves good supply at least

$$(1 + \alpha)(w - d/2 - k(1/2 + h)).$$

Subtracting the preceding demand gives the same scalar inequality as Eqs. (30)–(31) of the source:

$$w(\alpha - d) \geq kB(d) + \frac{d}{2}(1 + \alpha), \quad B(d) := \frac{\alpha}{2} + (k_{\text{good}} + 1)h + \alpha h - \frac{\epsilon_B}{2} - k_{\text{good}}\epsilon_B h - d. \quad (*)$$

For $k = 0$, the slack in $(*)$ is minimized at $w = 1$ and $d = d_0$, where

$$\alpha - d_0 - \frac{d_0}{2}(1 + \alpha) = \frac{d_0}{2}(1 - \alpha) > 0.$$

Now suppose $k \geq 1$ and define

$$S(w, k, d) := w(\alpha - d) - kB(d) - \frac{d}{2}(1 + \alpha).$$

Since

$$\frac{\partial S}{\partial d} = -w + k - \frac{1 + \alpha}{2} < 0,$$

the worst value of d is d_0 . At that value,

$$B(d_0) = -\frac{1 + 2k_{\text{good}}h}{2} 10^{-11} < -5 \cdot 10^{-12}, \quad S(1, 1, d_0) > 0.$$

Increasing w by one increases S by $\alpha - d_0 = d_0 > 0$, while increasing k by one increases S by $-B(d_0) > 0$. Hence $S(w, k, d) > 0$ for every $1 \leq k \leq w$. Thus every source–sink cut has the required capacity, and the matching exists with the required capacity inequality and conservation identity. Only the selected bad endpoints used the fractional discount; no atom with $b(u) = 0$ was discounted in the cut bounds. This complete cut verification replaces the source’s sufficient condition $\epsilon_B \geq 21h$; that condition is not an additional premise here. $\square$

Lemma 4.33 (Structural trichotomy). *Let v be a non-root hierarchy atom with parent S , let A, B, C be the degree partition of $\delta(v)$, and suppose $0 \leq d \leq d_0$. At least one of the following holds:*

- (a) *the x -mass of bad top edges in $\delta^{\rightarrow}(v)$ is at least $1/2 - h$;*
- (b) *the x -mass of top edges in $\delta^{\rightarrow}(v)$ that are 2–1–1 good with respect to v is at least $1/2 - h - d$;*
- (c) *there are two top half bundles $\mathbf{e}, \mathbf{f}$ in $\delta^{\rightarrow}(v)$ such that $x_{\mathbf{e}}(B) \leq \omega$, $x_{\mathbf{f}}(A) \leq \omega$, and their joint 2–2–2 happiness event has probability at least p .*

Proof. The small- and large-bundle estimates in Lemma 4.25 make every non-half top bundle 2–1–1 good. Inspecting the proof of [KKO21, Lemma 5.25], its only probabilistic inputs are the four individual happiness bounds supplied by the same lemma; the rest of its argument only partitions incident bundle mass. For completeness, at most four half bundles are incident to an atom, since $5(1/2 - h) > 2 + d_0$. If good but not 2–1–1-good mass on one degree side exceeds $1/2 + 4\omega$, two half bundles each contribute at least ω there: otherwise one contributes at most $1/2 + h$ and the other three contribute less than 3ω , totaling at most $1/2 + h + 3\omega \leq 1/2 + 4\omega$. For each of these two bundles, the direct mixed test forces opposite-side mass below ω . The same-side test then makes one 2–1–1 good, a contradiction. Thus the $1/2 + 4\omega$ bound holds for the replacement good class. An ordinary good half bundle has its endpoint-tree 2–2 event with probability at least $(1 - d_0)\gamma_{\text{good}}h > p$.

We now repeat the proof of [KKO21, Theorem 5.28]. If the first alternative fails, there is no bad half bundle, because every half bundle has mass at least $1/2 - h$. If there is at most one half bundle, all remaining mass is non-half; since $x(\delta^{\rightarrow}(v)) \geq 1 - d$, that mass is at least

$1 - d - (1/2 + h) = 1/2 - h - d$, and the second alternative follows. Otherwise choose two good half bundles $\mathbf{e}, \mathbf{f}$. If either is 2–1–1 good, its mass alone proves the second alternative. If neither is, the two-sided mixed-bundle estimate included in Lemma 4.25 implies that each bundle has at least one of its A - and B -masses at most ω , while each bundle has mass exceeding 2ω after removing the remainder, since $1/2 - h - 2\omega - 4r - 2d_0 > 0$. Lemmas 4.19 and 4.21 rule out the two small masses lying in the same degree part. After exchanging A, B or $\mathbf{e}, \mathbf{f}$ if necessary, this leaves $x_{\mathbf{e}}(B), x_{\mathbf{f}}(A) \leq \omega$. Lemma 4.31 then gives the third alternative. These are exactly the uses of [KKO21, Lemmas 5.21–5.24 and 5.27] in the source proof; after their replacement, no step uses the old inequality $12r \leq h$. $\square$

4.5 Parity and payment estimates

Definition 4.34 (Parity bootstrap at bad endpoints). Put

$$\rho := 20d_0, \quad q_- := 1/2 - h - \rho, \quad q_+ := 1/2 + h + \rho, \quad \nu := 3/2 - h - 3\rho,$$

$$b_0 := 1/2 - h - 2\rho, \quad \bar{b} := \gamma_{\text{good}}h/(1 - 2d_0), \quad M_0 := 2h + 4\rho.$$

For a real number x , let $\lceil x \rceil_{20}$ and $\lfloor x \rfloor_{20}$ denote upward and downward rounding to the 10^{-20} grid. For fixed inputs $a_*, b_* > 0$, $D_* \geq 0$, $s_* \in \mathbb{R}$, and a starting bound $B_0 \geq 0$, define eight stages, indexed by $j = 0, \dots, 7$:

$$\begin{aligned} Q_j &:= \lceil b_*/(a_*\psi(s_* - B_j)) \rceil_{20}, \\ C_j &:= \lceil \min\{B_j, Q_j/\nu\} \rceil_{20}, \\ \ell_j &:= \lfloor 1 + D_* - 3C_j - 2Q_j \rfloor_{20}. \end{aligned}$$

For $j < 7$, set

$$B_{j+1} := \lceil \min\{B_j, Q_j/\nu, Q_j^2/(2\ell_j)\} \rceil_{20}.$$

Each use below verifies $0 < s_* - B_j < 1$ and $\ell_j > 0$. The initial instance uses

$$(a_*, s_*, b_*, D_*, B_0) = (q_-, b_0, \bar{b}, 0, 0.226).$$

Put

$$\Delta_0 := \lceil \min\{1, (M_0 + 2B_7 + Q_7)/(1 - q_+)\} \rceil_{20}.$$

$$q_{\text{bad}} := q_+ + (1 - q_+)\Delta_* + 2d_0 < q_0.$$

Lemma 4.35 (Polygon-conditioned parity at bad endpoints). *Use Definitions 4.2 and 4.34. Let $\mathbf{e} = E(u, v)$ be a bad half bundle internal to a degree parent S , and let $\mathcal{P}$ be the polygon parent of an upward bottom edge out of u . For the polygon event $E_{\mathcal{P}}$ from Lemma 4.24, including its independent thinning to probability P , one has*

$$\Pr[\delta(u)_T \text{ odd} \mid E_{\mathcal{P}}] \leq q_{\text{bad}}.$$

The assertion is uniform for all hierarchy errors $0 \leq d \leq d_0$.

Proof. Laminarity gives $u, v \subseteq S \subsetneq \mathcal{P}$: a bottom parent above u cannot be its degree parent, and therefore contains all the children of S . All these cuts avoid the distinguished vertices u_0, v_0 . Write $\Omega := \{u, v, \mathcal{P} \text{ are trees}\}$.

We first control means under this conditioning. Conditioning an internal edge set to its maximum rank increases its internal marginals and decreases the external marginals, by [KKO21, Fact 2.12 and Lemma 2.23]. If the current rank deficit is ξ , the total increase is ξ ; homogeneity

makes the total decrease ξ as well. Thus the mean of any edge-set count changes by at most ξ , even when that set meets both sides. Initially the rank deficits of $u, v, \mathcal{P}$ are at most $d/2$. Conditioning successively on u , then v , then $\mathcal{P}$ being trees has deficits at most $d/2, d, 2d$, respectively. Every edge-set mean therefore changes by at most $7d/2$. These maximum-rank conditionings preserve homogeneous real stability. In particular, conditional on Ω ,

$$\mathbb{E}[\delta(u)_T], \mathbb{E}[\delta(v)_T] \in [2 - \rho, 2 + \rho], \quad q := \Pr[\mathbf{e}_T = 1] \in [q_-, q_+].$$

Here and until the transfer to $E_{\mathcal{P}}$, probabilities and expectations are conditional on Ω unless specified otherwise.

Since u and v are trees, a spanning tree contains at most one edge of $\mathbf{e}$. Identifying the variables of this parallel bundle therefore gives a binary coordinate $E := \mathbf{e}_T$ in a homogeneous multiaffine stable polynomial. Project this law off the coordinate E . Its two conditional laws have adjacent total ranks, and [KKO21, Theorem 2.10] gives a monotone coupling in which changing $E = 1$ to $E = 0$ adds exactly one remaining edge. In particular it adds at most one edge to any prescribed subset. This is the same projection argument used in [KKO21, proof of Lemma 2.27].

Each of the three tree events fails with probability at most $d/2$. Consequently

$$\Pr[\mathcal{P} \text{ is not a tree} \mid u, v \text{ are trees}] \leq \frac{d}{2(1-d)} \leq 2d_0.$$

Badness is defined conditional on u, v being trees. Adding the $\mathcal{P}$ -tree requirement can only decrease its numerator, so

$$b_\Omega := \Pr[\delta(u)_T = \delta(v)_T = 2] \leq \bar{b}.$$

This step does not assert independence of badness and the additional tree event.

Let $X := |(\delta(u) \setminus \mathbf{e}) \cap T|$ and $Y := |(\delta(v) \setminus \mathbf{e}) \cap T|$. They count disjoint edge sets, and $X + Y = |\delta(u \cup v) \cap T|$. Define

$$\Delta := \mathbb{E}[X + Y \mid E = 0] - \mathbb{E}[X + Y \mid E = 1].$$

The preceding coupling gives $0 \leq \Delta \leq 1$. For any subcount J of X , the same coupling satisfies

$$0 \leq \Delta_J := \mathbb{E}[J \mid E = 0] - \mathbb{E}[J \mid E = 1] \leq \Delta, \quad \Pr[J_0 \neq J_1] \leq \Delta_J.$$

Write $\mu_X := \mathbb{E}[X \mid E = 1]$, $\mu_Y := \mathbb{E}[Y \mid E = 1]$, and $m := \mu_X + \mu_Y$. The cut-mean bounds and the coupling imply

$$\begin{aligned} 1 - \rho &\leq \mu_X, \mu_Y \leq 2 + \rho - q \leq 3/2 + h + 2\rho, \\ 3 - 2\rho - q &\leq m \leq 4 + 2\rho - 2q - (1 - q)\Delta. \end{aligned}$$

For example, if $\Delta_X := \mathbb{E}[X \mid E = 0] - \mathbb{E}[X \mid E = 1] \leq 1$, then

$$\mu_X = \mathbb{E}[X] - (1 - q)\Delta_X \geq 2 - \rho - q - (1 - q) = 1 - \rho.$$

The total-mean identity subtracts exactly $(1 - q)\Delta$ and gives the second display.

Conditional on $E = 1$, the union $u \cup v$ is a connected proper set, so $Z := X + Y \geq 1$. Put $D := \max\{(1 - q)\Delta - M_0, 0\}$. The displayed identities give $m \geq 1 + \nu > 2$ and, if $D > 0$, $m \leq 3 - D < 3$. If $D = 0$, then $\Delta \leq M_0/(1 - q) \leq M_0/(1 - q_+) < \Delta_*$ directly. Suppose $D > 0$ and write $p_i := \Pr[Z = i \mid E = 1, \Omega]$. Here $m > 2$, so gapless rank support and $m < 3$ imply $p_2 > 0$. Since $\nu > 1.49$,

$$p_1 \leq e^{-\nu} < 0.226 = B_0;$$

the last inequality follows by taking the exponential Taylor sum through degree 16 at 1.49. The invariant for the recurrence in Definition 4.34 is the following. Suppose $p_1 \leq B_0$, $m \in [1 + \nu, 3 - D_*]$, and for every applicable bound $p_1 \leq B$,

$$b_* \geq a_* p_2 \psi(s_* - B).$$

Then $p_1 \leq B_j$ implies $p_2 \leq Q_j$. The odds bound gives $p_1 \leq C_j$; the rank-three estimate gives $p_3 \geq \ell_j$; and Newton gives $p_1 \leq Q_j^2/(2\ell_j)$. These are the inequalities in Lemma 4.11, and outward rounding preserves each one. Induction therefore proves $p_1 \leq B_j$ and $p_2 \leq Q_j$ at all eight stages.

For the initial instance, $m < 3$ and $\bar{b} \geq qp_2\psi(b_0 - B_j) \geq q_-p_2\psi(b_0 - B_j)$ by the same rank-split lemma. All split arguments belong to $(0, 1)$ and all ℓ_j are positive. Its last step gives

$$B_7 > 0, \quad Q_7 > 0,$$

$$D \leq 3 - m \leq 2p_1 + p_2 \leq 2B_7 + Q_7, \quad \Delta \leq \Delta_0 < 0.17.$$

No fixed-point convergence is used.

We now retain the actual bundle probability in both edge states and charge their two contributions to the same badness budget:

$$\bar{b} \geq q \Pr[X = Y = 1 \mid E = 1] + (1 - q) \Pr[X = Y = 2 \mid E = 0].$$

Suppose for contradiction that $\Delta \geq \Delta_*$. Cover $[q_-, q_+]$ by the 128 closed cells

$$a_i := q_- + (q_+ - q_-)i/128, \quad b_i := q_- + (q_+ - q_-)(i + 1)/128, \quad 0 \leq i < 128.$$

Fix a cell $[a_i, b_i]$ containing q . Under $E = 1$, the exact mean identity above gives

$$m \leq 4 + 2\rho - 2q - (1 - q)\Delta \leq 3 - D_i, \quad D_i := 2a_i - 1 - 2\rho + (1 - b_i)\Delta_*.$$

Under $E = 0$, the two proper cuts have compulsory successes. Thus $U := X - 1$, $V := Y - 1$ have a stable joint generating polynomial. Writing $\Delta_X := \mathbb{E}[X \mid E = 0] - \mathbb{E}[X \mid E = 1] \in [0, 1]$ gives

$$\mathbb{E}[X \mid E = 0] = \mathbb{E}[X] + q\Delta_X \in [2 - \rho - q, 2 + \rho].$$

The symmetric identity and the sum identity imply

$$\mathbb{E}[U], \mathbb{E}[V] \in [L_i, 1 + \rho], \quad L_i := 1 - \rho - b_i,$$

$$1 + \delta_i \leq \mathbb{E}[U + V] \leq 2 - 2a_i + 2\rho + b_i\Delta_0 < 1.2, \quad \delta_i := 1 - 2b_i - 2\rho + a_i\Delta_*.$$

Every D_i, δ_i is positive. Lemma 4.3 gives a rank-one atom greater than 0.36. By [KKO21, Lemma 2.21], the rank-two atom $q' := \Pr[U + V = 2 \mid E = 0]$ is at least

$$q_i := \delta_i(1 - \delta_i + \delta_i^2/2 - \delta_i^3/6).$$

Indeed, the one-forced branch is at least $\delta_i e^{-\delta_i}$, and the zero-forced branch is at least $e^{-1.2}/2 > \delta_i$. The individual nonzero tails exceed 0.39 and their lower tails at one exceed 0.70. If $q' \geq 0.1$, the up/down inequality and Lemma 4.4 give $\Pr[U = V = 1] > 0.025$. Otherwise apply Lemma 4.7 with $c = 0.36$ and $L_0 = L_i$. Denote its function by F_i . Every degree-one coefficient and denominator is positive on $[q_i, 0.1]$, and its W is nonnegative. Concavity gives

$$\Pr[X = Y = 2 \mid E = 0] = \Pr[U = V = 1 \mid E = 0] \geq z_i := \min\{0.025, F_i(q_i), F_i(0.1)\}.$$

The present-edge contribution to badness is therefore at most

$$\tilde{b}_i := \bar{b} - (1 - b_i)z_i > 0.$$

Apply the same eight-stage recurrence with inputs

$$(a_*, s_*, b_*, D_*, B_0) = (a_i, a_i - \rho, \tilde{b}_i, D_i, B_7),$$

and denote its outputs by $\tilde{B}_j, \tilde{Q}_j, \tilde{C}_j, \tilde{\ell}_j$. The rank-split argument now uses the sharper individual bound $\mu_X, \mu_Y \leq 2 + \rho - q$ and the lower bound $q \geq a_i$. Thus its split parameter is at least $a_i - \rho - \tilde{B}_j$. The rank-three lower bound uses $m \leq 3 - D_i$, while the odds bound still uses $m \geq 1 + \nu$. The recurrence invariant therefore gives $p_1 \leq \tilde{B}_j$ and $p_2 \leq \tilde{Q}_j$. Every displayed premise, including positivity of $\tilde{\ell}_j$ and membership of the split parameter in $(0, 1)$, is verified by exact rational substitution in every cell. The finite certificate gives

$$\min_{0 \leq i < 128} \{D_i - 2\tilde{B}_7 - \tilde{Q}_7\} > 9.9996 \cdot 10^{-7} > 0.$$

This contradicts $D_i \leq 3 - m \leq 2p_1 + p_2$. The cells share their endpoints exactly and cover every possible q ; this is an outward interval certificate, not point sampling. We conclude $\Delta < \Delta_*$.

To control parity inside $\mathcal{P}$, take

$$J := |((\delta(u) \cap E(\mathcal{P})) \setminus \mathbf{e}) \cap T|, \quad I := E + J.$$

Couple J_0, J_1 as above. On $J_0 = J_1$, the two possible values of $E + J$ have opposite parity, so either parity atom of their q -mixture is at most $\max\{q, 1 - q\}$. On a mismatch use the bound one. Hence, for either parity ς ,

$$\begin{aligned} \Pr[I \text{ has parity } \varsigma] &\leq \max\{q, 1 - q\} + \min\{q, 1 - q\} \Pr[J_0 \neq J_1] \\ &\leq q_+ + (1 - q_+) \Delta_*. \end{aligned}$$

For the last inequality, write $w := \max\{q, 1 - q\} \leq q_+$. The bound $w + (1 - w)\Delta_*$ increases with w , since $\Delta_* < 1$. Bounding both parity atoms is necessary because the external boundary count can flip parity.

Finally, conditional on $\mathcal{P}$ being a tree, its internal tree and the contracted tree in $G/\mathcal{P}$ are independent by [KKO21, Fact 2.8]. The two child-tree events are internal. All further requirements in $E_{\mathcal{P}}$, including the deleted C boundary, the accepted A/B pair, and its acceptance and thinning coins, belong to the contracted factor. Therefore

$$\mathcal{L}(T \cap E(\mathcal{P}) \mid E_{\mathcal{P}}, u, v \text{ trees}) = \mathcal{L}(T \cap E(\mathcal{P}) \mid \Omega).$$

Under the left conditioning, I remains independent of $K_{\text{ext}} := |\delta(u) \cap \delta(\mathcal{P}) \cap T|$. The equality $\delta(u)_T = I + K_{\text{ext}}$ and arbitrary convolution with K_{ext} preserve the bound on the larger internal parity atom. The same product factorization gives

$$\Pr[u \text{ or } v \text{ not a tree} \mid E_{\mathcal{P}}] = \Pr[u \text{ or } v \text{ not a tree} \mid \mathcal{P} \text{ tree}] \leq \frac{d}{1 - d/2} \leq 2d_0.$$

Restoring these exceptional outcomes gives

$$q_+ + (1 - q_+) \Delta_* + 2d_0 = q_{\text{bad}},$$

without dividing by the rare polygon-event probability. Lemma 4.26 gives the universal envelope q_0 , and exact substitution gives $q_{\text{bad}} < q_0$. $\square$

Lemma 4.36 (Synchronized reduction events). *An event of probability at least $\pi > 0$ can be thinned to probability π without changing its conditional tree law. Thus top events may be thinned to p and polygon events to P . The two reductions arising from one paired 2–2–2 event may moreover be chosen pointwise identical.*

Proof. For an event $\mathcal{H}$ of probability $q_{\mathcal{H}} \geq \pi$, use a tree-independent Bernoulli coin $C_{\mathcal{H}}$ with probability $\pi/q_{\mathcal{H}}$. Then $\mathcal{H} \cap C_{\mathcal{H}}$ has probability π and the same conditional tree law as $\mathcal{H}$, as in [KKO21, Section 7].

In Case 3 of [KKO21, Theorem 5.28], let $\mathcal{J}_{\mathbf{e},\mathbf{f},u}$ be the joint 2–2–2 happiness event for the fixed paired bundles $\mathbf{e}, \mathbf{f}$ at u . Define $q_{\mathcal{J}} := \Pr[\mathcal{J}_{\mathbf{e},\mathbf{f},u}] \geq p$, use one Bernoulli event $C_{\mathcal{J}}$ of probability $p/q_{\mathcal{J}}$ independent of the tree, and set

$$\mathcal{R}_{\mathbf{e},u} = \mathcal{R}_{\mathbf{f},u} := \mathcal{J}_{\mathbf{e},\mathbf{f},u} \cap C_{\mathcal{J}}$$

pointwise. Thus the simultaneous-reduction identity required in the proof of [KKO21, Lemma 7.8] is retained without biasing the conditional tree law. All reduction events below use this synchronized convention.

These auxiliary coins define only the payment vector used in the analysis. The passage from μ to μ_{λ} is made directly at the level of $T \mapsto c(T) + J(T)$ by Lemma 3.2; it does not require the reduction events or layered slack vectors to be reconstructed under μ_{λ} . □

Lemma 4.37 (Finite-rank polygon-child parity). *Let P be a polygon cut, S a child of P , and u a proper hierarchy descendant of S . Put*

$$t_u := x(\delta(u) \cap \delta(S)), \quad e_{\text{ch}} := \epsilon_M/2 + 40d_0.$$

For $y \in \{0.0025, 0.10, 0.15, 0.20, 0.25\}$ and $t_u \geq y$, the accepted and thinned event E_P satisfies

$$\Pr[\delta(u)_T \text{ odd} \mid E_P] \leq Q_2(y) := \frac{1 + (1 - y + e_{\text{ch}})^2 e^{-2(1-y)}}{2} + 14d_0.$$

The assertion is uniform for $0 \leq d \leq d_0$.

Proof. The cut identity and $x(\delta(S \setminus u)) \geq 2$ give $t_u \leq 1 + d$. Conditional on P being a tree, its internal tree factors from the contracted tree by [KKO21, Fact 2.8]. Acceptance and thinning use only the contracted boundary pair and independent coins. Therefore these conditions do not change the internal tree law. Also

$$\Pr[S \text{ not a tree} \mid E_P] \leq \frac{d}{2(1 - d/2)} \leq d.$$

Condition on S being a tree. Its internal tree factors again and is independent of every remaining condition imposed below. The count

$$Y := |(\delta(u) \cap E(S)) \cap T| - 1$$

is Poisson-binomial, since u is a proper nonempty subset of the connected set S and its internal count has a compulsory success. The S -tree condition may be imposed first. Its internal factor is then independent of the remaining condition that P is a tree, and of polygon acceptance. Thus the internal law is exactly the original S -tree internal law. Maximum-rank conditioning increases that mean by at most $d/2$, so

$$1 - t_u \leq \mu := \mathbb{E}[Y] \leq 1 + 3d/2 - t_u.$$

In particular, $0 \leq \mu \leq 1 + 3d_0/2 < 1.2$.

We construct, outside an event of conditional probability at most $14d$, a boundary count B that is a sum of at most two independent Bernoulli variables, is independent of Y , and has mean θ satisfying $|\theta - t_u| \leq e_{\text{ch}}$.

Suppose first that S is a boundary child, including either child of a triangle. Its exterior boundary is one accepted parent side, with the parent remainder deleted, and contains one selected edge. The subset version of Lemma 4.26 bounds the change in its contribution to B by $\epsilon_M/2 + 3d$. Let $H_S := |\delta(S) \cap E(P) \cap T|$. Connectivity gives $H_S \geq 1$. The hierarchy gives $x(\delta(S) \cap E(P)) \leq 1 + 2d$; conditioning P to be a tree adds at most $d/2$, and conditioning S to be a tree cannot increase this exterior count. Hence $\mathbb{E}[H_S] \leq 1 + 5d/2$. Condition further on $H_S = 1$. Its failure probability is at most $5d/2$, and

$$\mathbb{E}[H_S 1_{\{H_S \geq 2\}}] \leq 2(\mathbb{E}[H_S] - 1) \leq 5d.$$

For a subset F of this boundary, conditioning changes its mean by at most $5d/(1 - 5d/2) \leq 6d$. Indeed, the numerator of the mean difference is $\mathbb{E}[F_T] \Pr[H_S \neq 1] - \mathbb{E}[F_T 1_{\{H_S \neq 1\}}]$, and each nonnegative term is at most $5d$. The preceding P - and S -tree changes cost at most d . Thus the internal boundary mean changes by at most $7d$. The two boundary contributions are independent Bernoulli variables, because one belongs to the internal parent factor and the other to the contracted factor. They are also independent of the internal tree of S . Their mean error is at most $\epsilon_M/2 + 10d \leq e_{\text{ch}}$, and the total exceptional probability is at most $14d$.

Now suppose that S is an interior child. Its exterior part lies in the parent remainder and is deleted; its original mass is at most d . Let S_-, S_+ be its neighboring non-root children and put $F_- = E(S_-, S)$, $F_+ = E(S, S_+)$. Each link has mass at least $1 - d$. The adjacent union sides have cut value at most $2 + 4d$, by the cut identity and [KKO21, Definition 4.31]. Applying the forest bound [KKO21, Lemma 2.23] to the two endpoint children and their union gives $\Pr[(F_{\pm})_T \neq 1] \leq 3d$ under the original law. Conditional on P being a tree, each failure probability is at most $4d$. The remaining internal boundary has original mass at most $3d$ and gains at most $d/2$. Therefore

$$\Pr[H_S \neq 2 \mid E_P] \leq 4d + 4d + 7d/2 \leq 12d.$$

After the S -tree condition this failure is at most $12d/(1 - d) \leq 13d$. The mean of H_S is at most $2 + 2d$, and hence

$$\mathbb{E}[H_S 1_{\{H_S \neq 2\}}] = \mathbb{E}[H_S] - 2 + 2\Pr[H_S \neq 2] \leq 28d.$$

For every boundary subset, the same mean-difference identity now bounds its change upon conditioning $H_S = 2$ by $28d/(1 - 13d) \leq 29d$. The preceding tree conditions cost at most d , and the deleted exterior part has original mass at most d . The total error is at most $31d \leq e_{\text{ch}}$.

After the P - and S -tree conditions, project the contracted internal law to the full boundary of S , then condition this full projected rank to two. This preserves stability. Any subcount of that rank-two law is Poisson-binomial of degree at most two. Take B to be its subcount on $\delta(u)$. Its Bernoulli factorization is independent of Y , by the S -internal factorization; no independence of the two actual boundary edges is asserted. The union of the exceptional events has probability at most $14d$. All adjacent sets used in the forest bounds are proper and avoid the distinguished vertices; no forest bound is applied to a root atom.

On either good conditioning, write $B = B_1 + B_2$ with independent Bernoulli parameters b_1, b_2 , padding by a zero parameter if needed. The degree parity is that of $1 + Y + B$. Put

$$M_Y := \mathbb{E}[(-1)^Y], \quad M_B := (1 - 2b_1)(1 - 2b_2), \quad \theta = b_1 + b_2.$$

Then its odd probability is $(1 + M_Y M_B)/2$. For a Poisson-binomial variable of mean at most 1.2, a nonnegative parity moment is at most $e^{-2\mu}$. If no parameter exceeds $1/2$, this follows factor by factor. If two exceed $1/2$, their product is at most $(\mu - 1)^2 \leq 0.04 < e^{-2.4} \leq e^{-2\mu}$; a larger even number is impossible. A zero parity factor is immediate. A negative parity moment has absolute value at most $2\mu - 1$, because exactly one parameter exceeds $1/2$.

If $M_Y M_B \leq 0$, there is nothing to prove. If both factors are nonnegative, then $M_B \leq (1 - \theta)^2$. For $t_u \leq 1$ this gives

$$M_Y M_B \leq (1 - t_u + e_{\text{ch}})^2 e^{-2(1-t_u)} \leq (1 - y + e_{\text{ch}})^2 e^{-2(1-y)}.$$

The final inequality holds because $(z + e_{\text{ch}})^2 e^{-2z}$ increases for $0 \leq z \leq 1 - y$ when $y \geq e_{\text{ch}}$. For $t_u > 1$, use $M_Y M_B \leq (e_{\text{ch}} + d_0)^2$ instead. If both factors are negative, their means imply $1/2 - e_{\text{ch}} < t_u < 1/2 + 3d_0/2$, and

$$|M_Y M_B| \leq (2e_{\text{ch}} + 3d_0)^2.$$

For each of the five stated values of y , the exact exponential certificate verifies

$$e_{\text{ch}} < y < 1, \quad \max\{(e_{\text{ch}} + d_0)^2, (2e_{\text{ch}} + 3d_0)^2\} < (1 - y + e_{\text{ch}})^2 e^{-2(1-y)}.$$

Restoring the exceptional probability at most $14d_0$ proves the lemma. Every conditioning probability above is bounded explicitly away from zero, and the internal law is never divided by the rare probability of E_P . $\square$

Lemma 4.38 (Ancestor estimate). *Let h, r, d_0, p, K, ω be as in Definition 4.2, and use the parameters in Definition 4.34. Fix a KKO hierarchy with error $0 \leq d \leq d_0$ and $\beta > 0$, and define $\tau := t\beta$. Use the synchronized top reduction events with probability p , the polygon events with probability $P = sp$, and the matching from Lemma 4.32. For a non-root hierarchy atom u , define $U := x(\delta^\dagger(u))$ and partition its upward edges into good top, bad, and bottom sets $\mathcal{G}_u, \mathcal{D}_u, \mathcal{L}_u$. For a good top edge $g \in f = (u', v')$, define*

$$h_u(g) := \frac{\Pr[\delta(u)_T \text{ odd} \mid \mathcal{R}_{f,u'}] + \Pr[\delta(u)_T \text{ odd} \mid \mathcal{R}_{f,v'}]}{2}.$$

For a bottom edge g with polygon parent Q , put $q_u(g) := \Pr[\delta(u)_T \text{ odd} \mid E_Q]$. The bad set includes the edges incident to u_0 or v_0 , which are never reduced. Set

$$B_u := \tau \sum_{g \in \mathcal{G}_u} x_g h_u(g) + s\beta \sum_{g \in \mathcal{L}_u} x_g q_u(g).$$

For $U \geq \sigma$,

$$B_u \leq \tau(1 - \chi)F_u U,$$

where F_u is the localized matching factor of Lemma 4.32: it equals $1 - \epsilon_B$ only when U is ϵ_F -fractional and u is incident to an internal bad bundle of its degree parent, and equals one otherwise. For an atom without a degree parent, set $b(u) = 0$ and $F_u = 1$.

Proof. We use the conditional parity estimates underlying [KKO21, Lemma 7.3] with the present parameters. By Lemma 4.36, thinning to the separate rates p and P preserves every conditional tree law. The event E_S from the proof of Lemma 4.25, Lemma 4.26, and [KKO21, Corollary 2.17] give, for every bottom edge,

$$q_u(g) \leq \frac{1 + \exp(-2(1 - \epsilon_M/2 - 3d_0))}{2} < q_0.$$

The last inequality follows from the exact rational comparison

$$\sum_{j=0}^{18} \frac{(2(1 - \epsilon_M/2 - 3d_0))^j}{j!} - \frac{1}{2q_0 - 1} > 0.$$

Define $c := 1 - sq_0/t$ and $c_{\text{bad}} := 1 - sq_{\text{bad}}/t$. Let $b(u)$ be the bad-incidence indicator in Lemma 4.32, and put $c_u = c_{\text{bad}}$ if $b(u) = 1$, and $c_u = c$ otherwise. Every bottom term is bounded by the corresponding parity estimate: universally by q_0 , and at a bad endpoint by Lemma 4.35. The normalization in B_u divides the actual expected upward burden by p : its top terms have factor $p\tau$, while its bottom terms have factor $P\beta = sp\beta$. Subtracting B_u from τU shows that it is sufficient to prove

$$x(\mathcal{D}_u) + \sum_{g \in \mathcal{G}_u} x_g(1 - h_u(g)) + \sum_{g \in \mathcal{L}_u} x_g(1 - sq_u(g)/t) \geq (1 - (1 - \chi)F_u)U. \quad (16)$$

Call the left side the available saving. It is additive over disjoint sets of upward edges, and every contribution is nonnegative. Each bottom coefficient is at least $c_u \geq c$, so the universal coefficient c remains available in every case below. We use c_{bad} only in the discounted fractional case. Also $c > 0$ gives the crude bound $B_u \leq \tau U$ for every U , without the lower restriction $U \geq \sigma$.

We first record the level estimates used in the two large-mass cases. Let $S_1 = \mathbf{p}(u), S_2, \dots, S_m = R$ be its successive ancestors, where $R = V \setminus \{u_0, v_0\}$ is the hierarchy root. Use the level sets from [KKO21, proof of Lemma 7.3], with an explicit terminal convention:

$$\delta^{\geq i} := \delta(u) \cap \delta(S_i), \quad \delta^{\geq m+1} := \emptyset,$$

$$\delta^i := \delta^{\geq i} \setminus \delta^{\geq i+1}, \quad t_i := x(\delta^{\geq i}).$$

Thus $t_1 = U \leq 1 + d$, $t_{m+1} = 0$, and $x(\delta^i) = t_i - t_{i+1}$. For $i < m$, each level consists entirely of top edges or entirely of bottom edges. The terminal level $\delta^m = \delta(u) \cap \delta(R)$ consists of bad edges incident to u_0 or v_0 and saves one full unit per unit mass.

Recall s_0, w_0 from Definition 4.2, and put

$$w_1 := r(1 - r),$$

$$b_d := \frac{1}{2} + 4\omega + 2r + d, \quad k_d := \frac{1 - r - 2d}{2}.$$

For any $v \in [r, s_0]$, retain a level only when $v + 2d \leq t_i \leq 1 - v$. The conditioning in [KKO21, Claim 7.5] applies to its good top edges and gives

$$h_u(g) \leq 1 - v + \max\{2d, v^2\} = 1 - v + v^2.$$

Indeed, each endpoint reduction event makes the relevant ancestor a tree; the internal and contracted trees factor by [KKO21, Fact 2.8]. The claim is uniform in the boundary count revealed by the event, and its intersection-mass interval includes both endpoints. The retained levels satisfy that interval, and $v^2 \geq r^2 > 2d_0 \geq 2d$. A bad edge saves one unit, while a bottom edge saves c . The exact parameter comparisons give

$$r < s_0, \quad c > w_0, \quad k_d > w_0.$$

Since $v(1 - v)$ is increasing for $v \in [r, s_0]$, every unit of retained mass, of any of the three edge types, saves at least $v(1 - v)$.

An atom incident to a bad bundle has $U \leq 1/2 + k_{\text{good}}h < 1 - \epsilon_F$. Thus neither of the following large-mass cases is discounted. Suppose first that $U \geq 1 - r$, so $F_u = 1$. Let j be the last level with $t_j \geq 1 - r$, and put

$$z := t_{j+1} < 1 - r, \quad v := x(\delta^j) = t_j - z \geq 1 - r - z.$$

For a real number a , write $(a)_+ := \max\{a, 0\}$. If $z \geq 1 - s_0$, choose $v := 1 - z$, so $r < v \leq s_0$. Retain all levels after j whose tail mass is at least $v + 2d$. Their total mass is at least

$$z - v - 2d = 1 - 2v - 2d.$$

Their tail masses are at most $z = 1 - v$, so the preceding level estimate applies. Hence the available saving is at least

$$F_d(v) := v(1 - v)(1 - 2v - 2d).$$

Its derivative satisfies

$$F'_d(v) = 1 - 2d - (6 - 4d)v + 6v^2 \geq 1 - 2d_0 - 6s_0 > 0.$$

It follows that $F_d(v) \geq F_d(r)$. Since $U \leq 1 + d$, the exact margin

$$r(1 - r)(1 - 2r - 2d_0) - \chi(1 + d_0) > 0$$

proves Eq. (16) in this subcase.

For a bottom level $i < m$, its polygon parent is S_{i+1} and S_i is a child of that polygon. Lemma 4.37 applies with the starting tail t_i , not t_{i+1} . For $y \in \{0.0025, 0.10, 0.15, 0.20, 0.25\}$, let $\hat{Q}_2(y)$ be the rational upper enclosure obtained from the degree-40 exponential bounds, and put $c_y := 1 - s\hat{Q}_2(y)$; recall $t = 1$. Exact comparisons give

$$c_{0.0025} > w_0, \quad c_{0.25} > 0.0099.$$

For the last top level, the degree partition gives $x(C) \leq 2r + d$, $B \cap \delta(u) = \emptyset$, and $x(A \cap \delta^j) \geq v - 2r - d$. At most $1/2 + 4\omega$ of this mass is good but not 2-1-1 good, by Lemma 4.33. The remaining mass, at least $(v - b_d)_+$, is bad or 2-1-1 good. For each such good edge, [KKO21, Claim 7.4] bounds the odd probability under one endpoint event by $r + 2d$; the other probability is at most one. Its average saving is at least k_d , and a bad edge saves one.

For $z < 1 - s_0$, use the following three closed ranges.

If $0 \leq z \leq 0.35$, a top last level has saving at least

$$\frac{1 - r - 2d_0}{2}(1/2 - 4\omega - 3r - 0.35 - d_0) > \chi(1 + d_0).$$

This is the same $1/2 + 4\omega$ mass bound and one-endpoint happiness estimate from Lemma 4.33. If the last level is bottom, its starting tail is at least $1 - r > 0.25$, so it saves at least $c_{0.25}(1 - r - 0.35) > \chi(1 + d_0)$. If it is terminal, its root-bad mass saves $1 - r - 0.35$, which also suffices.

If $0.35 \leq z \leq 0.99$, use the five thresholds

$$(y_1, y_2, y_3, y_4, y_5) := (0.0025, 0.10, 0.15, 0.20, 0.25).$$

Set $\kappa_0 := 0$ and

$$\varepsilon_l := \min\{y_l, 0.01\}, \quad \kappa_l := \min\{\varepsilon_l(1 - \varepsilon_l), c_{y_l}\}, \quad 1 \leq l \leq 5.$$

These coefficients satisfy $0 < \kappa_1 \leq \dots \leq \kappa_5$. A later level with $t_i \geq y_l + 2d$ saves at least κ_l per unit: top edges satisfy Claim 7.5 with ε_l , since $\varepsilon_l + 2d \leq t_i \leq 0.99 \leq 1 - \varepsilon_l$ and $\varepsilon_l^2 > 2d_0$; bottom edges save at least c_{y_l} ; bad edges save one.

For each l , these levels form a prefix of the later levels. Summing $t_i - t_{i+1}$, including the entire level that crosses the threshold, gives total mass at least $(z - y_l - 2d)_+$. At each level, sum the increments $\kappa_l - \kappa_{l-1}$ for which $t_i \geq y_l + 2d$; their sum is at most that level's saving coefficient. Interchanging the two finite sums proves the bound. The available saving is therefore at least

$$\sum_{l=1}^5 (\kappa_l - \kappa_{l-1})(z - y_l - 2d)_+.$$

This counts each level with one coefficient, not with overlapping full credits. The bound increases with z and decreases with d . The exact endpoint check gives

$$\sum_{l=1}^5 (\kappa_l - \kappa_{l-1})(0.35 - y_l - 2d_0) > \chi(1 + d_0).$$

If $0.99 \leq z \leq 1 - s_0$, retain later levels with starting tail at least $0.0025 + 2d$. Their mass is at least $0.9875 - 2d$. Good top edges satisfy the source claim with parameter $s_0 < 0.0025$, and save at least w_0 . Bottom edges save at least $c_{0.0025} > w_0$; bad edges save one. The exact final comparison is

$$w_0(0.9875 - 2d_0) > \chi(1 + d_0).$$

These ranges share their endpoints and exhaust $z < 1 - s_0$.

Suppose next that $1 - \epsilon_F < U < 1 - r$, again with $F_u = 1$. Retain all levels with $t_i \geq r + 2d$. Their total mass is at least $U - r - 2d$, and every unit saves at least w_1 . Since $U > 1 - \epsilon_F = 0.9$, division by U gives saving per unit at least

$$w_1(1 - (r + 2d)/0.9).$$

The exact inequality

$$w_1(1 - (r + 2d_0)/0.9) - \chi > 0$$

proves Eq. (16) in this case.

For $\epsilon_F \leq U \leq 1 - \epsilon_F$, first bound the good top terms without using bad incidence. Let $u' := p(u)$, fix a good top edge $g \in f = (u'', v'')$, and condition on either endpoint event $\mathcal{R}_{f,z}$, $z \in \{u'', v''\}$. The auxiliary thinning coin is independent of the tree. After the endpoint u'' is made a tree, [KKO21, Fact 2.8] factors its internal tree from the contracted tree. Relative to $E(u')$, the other tree and boundary-count conditions in the happiness event depend only on the contracted tree and on $(\delta(u) \cap \delta(u'))_T$; they reveal no other internal edges of u' . Conditioning further on u' being a tree and factoring once more leaves the tree-conditioned law inside u' used in [KKO21, Claim 7.5], possibly conditioned on that boundary count. The claim is uniform in its value. Thus this is a factorization after contraction, not independence of overlapping tree events. Together with [KKO21, Lemma 2.23], it gives

$$\Pr[u' \text{ is not a tree} \mid \mathcal{R}_{f,z}] \leq d/2.$$

Since $x(\delta(u) \cap \delta(u')) = U \in [\epsilon_F, 1 - \epsilon_F]$, [KKO21, Claim 7.5], uniformly in the boundary count, gives

$$\Pr[\delta(u)_T \text{ odd} \mid \mathcal{R}_{f,z}] \leq 1 - \epsilon_F + \epsilon_F^2 + d/2 \leq 1 - \epsilon_F + 2\epsilon_F^2.$$

Thus each good top edge saves at least $\epsilon_F - 2\epsilon_F^2 = 0.08$ per unit, and every bad edge saves one. If $b(u) = 0$, then $F_u = 1$ and the universal bottom coefficient suffices, because $0 < c < 0.08$ and

$$c - \chi > 0.$$

If $b(u) = 1$, use Lemma 4.35 for every upward bottom edge. Its polygon parent contains both endpoints of the internal bad bundle, as required by that lemma. The exact comparisons are

$$c_{\text{bad}} - (\chi + \epsilon_B - \chi\epsilon_B) > 0,$$

$$0.08 - (\chi + \epsilon_B - \chi\epsilon_B) > 0.$$

Thus every unit supplies at least $\chi + \epsilon_B - \chi\epsilon_B$, which proves the discounted bound. This also suffices at either boundary of the fractional interval. Only this case uses the improved parity estimate; all retained-level and small-mass arguments use c .

Finally, let $\sigma \leq U < \epsilon_F$, so $F_u = 1$. Condition on the parent of u as above and apply [KKO21, Claim 7.5] with its actual parameter U . The conditional failure of the parent-tree event contributes at most $d_0/2$, so a good top edge saves at least $U - U^2 - d_0/2$. This expression is increasing on $[\sigma, \epsilon_F]$, and

$$\sigma^2 - 2d_0 > 0, \quad \sigma - \sigma^2 - d_0/2 - \chi > 0.$$

A bad edge saves one unit and a bottom edge saves $c > \chi$. Every unit therefore supplies the required saving χ , completing all cases of Eq. (16). $\square$

Lemma 4.39 (Polygon charges on the enlarged domain). *Use Definition 4.2 and $0 \leq d \leq d_0$. For a polygon cut S with polygon parent $\hat{S}$, the expected horizontal charge is at most $0.31P\beta$ if S is a boundary atom, and at most $0.85P\beta$ if it is an interior atom.*

Proof. First suppose S is a boundary atom. Work conditional on the accepted parent event, including its independent thinning. Let A, B, C be the polygon partition of $\delta(S)$; the arrows refer to $\hat{S}$. Put

$$a := x(A^\rightarrow), \quad b := x(B^\rightarrow), \quad a_\uparrow := x(A^\uparrow), \quad b_\uparrow := x(B^\uparrow).$$

The internal parent tree is independent of the accepted external boundary pair. The external part of $\delta(S)$ meets exactly one parent boundary side, so it contains one selected edge. The separate boundary marginal budget in Lemma 4.24, the parent-tree change, and remainder deletion show that its $A^\uparrow$ and $B^\uparrow$ probabilities differ from the original masses by at most $e_M := \epsilon_M + 2d$. This is the internal/external factorization used in [KKO21, Lemma 7.10]; no strong-Rayleigh property is asserted for the entire accepted law.

Let H_S count the internal parent-tree edges leaving S . Connectivity gives $H_S \geq 1$. The hierarchy bounds and the parent-tree rank deficit give

$$x(\delta^\rightarrow(S)) \leq 1 + 2d, \quad \mathbb{E}[H_S] \leq 1 + 5d/2.$$

Consequently

$$\mathbb{E}[H_S \mathbf{1}_{\{H_S \geq 2\}}] \leq 2(\mathbb{E}[H_S] - 1) \leq 5d.$$

Internal marginals increase on conditioning the parent to be a tree. Thus the probabilities that the unique internal edge belongs to $A^\rightarrow$ and $B^\rightarrow$ are at least $(a - 5d)_+$ and $(b - 5d)_+$, respectively, where $z_+ := \max\{z, 0\}$. Opposite internal and external single edges make S happy. Independence and $(u - e)_+(v - f)_+ \geq uv - ev - fu$ for nonnegative u, v, e, f therefore give

$$\Pr[S \text{ happy}] \geq a_\uparrow b + b_\uparrow a - e_M(a + b) - 5d(a_\uparrow + b_\uparrow).$$

The polygon-side lower bounds imply

$$a_{\uparrow}b + b_{\uparrow}a \geq (1-d)(a+b) - 2ab.$$

Also $1 - 2d \leq a + b \leq 1 + 2d$, $0 \leq a \leq 1 + 2d$, and $a_{\uparrow} + b_{\uparrow} \leq 1 + d$. Writing $a + b = 1 + t$, $|t| \leq 2d$, gives

$$(1-d)(a+b) - 2ab \geq a^2 + (1-a)^2 - 3d - 10d^2.$$

Substitution yields

$$\begin{aligned} \Pr[S \text{ happy}] &\geq a^2 + (1-a)^2 - \epsilon_M - 10d - 2d\epsilon_M - 19d^2 \\ &\geq a^2 + (1-a)^2 - 2\epsilon_M - 20d. \end{aligned}$$

The last step uses $\epsilon_M(1 - 2d) + d(10 - 19d) \geq 0$, valid throughout $0 \leq d \leq d_0 = 2 \cdot 10^{-8}$ and $\epsilon_M < 0.005$. Assume $a = \max\{a, b\}$. Taking expectations in the pointwise payment formula bounds the horizontal charge by

$$(1+d)P\beta(a \Pr[S \text{ not happy}] + x(C^{\rightarrow})).$$

Since $x(C^{\rightarrow}) \leq d$ and $2a^2(1-a) \leq 8/27$ for $a \geq 0$, this is at most $J_{\partial}P\beta$, where

$$J_{\partial} := (1+d_0)(8/27 + (2\epsilon_M + 20d_0)(1+2d_0) + d_0) < 0.31.$$

This proves the boundary estimate directly on the new domain, without invoking the parameter-restricted conclusion of [KKO21, Lemma 7.9].

For an interior atom, its upward boundary lies in the parent remainder and is deleted by the accepted parent event. Thus only the internal parent-tree factor remains; the size of ϵ_M is irrelevant. Condition also on $C_T = 0$, whose failure probability is at most $2d$. The unrounded calculation in [KKO21, Lemma 7.11] gives both remaining side means in $[1 - 2d, 1 + 3d]$. Their nonzero tails are at least $1 - e^{-1+2d}$ and their tails at most one exceed 0.495 by Markov. The up/down split therefore gives central probability at least 0.155 at total rank two. The three link failures cost at most $12d$, so the internal happiness probability is at least $0.155(1 - 12d) > 0.153$. Restoring deletion gives $\Pr[S \text{ happy}] > 0.153(1 - 2d) > 0.152$. The charge is consequently at most

$$(1+d)P\beta(1 - 0.152)(1 + 2d) < 0.85P\beta.$$

All scalar comparisons hold at $d = d_0$ and hence throughout the interval. $\square$

Lemma 4.40 (Payment theorem). *Use Definitions 4.2 and 4.34. Fix a hierarchy with error $0 \leq d \leq d_0$ and let $\beta > 0$. Set $\tau := t\beta$, take the common events from Lemma 4.25, and use the matching from Lemma 4.32. Use the KKO reduction and increase formulas in [KKO21, Section 7]. Their expected-decrease coefficient is*

$$a = \zeta r p t > 1.37147 \cdot 10^{-8}.$$

The bottom coefficient is

$$a_{\text{bot}} > 4.43511 \cdot 10^{-8}.$$

More precisely, the construction produces a partition $E = E_g \dot{\cup} E_b$ and a payment vector s^{pay} satisfying the following:

- (a) *every bottom edge is in E_g ;*
- (b) *$s_e^{\text{pay}} \geq -\beta x_e$ on E_g and $s_e^{\text{pay}} = 0$ on E_b ;*

(c) $\mathbb{E}[s_e^{\text{pay}}] \leq -a\beta x_e$ on E_g , while every bottom edge satisfies $\mathbb{E}[s_e^{\text{pay}}] \leq -a_{\text{bot}}\beta x_e$;

(d) the deterministic cut inequalities in Lemma 4.41 hold.

Proof. Use Lemma 4.36 to thin the top events to p and the polygon events to P , preserving the synchronized pairs. The raw probability bounds come from Lemma 4.25. Use the matching from Lemma 4.32.

The reductions themselves retain their pointwise formulas: a bottom edge has reduction βx_e times its polygon indicator, and a top edge in bundle $\mathbf{e} = (u, v)$ has reduction $\tau x_e(\mathcal{R}_{\mathbf{e},u} + \mathcal{R}_{\mathbf{e},v})/2$. Their expected reductions are therefore $P\beta x_e$ and $p\tau x_e$, respectively. The payment increases depend on these realized reductions and the matching, not on the indicators' common frequency. Thus the change from p to P for polygon events changes neither the support nor the coordinate bound, and leaves the pointwise proof of Lemma 4.41 applicable. Since $0 \leq d \leq d_0 < 1$, the deterministic polygon calculation also has $(1+d)(1-d/2) \geq 1$. Thus Items (a), (b), and (d) follow from the pointwise construction and Lemma 4.41. We verify the expectation inequalities below.

The present parameters satisfy

$$h < 0.0055, \quad d_0 < h^2,$$

$$\epsilon_M < 0.005, \quad d_0 < \epsilon_M^2,$$

as well as $q_0 < t = 1$ and $\tau = \beta$. Lemma 4.26 gives the conditional polygon estimate

$$\Pr[u \text{ not left happy} \mid E_S] \leq q_{\text{unh}} < t/s,$$

with the same bound on the right. Its proof uses the half marginal budget for the side count and coordinate domination for the small remainder. The exact strict margin is $t - sq_{\text{unh}} > 0$. An upward bottom edge therefore has expected unhappy reduction at most $P\beta q_{\text{unh}}x_e < p\tau x_e$.

For completeness, the dependencies in the payment argument are as follows. The separate polygon and top probabilities and the synchronized top-event identities come from Lemmas 4.25 and 4.36. The weighted matching capacity and conservation identities come from Lemma 4.32, with fixed $\alpha = 2d_0$ and the present ϵ_B . The large-endpoint burden is supplied by Lemma 4.38; the smaller endpoints are treated explicitly below. The three bottom-parent configurations use Lemma 4.33 and the conditional polygon estimates just proved. The final matching combination and deterministic cut inequalities use only these identities and the pointwise support and coordinate bounds. In particular, no old parameter-dependent payment conclusion or hypothesis $12r \leq h$ is invoked.

Suppose first that both endpoints of a good top bundle have upward mass at least σ . The ancestor bound, matching conservation, and weighted capacity give expected total burden at most $(1 - \chi)(1 + 2d_0)p\tau$ times the bundle mass. Subtracting from its expected reduction, the decrease normalized by $r p \tau$ times its mass is at least

$$\frac{1 - (1 - \chi)(1 + 2d_0)}{r} > \zeta.$$

If an endpoint has upward mass below σ and its degree parent has at least four atoms, the matching factor $Z_u = 2$ halves its crude burden. Since $1/2 < 1 - \chi$, the same bound applies. This also covers the case when both endpoints have the factor two. Zero upward mass causes zero endpoint burden, according to the zero-mass convention in the construction.

It remains to check a three-atom degree parent, with atoms u, v, w , where $U := x(\delta^\uparrow(u)) < \sigma$. All three internal bundles are good, as proved in Lemma 4.32. Write

$$e := x(E(u, v)), \quad f := x(E(u, w)), \quad g := x(E(v, w)),$$

$$V := x(\delta^\uparrow(v)), \quad W := x(\delta^\uparrow(w)).$$

The degree and parent-cut equations give

$$|f - V| \leq d, \quad |e - W| \leq d, \quad |g - U| \leq d. \quad (6)$$

The nested-cut bound gives $V, W \leq 1 + d$, so $e, f \leq 1 + 2d$ and $g \leq \sigma + d$. Since $e + f + U = x(\delta(u)) \geq 2$, we also have

$$e, f \geq 1 - \sigma - 2d, \quad V, W \geq 1 - \sigma - 3d > 1 - \epsilon_F.$$

Thus $F_u = F_v = F_w = 1$ and $Z_v = Z_w = 1$. Use the fixed matching constant $D := 1 + 2d_0$. Conservation at v, w followed by capacity for the f and g bundles gives

$$V + W = m_{e,v} + m_{g,v} + m_{f,w} + m_{g,w} \leq m_{e,v} + D(f + g).$$

Also $V + W \geq e + f - 2d$. Therefore

$$\begin{aligned} m_{e,v} &\geq e - (\sigma + 3d + 2d_0 + 2d_0\sigma + 6d_0d) \\ &\geq e - (\sigma + 5d_0 + 2d_0\sigma + 6d_0^2). \end{aligned}$$

Recall

$$\rho_{\text{sm}} := 1 - \frac{\sigma + 5d_0 + 2d_0\sigma + 6d_0^2}{1 - \sigma - 2d_0}.$$

Then $m_{e,v}/e \geq \rho_{\text{sm}}$. Use the crude burden at u and Lemma 4.38 at v . Weighted matching capacity bounds the sum of the crude endpoint burdens by $Dp\tau e$, and the ancestor saving at v is at least $\chi p\tau m_{e,v}$. The expected net decrease is consequently at least

$$p\tau(\chi m_{e,v} - 2d_0e) \geq rp\tau e(\vartheta\rho_{\text{sm}} - 2d_0/r).$$

The exact margin

$$\vartheta\rho_{\text{sm}} - 2d_0/r - \zeta = 10^{-9} > 0$$

proves the required bound. Interchanging v and w proves it for the other bundle incident to the small endpoint. Both endpoints of the remaining bundle exceed σ , so the ordinary case already applies. Thus every good top edge has expected decrease at least $\zeta rp\tau x_e = a\beta x_e$.

We now bound bottom-edge payments, starting with a polygon cut S whose parent is a degree cut. Let A, B, C be its polygon partition and A', B', C' its degree partition. The partition bounds give

$$x(\delta(S)) \leq 2 + d, \quad x(A), x(B) \geq 1 - d, \quad x(A'), x(B') \geq 1 - r,$$

and $A' \subseteq A, B' \subseteq B$ by [KKO21, Remark 5.20]. The polygon-side lower bounds also hold for a triangle: if its two non-root atoms are a, b , the cut identity gives

$$x(E(a, \bar{S})) = \frac{x(\delta(a)) + x(\delta(S)) - x(\delta(b))}{2} \geq 1 - d/2 \geq 1 - d,$$

and the other side is symmetric. Here $\bar{S}$ denotes the complement of S in the vertex set.

The polygon partition also satisfies $x(C) \leq d$, and the degree partition satisfies $x(C') \leq 2r + d$. In the paired alternative of Lemma 4.33, Item (c), write $\mathbf{e} = (S, v)$ and $\mathbf{f} = (S, w)$. They have masses at least $1/2 - h$ and cross bounds $x_{\mathbf{e}}(B') \leq \omega, x_{\mathbf{f}}(A') \leq \omega$. For $D \in \{A, B, C\}$, put

$$e_D := x_{\mathbf{e}}(D), \quad f_D := x_{\mathbf{f}}(D).$$

Since $A' \subseteq A$, $B' \subseteq B$, and A', B', C' partition the boundary, we have $B \setminus B' \subseteq C'$ and $A \setminus A' \subseteq C'$. Thus

$$e_A \geq \frac{1}{2} - h - \omega - 2r - d, \quad e_B \leq \omega + 2r + d.$$

The first inequality follows already by retaining the part of $\mathbf{e}$ in A' . Since

$$\frac{1}{2} - h - 2\omega - 4r - 2d_0 > 0,$$

we have $e_A > e_B$. Symmetrically, $f_B > f_A$.

Let R_0 be the common synchronized indicator $\mathcal{R}_{\mathbf{e},S} = \mathcal{R}_{\mathbf{f},S}$, and let $R_1 := \mathcal{R}_{\mathbf{e},v}$ and $R_2 := \mathcal{R}_{\mathbf{f},w}$. Each has expectation p ; no independence is required. Use the entire paired set $D := \mathbf{e} \dot{\cup} \mathbf{f}$, and write $\mathbf{e} := (e_A, e_B, e_C)$ and $\mathbf{f} := (f_A, f_B, f_C)$. For a nonnegative vector $\mathbf{z} = (z_A, z_B, z_C)$, define

$$N(\mathbf{z}) := \max\{z_A, z_B\} + z_C.$$

This function is positively homogeneous and subadditive. The boundary totals of the reduction vector on D are

$$\mathbf{r}_D = \frac{\tau}{2}((\mathbf{e} + \mathbf{f})R_0 + \mathbf{e}R_1 + \mathbf{f}R_2).$$

Here $\mathbf{r}_D$ denotes reductions, not the scalar partition parameter r . Let $I_S(D)$ be the polygon charge attributable to reductions in D , as defined in [KKO21, Eq. (48)]. This charge is subadditive over D and its complement. On D , discarding the unhappiness indicators only increases that charge. Subadditivity of N therefore gives

$$\mathbb{E}[I_S(D)] \leq \frac{(1+d)p\tau}{2}(N(\mathbf{e} + \mathbf{f}) + N(\mathbf{e}) + N(\mathbf{f})).$$

The complementary top edges keep their trivial bound. Compare this display with the trivial contribution $(1+d)p\tau(x(\mathbf{e}) + x(\mathbf{f}))$. The opposite-side inequalities imply

$$N(\mathbf{e}) = x(\mathbf{e}) - e_B, \quad N(\mathbf{f}) = x(\mathbf{f}) - f_A,$$

$$N(\mathbf{e} + \mathbf{f}) = x(\mathbf{e}) + x(\mathbf{f}) - \min\{e_A + f_A, e_B + f_B\}.$$

Consequently the normalized saving on the entire paired set is at least

$$\begin{aligned} \frac{\min\{e_A + f_A, e_B + f_B\} + e_B + f_A}{2} &= \frac{\min\{e_A + e_B + 2f_A, f_A + f_B + 2e_B\}}{2} \\ &\geq \frac{\min\{x(\mathbf{e}) - e_C, x(\mathbf{f}) - f_C\}}{2} \\ &\geq \frac{1}{4} - \frac{h}{2} - \frac{d}{2}. \end{aligned}$$

The last line uses $e_C, f_C \leq x(C) \leq d$.

Define

$$\Delta_{\text{bot}}(d) := \frac{1}{4} - \frac{h}{2} - \frac{d}{2}.$$

In the first alternative of [KKO21, Lemma 7.8], the bad mass is at least $1/2 - h$ and is never reduced. Its saving is therefore at least $1/2 - h > \Delta_{\text{bot}}(d)$. In the second alternative, the 2–1–1-good mass is at least $1/2 - h - d$. Its reduction event at S makes S happy, so at most half the trivial charge remains. Its saving is at least $(1/2 - h - d)/2 = \Delta_{\text{bot}}(d)$. Thus the same coefficient applies in all three alternatives.

For the upward boundary, a top edge has expected reduction $p\tau x_e$. An upward bottom edge in $A \cup B$ has expected charge at most $(1+d)P\beta q_{\text{unh}}x_e < (1+d)p\tau x_e$ by the conditional polygon estimate above. For the remaining upward edges in C , use the crude bound $(1+d)P\beta x_e$. This also dominates a top edge's crude reduction, since $P \geq p$ and $\tau \leq \beta$. Since $x(C) \leq d$ and $d \leq 1$, the excess over the $p\tau$ envelope is at most $2dP\beta = 2dsp\beta$. The synchronized saving just proved involves only top events and retains the factor $p\tau$. Hence the degree-parent case satisfies

$$\mathbb{E}[I_S] \leq (1+d)p\tau(2+d-\Delta_{\text{bot}}(d)) + 2dP\beta.$$

The three normalized increases for a degree parent, a boundary atom with polygon parent, and an interior atom are therefore bounded by

$$J_1(d) := (1+d)t(2+d-\Delta_{\text{bot}}(d)) + 2ds,$$

$$J_2(d) := (1+d)(t(1+d) + 0.31s) + 2ds, \quad J_3(d) := (1+d)(td + 0.85s) + 2ds.$$

For the last two bounds, Lemma 4.39 supplies the horizontal bottom contributions $0.31P\beta$ and $0.85P\beta$. The upward contributions retain their pointwise bounds $p\tau(1+d)$ for a boundary atom and $p\tau d$ for an interior atom. The factor $1+d$ and remainder $2dP\beta$ give exactly $J_2(d)$ and $J_3(d)$ after division by $p\beta$. Every horizontal bottom term and C -side remainder is therefore scaled by s ; no equality $P = p$ is used. Each $J_i(d)$ is increasing on $[0, d_0]$. Define

$$\Delta_{\text{bot}} := \Delta_{\text{bot}}(d_0) = 1/4 - h/2 - d_0/2, \quad J_i := J_i(d_0) \quad (1 \leq i \leq 3).$$

Direct rational substitution gives

$$J_1 - J_2 > 0, \quad J_1 - J_3 > 0.$$

Every normalized increase is therefore at most J_1 . Since each bottom edge has expected reduction $P\beta x_e = sp\beta x_e$, define

$$a_{\text{bot}} := p(s - J_1) = p(s - (1+d_0)t(2+d_0-\Delta_{\text{bot}}) - 2d_0s).$$

The exact rational evaluation gives

$$a_{\text{bot}} > 4.43511 \cdot 10^{-8},$$

which also shows $a_{\text{bot}} > a$. This proves Item (c) on every edge. All numerical comparisons in the proof use rational quantities with positive denominators; in particular $1 - \sigma - 2d_0 > 0$, $1 - r > 0$, and $2q_0 - 1 > 0$. They therefore follow by clearing denominators and comparing integers. $\square$

Lemma 4.41 (Deterministic payment inequalities). *Fix a KKO hierarchy with uniform error envelope $0 \leq d \leq 1$ and a parameter $\beta > 0$, and let s^{pay} be the payment vector from the synchronized construction for these data. For a polygon cut P , including a triangle cut viewed as a degenerate polygon with $C = \emptyset$, let A, B, C be its polygon partition and call a set F admissible for P if $p(e) = P$ for every $e \in F$ and $x(F) \geq 1 - d/2$.*

(a) *if P is not left happy, then for every admissible edge set F for P , $s^{\text{pay}}(A) + s^{\text{pay}}(F) + (s^{\text{pay}})^-(C) \geq 0$; the symmetric inequality holds when P is not right happy;*

(b) *if S has a degree-cut parent and $\delta(S)_T$ is odd, then $s^{\text{pay}}(\delta(S)) \geq 0$.*

Proof. The proof uses no numerical payment estimate. Use the zero-mass convention fixed after Definition 4.2. If a polygon cut P (possibly a triangle) with partition A, B, C is not left happy, then its reduction event does not occur and the reduction is zero on every admissible set F . Writing r and I_P for the unchanged KKO reduction and increase vectors, the definition of I_P gives

$$s^{\text{pay}}(A) + s^{\text{pay}}(F) + (s^{\text{pay}})^-(C) \geq -r(A) + (1+d)(r(A) + r(C))(1-d/2) - r(C) \geq 0.$$

The right-happy inequality is symmetric. If S has a degree-cut parent and $\delta(S)_T$ is odd, the horizontal reductions vanish. If $x(\delta^\uparrow(S)) = 0$, there is no upward reduction to compensate and all remaining increases are nonnegative, so the conclusion is immediate. Otherwise, matching conservation gives $\sum_{f \in \delta \rightarrow (S)} m_{f,S} = Z_S x(\delta^\uparrow(S)) > 0$, and hence

$$s^{\text{pay}}(\delta(S)) \geq - \sum_{g \in \delta^\uparrow(S)} r_g + \sum_{e \in \delta \rightarrow (S)} \sum_{g \in \delta^\uparrow(S)} r_g \frac{m_{e,S}}{\sum_{f \in \delta \rightarrow (S)} m_{f,S}} = 0.$$

These are precisely [KKO21, Theorem 4.33(iii)–(iv)]. This proves both conclusions. $\square$

4.6 Hierarchy and separated repairs

Lemma 4.42 (Hierarchy and repair on the required domain). *For $0 < \eta \leq 1.4 \cdot 10^{-9}$, the hierarchy construction in the preliminaries remains valid, with added cuts 7η -near-minimum and the same exhaustive one-sided and two-sided classification. Under the deterministic payment hypotheses of Lemma 4.47, its separated repair conclusions remain valid on this domain, with*

$$R_1(\eta) := 21 \frac{2+\eta}{1-7\eta}, \quad R_2(\eta) := 5 \frac{2+\eta}{1-\eta}.$$

Proof. We verify the hierarchy domain explicitly. The construction in [KKO22, Appendix B] retains singleton one-sided components and, for each non-singleton component, inserts its non-root atoms and their union. The one-sided component has no inside atoms. Its polygon structure gives adjacent-bundle mass at least $1 - 7\eta$, atom-boundary mass at most $2 + 7\eta$, and remainder mass at most 7η [KKO22, Theorem 6.2]. The inserted union is the complement of the root atom and has the same boundary. The geometric facts giving laminarity and the valid-hierarchy conclusion of [KKO22, Fact B.4] do not use $\eta \leq 10^{-12}$: their polygon domain is $\eta \leq 1/10$. The two-child convention gives the same triangle parents. Thus the construction and its cut classification apply on the asserted domain.

Separate this geometric construction from the source's payment theorem. At the present larger threshold, 7η can exceed its printed payment-error cutoff 10^{-10} , so that theorem is not invoked. Instead use the probability, matching, ancestor, and payment estimates proved above, with conservative hierarchy error 14η . Their domain holds because

$$14(1.4 \cdot 10^{-9}) = 1.96 \cdot 10^{-8} < d_0.$$

The geometric hierarchy and deterministic cut classification are unchanged. In particular, no extension of a parameter-specialized source payment conclusion is assumed.

The component repair lemmas allow $\eta \leq 1/10$. Lemma 4.47 combines them on the required domain; their amplitudes give exactly R_1 and R_2 . $\square$

Lemma 4.43 (Outside witnesses for finitely many cuts). *Let P represent a non-singleton connected component of η -near-minimum cuts, where $0 < \eta \leq 2/5$. If $C_1, \dots, C_k$ are cuts in this component and $1 \leq k < 1/\eta$, then every atom a has an outside atom a' such that*

$$a \subseteq C_i \iff a' \subseteq C_i \quad (1 \leq i \leq k).$$

This includes the root atom, which need not be an outside atom.

Proof. Apply the polygon representation to x^0 , including e_0 . This vector is fractionally 2-edge-connected. Every component cut avoids the root atom containing both endpoints of e_0 , so its capacity under x^0 equals its capacity under x .

For each representing diagonal choose the half-polygon containing the cell of a . Their intersection contains the whole cell and therefore has positive area. If a is outside, take $a' = a$. Otherwise, if the intersection contained no outside atom, [KKO22, Lemma 4.30] would imply that it contained no inside atom either, a contradiction. An outside atom in the intersection has exactly the chosen membership pattern. Atom cells have positive area and lie wholly on the prescribed sides of the representing diagonals; no witness is placed at a diagonal intersection. The same argument applies to the root atom. $\square$

Lemma 4.44 (Two directional increase sets per edge). *For $0 < \eta \leq 1/10$, every edge belongs to at most one right-event increase set and at most one left-event increase set in the KKO two-sided repair construction.*

Proof. By [KKO22, Fact 4.9], an edge can receive a two-sided increase in at most one polygon. Fix that polygon, and let a, b be the atoms containing the endpoints of an edge e . Suppose e belongs to the right-event increase sets at two distinct polygon points p, q . Using the notation of [KKO22, Section 5], put

$$L_1 = L(p), \quad R_1 = L(p)_R, \quad L_2 = L(q), \quad R_2 = L(q)_R.$$

The increase set at p is contained in $E(L_1 \cap R_1, R_1 \setminus L_1)$, and the analogous containment holds at q . If an endpoint belongs to both L_1, L_2 , then [KKO22, Lemma 5.4] gives a contradiction. Otherwise, after interchanging a, b , the memberships are

$$\begin{aligned} a &\subseteq L_1 \cap R_1, & b &\subseteq R_1 \setminus L_1, \\ b &\subseteq L_2 \cap R_2, & a &\subseteq R_2 \setminus L_2. \end{aligned}$$

All four sets are component cuts, and $4 < 1/\eta$. Lemma 4.43 supplies outside atoms a', b', r' matching, on all four cuts, the respective membership patterns of a, b and the actual root. In particular, r' belongs to none of the four cuts. The three witnesses are distinct because their membership patterns differ.

Start the counterclockwise order of outside atoms immediately after r' . All four cuts are non-wrapping intervals in this single linear order. This change of origin does not change a crossing orientation: the union of a crossing pair avoids r' , and rotating the origin within its complement preserves the order of its ends. Since R_1 crosses L_1 on the right, every outside atom of $L_1 \cap R_1$ precedes every outside atom of $R_1 \setminus L_1$, giving $a' < b'$. The right crossing of R_2 and L_2 gives $b' < a'$, a contradiction. Thus at most one right-event increase set contains e . Reversing the polygon orientation proves the left-event assertion. Only these four membership patterns have been transferred to outside atoms; the polygon may still have inside atoms. $\square$

We retain the KKO repair vector and its pointwise guarantee. Exact cut-excess accounting bounds each directional bad event, and the preceding lemma bounds how many such events can increase an edge.

Lemma 4.45 (Sharpened two-sided KKO repair). *Let x^0 be a feasible subtour-LP solution with support $E \cup \{e_0\}$, let x be its restriction to E , and let μ be any spanning-tree distribution with marginals x . Let $0 < \eta \leq 1/10$. For every $\alpha > 0$, there is a random vector $r^{(2)} : E \rightarrow \mathbb{R}_{\geq 0}$ such that:*

(a) if an η -near-minimum cut S is crossed on both sides and $\delta(S)_T$ is odd, then $r^{(2)}(\delta(S)) \geq \alpha(1 - \eta)$;

(b) for every edge e , $\mathbb{E}[r_e^{(2)}] \leq 5\alpha\eta x_e$.

Proof. We use the KKO construction from [KKO22, Section 5]. Fix a cut L crossed on both sides, and let $Q = L^L$ and $R = L^R$ be the left and right crossing cuts chosen by that construction. Define

$$A := L \cap R, \quad B := R \setminus L, \quad C := L \cap Q, \quad D := Q \setminus L,$$

and, for every proper cut side X , define

$$\epsilon_X := x(\delta(X)) - 2.$$

The cuts L, Q, R are η -near-minimum cuts. The four cut sides A, B, C, D are proper and avoid the endpoints of the added edge; in particular,

$$\epsilon_A, \epsilon_B, \epsilon_C, \epsilon_D \geq 0.$$

They are also 2η -near-minimum cuts by [KKO22, Lemma 2.7].

The KKO edge sets partition $\delta(L)$ as

$$\delta(L) = E^{\leftarrow}(L) \dot{\cup} E^{\rightarrow}(L) \dot{\cup} E^{\circ}(L),$$

where

$$E^{\rightarrow}(L) = E(A, B), \quad E^{\leftarrow}(L) = E(C, D).$$

The disjointness is also recorded in [KKO22, Corollary 5.8]. For disjoint cut sides X, Y , the cut identity gives

$$x(E(X, Y)) = 1 + \frac{\epsilon_X + \epsilon_Y - \epsilon_{X \cup Y}}{2}.$$

Consequently,

$$x(E^{\rightarrow}(L)) = 1 + \frac{\epsilon_A + \epsilon_B - \epsilon_R}{2}, \quad x(E^{\leftarrow}(L)) = 1 + \frac{\epsilon_C + \epsilon_D - \epsilon_Q}{2},$$

and hence

$$x(E^{\circ}(L)) = \epsilon_L + \frac{\epsilon_Q + \epsilon_R - \epsilon_A - \epsilon_B - \epsilon_C - \epsilon_D}{2}.$$

The right bad event is

$$\mathcal{B}^{\rightarrow}(L) := \{|E^{\rightarrow}(L) \cap T| \neq 1 \text{ or } |E^{\circ}(L) \cap T| \neq 0\}.$$

By [KKO22, Corollary 2.12],

$$\Pr[|E^{\rightarrow}(L) \cap T| \neq 1] \leq \frac{\epsilon_A + \epsilon_B + \epsilon_R}{2}.$$

By [KKO22, Fact 2.13],

$$\Pr[|E^{\circ}(L) \cap T| \neq 0] \leq x(E^{\circ}(L)).$$

Adding these bounds and substituting the exact expression for $x(E^{\circ}(L))$ gives

$$\Pr[\mathcal{B}^{\rightarrow}(L)] \leq \epsilon_L + \epsilon_R + \frac{\epsilon_Q - \epsilon_C - \epsilon_D}{2} \leq \frac{5}{2}\eta.$$

Here we used $\epsilon_L, \epsilon_Q, \epsilon_R < \eta$ and $\epsilon_C, \epsilon_D \geq 0$. The symmetric calculation gives

$$\Pr[\mathcal{B}^{\leftarrow}(L)] \leq \frac{5}{2}\eta.$$

Keep the increase sets of [KKO22, Theorem 5.2], and set $r_e^{(2)} = \alpha x_e$ if any directional bad event whose increase set contains e occurs, and $r_e^{(2)} = 0$ otherwise. Lemma 5.3 there triggers the relevant event, Lemma 5.1 places its increase set inside the cut boundary, and Lemma 5.6 supplies mass at least $1 - \eta$. These facts give (a).

By Lemma 4.44, at most two directional bad events can increase any fixed edge. Thus, without any independence assumption, the union bound gives

$$\mathbb{E}[r_e^{(2)}] \leq 2\frac{5}{2}\eta\alpha x_e = 5\alpha\eta x_e.$$

This proves (b). □

For one-sided cuts, exact cut-excess accounting and a fractional allocation to the two boundary bundles give a stronger repair estimate.

Lemma 4.46 (Fractional endpoint repair for one-sided cuts). *In the setting of [KKO22, Theorem A.12], for $0 < \eta \leq 1/10$ and $\alpha > 0$, there is a random repair vector $s^* : E \rightarrow \mathbb{R}_{\geq 0}$ that has all the pointwise cut guarantees of that theorem and satisfies*

$$\mathbb{E}[s_e^*] \leq 21\alpha\eta x_e$$

for every edge e . This also covers the triangle construction of [KKO22, Lemma A.13]. Both constructions remain supported only on internal bundles between consecutive non-root atoms, and hence only on bottom edges.

Proof. For every nonempty $X \subseteq V \setminus \{u_0, v_0\}$, put $\epsilon_X := x(\delta(X)) - 2 \geq 0$. The degree equations and the forest identity give

$$\Pr[T[X] \text{ is not a tree}] \leq |X| - 1 - \mathbb{E}[|T \cap E(X)|] = \epsilon_X/2.$$

Let S be such a cut side and let $E_j = E(X_j, Y_j) \subseteq \delta(S)$, $j \in \{1, 2\}$, be disjoint edge sets. Suppose that X_j, Y_j are disjoint nonempty subsets of $V \setminus \{u_0, v_0\}$, and put $P_j = X_j \cup Y_j$. If $T[X_j], T[Y_j], T[P_j]$ are all trees, then $|(E_j)_T| = 1$. Consequently,

$$\Pr[|(E_j)_T| \neq 1] \leq (\epsilon_{X_j} + \epsilon_{Y_j} + \epsilon_{P_j})/2,$$

while the cut identity gives

$$x(E_j) = 1 + (\epsilon_{X_j} + \epsilon_{Y_j} - \epsilon_{P_j})/2.$$

Writing $D = \delta(S) \setminus (E_1 \cup E_2)$, the event $|(E_1)_T| = |(E_2)_T| = 1$ and $D_T = 0$ implies $\delta(S)_T = 2$. The union bound and Markov's inequality therefore give the cancellation

$$\begin{aligned} \Pr[\delta(S)_T \neq 2] &\leq \sum_{j=1}^2 (\epsilon_{X_j} + \epsilon_{Y_j} + \epsilon_{P_j})/2 + x(D) \\ &= \epsilon_S + \epsilon_{P_1} + \epsilon_{P_2}. \end{aligned} \tag{7}$$

No independence assumption is used.

Fix a polygon from a non-singleton connected component of one-sided crossing cuts, with atoms $a_0, \dots, a_{m-1}$ and root atom a_0 . Such a polygon has no inside atoms, as in [KKO22, Appendix A]. The relevant sets are its component cuts and those non-root atoms that are themselves η -near-minimum cuts. An interior relevant set triggers when its tree boundary is odd; a boundary relevant set triggers when it is not happy. We first bound these trigger probabilities.

Consider an interior non-atom component cut A . After exchanging left and right if necessary, the geometry in [KKO21, proof of Lemma 4.26] supplies its strict parent B in the right laminar family and a component cut C crossing both A, B on the left. The two boundary links are

$$E_1 = E(A \cap C, C \setminus A), \quad E_2 = E(A \setminus C, B \setminus (A \cup C)).$$

Their sides are nonempty and avoid the distinguished endpoints, and the links are disjoint subsets of $\delta(A)$. Their union sides are $P_1 = C$ and $P_2 = B \setminus C$. Since A, B, C are η -near-minimum cuts, uncrossing B, C gives $\epsilon_{B \setminus C} \leq \epsilon_B + \epsilon_C \leq 2\eta$. Thus Eq. (7) yields

$$\Pr[\delta(A)_T \neq 2] \leq 4\eta.$$

For a relevant interior atom a_i , relevance gives $\epsilon_{a_i} \leq \eta$. The adjacent-pair estimate in [KKO21, Lemma 4.18], with the parameter halved as in [KKO22, Appendix A], gives

$$\epsilon_{a_{i-1} \cup a_i} \leq 6\eta, \quad \epsilon_{a_i \cup a_{i+1}} \leq 6\eta.$$

Indeed, its proof constructs two intervals containing the adjacent pair, each with excess at most 3η , whose intersection is that pair; uncrossing gives 6η . These are the same homogeneous uncrossing inequalities as in the cited proof, whose input cuts have excess at most 2η and whose displayed pair bound is 12η . Both neighbors of an interior atom are non-root atoms. Applying Eq. (7) to its two adjacent links now gives

$$\Pr[\delta(a_i)_T \neq 2] \leq 13\eta.$$

For a leftmost cut or atom S , define $F = E(S, V \setminus (S \cup a_0))$ and $G = E(S, a_0)$. These sets partition $\delta(S)$, and S is happy exactly when $F_T = 1$. For a link $E(X, Y) \subseteq F$ with disjoint nonempty sides avoiding u_0, v_0 , put $P = X \cup Y$. The same forest bound and cut identity give

$$\begin{aligned} \Pr[F_T \neq 1] &\leq (\epsilon_X + \epsilon_Y + \epsilon_P)/2 + x(F) - x(E(X, Y)) \\ &= 1 + \epsilon_S + \epsilon_P - x(G). \end{aligned}$$

For a leftmost component cut L , let L_R be the crossing cut used in [KKO21, proof of Lemma 4.30] and use the link $E(L_R \cap L, L_R \setminus L)$. Here $\epsilon_L, \epsilon_{L_R} \leq \eta$, while

$$x(G) \geq x(E(a_1, a_0)) \geq 1 - \eta$$

by [KKO21, Lemma 4.17], with the same parameter halving. Hence $\Pr[L \text{ is not happy}] \leq 3\eta$. For a relevant boundary atom a_1 , use $X = a_1, Y = a_2$. The bounds $\epsilon_{a_1} \leq \eta, \epsilon_{a_1 \cup a_2} \leq 6\eta$, and the same root-neighbor estimate give $\Pr[a_1 \text{ is not happy}] \leq 8\eta$. The rightmost cases are symmetric. No forest bound is applied to the root atom a_0 .

Let $\mathcal{H}_S$ denote the trigger for a relevant set S . Assign demand $c_S = 4$ to every component cut, $c_S = 13$ to every relevant interior atom, and $c_S = 8$ to every relevant boundary atom. The preceding estimates give $\Pr[\mathcal{H}_S] \leq c_S \eta$; using demand 4 only weakens the boundary component-cut bound 3.

Label the arcs by $1, \dots, m$, so that the internal arc i joins a_{i-1} to a_i , and put $I = \{2, \dots, m-1\}$. For a relevant interval $S = a_{\ell(S)} \cup \dots \cup a_{r(S)-1}$, define its allowed internal endpoints by

$$D(S) := \{\ell(S), r(S)\} \cap I.$$

Every $D(S)$ is nonempty. Indeed, a component cut crosses another member of its connected component, so it is neither an atom nor the union of all non-root atoms. The absence of inside atoms also makes each component cut uniquely determined by its endpoints. The component cuts partition into the laminar families $\mathcal{L}, \mathcal{R}$ of [KKO21, Definition 4.10 and Fact 4.11]. Every member of $\mathcal{L}$ is crossed on the right and hence has $r(S) < m$; every member of $\mathcal{R}$ has $\ell(S) > 1$.

We claim the weighted Hall inequality

$$\sum_{S: D(S) \subseteq W} c_S \leq 21|W| \tag{8}$$

for every $W \subseteq I$. It is immediate when $W = \emptyset$. Otherwise put $n = |W|$, let q count pairs $i, i+1 \in W$, and set

$$b_L = \mathbf{1}_{\{2 \in W\}}, \quad b_R = \mathbf{1}_{\{m-1 \in W\}}, \quad b = b_L + b_R.$$

Then $q \leq n-1$ and $b \leq 2$. At most q relevant interior atoms and b relevant boundary atoms have all their allowed endpoints in W , giving total atom demand at most $13q + 8b$.

Draw the $\mathcal{L}$ -cuts with $D(S) \subseteq W$ as chords on the $n+1$ points $W \cup \{1\}$ in convex position, respecting their cyclic order. Laminarity makes the chords noncrossing, and the graph is simple because endpoints determine the cut. Add the q edges between consecutive original internal arc indices in W and, if $b_L = 1$, the edge $\{1, 2\}$. These are boundary edges of the convex polygon, so they cross no chord. None was present: each would represent an atom, whereas no component cut is an atom. The simple outerplanar edge bound gives

$$N_L + q + b_L \leq 2(n+1) - 3 = 2n - 1,$$

where N_L counts the selected $\mathcal{L}$ -cuts. This also holds when $n = 1$, since a simple graph on two vertices has at most one edge. The symmetric drawing for $\mathcal{R}$ on $W \cup \{m\}$ gives $N_R + q + b_R \leq 2n - 1$. Therefore

$$\begin{aligned} \sum_{S: D(S) \subseteq W} c_S &\leq 4(4n - 2 - 2q - b) + 13q + 8b \\ &= 16n - 8 + 5q + 4b \leq 21n - 5 < 21n. \end{aligned}$$

This proves Eq. (8).

Use source-to-cut capacities c_S , cut-to-endpoint edges for $i \in D(S)$ of capacity $\sum_S c_S$, and endpoint-to-sink capacity 21. For any set of cut nodes, let W be its endpoint neighborhood. Its demand is at most the left side of Eq. (8), so max-flow/min-cut supplies nonnegative allocations $f_{S,i}$ with

$$\sum_i f_{S,i} = c_S, \quad \sum_S f_{S,i} \leq 21, \quad f_{S,i} = 0 \quad \text{if } i \notin D(S).$$

Set $w_{S,i} = f_{S,i}/c_S$. For $e \in E_i := E(a_{i-1}, a_i)$, define

$$s_e^* := \alpha x_e \max\{w_{S,i} \mathbf{1}_{\mathcal{H}_S} : i \in D(S)\},$$

where an empty maximum is zero. Apply this definition to every one-sided polygon and set $s_e^* = 0$ on edges outside their internal bundles, except for the separate triangle construction below. For a

triggered set S , the bundles E_i , $i \in D(S)$, are disjoint subsets of $\delta(S)$, and each has mass at least $1 - 7\eta$ by the near-cycle estimates used in [KKO22, Theorem A.12]. Since $\sum_i w_{S,i} = 1$,

$$s^*(\delta(S)) \geq \alpha \sum_{i \in D(S)} w_{S,i} x(E_i) \geq \alpha(1 - 7\eta).$$

This gives the interior odd-cut guarantee. If a boundary cut is odd and the corresponding polygon side is happy, that cut is not happy and therefore triggers, giving the boundary happy-case guarantee. Also $0 \leq s_e^* \leq \alpha x_e$ on these internal bundles. Bounding the maximum by the sum gives, without independence,

$$\mathbb{E}[s_e^*] \leq \alpha x_e \sum_S w_{S,i} \Pr[\mathcal{H}_S] \leq \alpha \eta x_e \sum_S f_{S,i} \leq 21\alpha \eta x_e.$$

The separate triangle construction has expectation at most $(3/2)(7\eta)\alpha x_e = (21/2)\alpha \eta x_e$ by [KKO22, Lemma A.13], so it satisfies the same bound. An edge belongs to the internal bundles of at most one one-sided polygon, and these bundles join consecutive non-root atoms. Thus there is no further multiplicity, and the bottom-edge support is unchanged. $\square$

We keep the two repair components separate so that the one-sided vector remains supported on bottom edges and can be absorbed by their stronger payment decrease.

Lemma 4.47 (Separated KKO repair vectors). *Fix the exact max-entropy tree law with marginals x and the KKO hierarchy for the η -near-minimum cuts, where $0 < \eta \leq 1.4 \cdot 10^{-9}$, and let $\beta > 0$. Let $E = E_g \dot{\cup} E_b$ be a partition in which every bottom edge belongs to E_g , and let s^{pay} satisfy the deterministic inequalities in Lemma 4.41. Assume in addition that*

$$s_e^{\text{pay}} \geq -\beta x_e \quad (e \in E_g), \quad s_e^{\text{pay}} = 0 \quad (e \in E_b).$$

Then there are nonnegative random vectors $r^{(2)}, r^{(1)} : E \rightarrow \mathbb{R}_{\geq 0}$ such that

(a) if an η -near-minimum cut S is crossed on both sides and $\delta(S)_T$ is odd, then

$$r^{(2)}(\delta(S)) \geq (2 + \eta)\beta;$$

(b) if an η -near-minimum cut S in the KKO one-sided classification is not the root and $\delta(S)_T$ is odd, then

$$s^{\text{pay}}(\delta(S)) + r^{(1)}(\delta(S)) \geq 0;$$

(c) for every edge,

$$\mathbb{E}[r_e^{(2)}] \leq 5 \frac{2 + \eta}{1 - \eta} \eta \beta x_e, \quad \mathbb{E}[r_e^{(1)}] \leq 21 \frac{2 + \eta}{1 - 7\eta} \eta \beta x_e;$$

(d) pointwise, $r^{(1)}$ is supported only on bottom edges, and hence only on E_g .

Proof. Apply Lemma 4.45 with parameter

$$\alpha_2 := \frac{2 + \eta}{1 - \eta} \beta$$

and apply Lemma 4.46, with its fractional endpoint allocation and separate triangle construction, with parameter

$$\alpha_1 := \frac{2 + \eta}{1 - 7\eta} \beta$$

to obtain $r^{(2)}$ and $r^{(1)}$. Lemma 4.45 gives (a) and the first expectation bound in (c). Lemma 4.46 retains the pointwise one-sided guarantees and gives

$$\mathbb{E}[r_e^{(1)}] \leq 21\alpha_1\eta x_e,$$

which is the second expectation bound in (c).

We verify (b) without invoking [KKO22, Theorem B.3] as a black box. If S is a hierarchy cut whose parent is a degree cut, Lemma 4.41-(b) gives

$$s^{\text{pay}}(\delta(S)) \geq 0$$

whenever $\delta(S)_T$ is odd.

For an interior cut or atom of a one-sided near-cycle component, Lemma 4.46 gives

$$r^{(1)}(\delta(S)) \geq \alpha_1(1 - 7\eta) = (2 + \eta)\beta \geq -s^{\text{pay}}(\delta(S)).$$

The last inequality is exactly where the coordinate hypothesis is used:

$$s^{\text{pay}}(\delta(S)) \geq -\beta x(\delta(S)) > -(2 + \eta)\beta.$$

The same conclusion holds for a leftmost or rightmost cut when the corresponding side is happy. Suppose instead, without loss of generality, that S is leftmost and its polygon P is not left happy. Let S' be the union of the non-root atoms of P and define

$$F := \delta(S) \setminus \delta(S').$$

By [KKO22, Lemma 2.10], $x(F) \geq 1 - d/2$. Every edge $e \in F$ joins distinct children of P , and hence satisfies $p(e) = P$. Thus Lemma 4.41-(a) applies. Since $A \cup F \subseteq \delta(S)$ and the worst possible contribution from the remaining C -edges is $(s^{\text{pay}})^-(C)$,

$$s^{\text{pay}}(\delta(S)) + r^{(1)}(\delta(S)) \geq s^{\text{pay}}(A) + s^{\text{pay}}(F) + (s^{\text{pay}})^-(C) \geq 0.$$

The rightmost case is symmetric. If the parent is a triangle cut P with children a_1, a_2 , use $F = E(a_1, a_2)$. Lemma A.13 of [KKO22] gives the happy-case repair. For the non-happy case, the identity

$$x(\delta(a_1)) + x(\delta(a_2)) = 2x(F) + x(\delta(P))$$

and the subtour constraints for a_1, a_2 , together with $x(\delta(P)) \leq 2 + d$, give $x(F) \geq 1 - d/2$. Therefore Lemma 4.41-(a) applies by the same displayed calculation. These are Types 4 and 5 in the proof of [KKO22, Theorem B.3], and together with the degree-parent and interior cases they exhaust the one-sided classification.

For (d), the fractional allocation in Lemma 4.46 assigns repair only to sets $E(a_{i-1}, a_i)$ between consecutive non-root atoms; the triangle construction assigns repair only to $E(a_1, a_2)$. In each case the atoms are distinct children of the same near-cycle or triangle cut. Their smallest common hierarchy ancestor is therefore their parent, so these edges are bottom edges. The hypothesis of the lemma makes every such edge good. $\square$

Root parity. The root $R := V \setminus \{u_0, v_0\}$ is handled by parity. Indeed, $x^0(\delta(u_0)) = x^0(\delta(v_0)) = 2$ and $x_{e_0}^0 = 1$, so $x(\delta(u_0)) = x(\delta(v_0)) = 1$. Every spanning tree has positive degree at both vertices, so the two degree random variables, each having expectation one, are equal to one almost surely. Connectivity prevents their two unique tree edges from joining u_0 directly to v_0 , since that would isolate $\{u_0, v_0\}$ from the remaining vertices. Hence the two unique incident edges both cross $\delta(R)$, and $\delta(R)_T = 2$ almost surely. Thus the root never creates an odd-cut constraint.

4.7 Cutwise payment credit and centering

Lemma 4.48 (Aggregate expected credit on one-sided cuts). *Fix one row in Definition 4.2, a threshold $0 < u \leq H$, its hierarchy with error $d \leq d_0$, and amplitude $\beta > 0$. Put*

$$\begin{aligned} \alpha &:= 2d_0, & c &:= 1 - \chi, & a^{(4)} &:= p(1 - c(1 + \alpha)), \\ b(u) &:= a_{\text{bot}} - R_1(u)u, & m(u) &:= \min\{a, b(u)\}, & V &:= 1/2 + k_{\text{good}}h, \end{aligned}$$

and suppose $b(u) > 0$. Define

$$\begin{aligned} A^{(0)}(u) &:= \frac{m(u)(2 - \alpha(2 + d_0))}{1 + m(u)/p}, \\ A^{(1)}(u) &:= a^{(4)}(3/2 - h - V) + m(u)V, \\ A(u) &:= \min\{A^{(0)}(u), A^{(1)}(u)\}, \quad B(u) := (1 - d_0)b(u). \end{aligned}$$

Let $W := s^{\text{pay}} + r^{(1)}$. On good edges, $m(u)\beta x_e \leq -\mathbb{E}[W_e] \leq \beta x_e$, while $W_e = 0$ on bad edges. Every degree-parent cut has expected credit at least $\beta A(u)$; every other relevant non-root one-sided cut has expected credit at least $\beta B(u)$.

Proof. Write $m = m(u)$. The payment bound and bottom support of $r^{(1)}$ give credit at least a on good top edges and $b(u)$ on bottom edges, in units of βx_e . The coordinate bound gives the upper bound one. Bad edges have neither payment nor one-sided repair. Zero-mass edges may be omitted.

For a child v of a degree parent, write

$$U_v := x(\delta^\uparrow(v)), \quad D_v := x(\delta^\uparrow(v) \cap E_b), \quad I_v := x(\delta^\rightarrow(v) \cap E_g), \quad \ell_v := \frac{B_v}{\beta U_v Z_v}.$$

Here B_v is the normalized upward burden in Lemma 4.38; its actual expectation is pB_v . When $U_v = 0$, set $\ell_v = 0$. Matching conservation then makes every corresponding $m_{\mathbf{e},v}$ zero. Bad upward mass includes distinguished-endpoint edges, which are never reduced.

For every child, $\ell_v \leq F_v$. If $U_v \geq \sigma$, this follows from the ancestor estimate. Otherwise $F_v = 1$ and $B_v \leq \beta U_v$, since $sq_0 < 1$. If the parent has at least four children, the stronger bound $\ell_v \leq cF_v$ holds: the ancestor estimate handles $U_v \geq \sigma$, and $U_v < \sigma < \epsilon_F$ gives $Z_v = 2$ and $\ell_v \leq 1/2 < c$.

Suppose that v is incident to an internal bad bundle. Its parent has at least four children by Lemma 4.32, and $U_v \leq V < 1 - \epsilon_F$. We claim, for $U_v > 0$, that

$$\ell_v \leq cF_v(1 - D_v/U_v).$$

If $U_v < \epsilon_F$, then $F_v = 1$, $Z_v = 2$, and $B_v \leq \beta(U_v - D_v)$. Otherwise $F_v = 1 - \epsilon_B$. The fractional case in the ancestor proof bounds each top conditional odd probability by 0.92, and Lemma 4.35 bounds every upward bottom probability by q_{bad} . The exact row comparisons

$$0.92 \leq c(1 - \epsilon_B), \quad sq_{\text{bad}} \leq c(1 - \epsilon_B), \quad pc(1 - \epsilon_B) \geq a, \quad a^{(4)} \geq a$$

give $B_v \leq \beta cF_v(U_v - D_v)$. Division by $U_v Z_v$, with $Z_v \geq 1$, proves the claim. This includes $U_v = \epsilon_F$.

For a good horizontal bundle $\mathbf{e} = (v, w)$, the exact payment formulas in [KKO21, Eqs. (33) and (37)] and localized matching capacity give

$$\begin{aligned} -\frac{\mathbb{E}[s^{\text{pay}}(\mathbf{e})]}{\beta} &= p(x_{\mathbf{e}} - m_{\mathbf{e},v}\ell_v - m_{\mathbf{e},w}\ell_w) \\ &\geq a^{(4)}x_{\mathbf{e}} + pcF_v \frac{D_v}{U_v} m_{\mathbf{e},v} \end{aligned}$$

when v is incident to a bad bundle. We retain both the baseline credit and the additional bad-mass credit. Summing over its incident good bundles and using $\sum_{e \ni v} m_{e,v} = U_v Z_v$ gives

$$-\frac{\mathbb{E}[s^{\text{pay}}(\delta^{\rightarrow}(v) \cap E_g)]}{\beta} \geq a^{(4)} I_v + pc F_v Z_v D_v. \quad (9)$$

For $U_v = 0$, the second term is zero and the same bound follows from the baseline bound. No independence between reduction and increase events is used.

Let $T(v) := -\mathbb{E}[W(\delta(v))]/\beta$. In the bad-incidence case, internal bad mass is at most $1/2 + h$, so $I_v \geq 3/2 - h - U_v$. The repair is zero on horizontal top bundles; the upward good mass has credit at least $m(U_v - D_v)$. Hence

$$\begin{aligned} T(v) &\geq a^{(4)} I_v + m(U_v - D_v) + pc F_v Z_v D_v \\ &\geq a^{(4)} (3/2 - h - U_v) + m U_v \geq A^{(1)}(u). \end{aligned}$$

The second step uses $F_v Z_v \geq 1 - \epsilon_B$ and the displayed row comparisons. The last uses $U_v \leq V$ and $a^{(4)} \geq a \geq m$.

If v has no internal bad bundle, its good boundary mass is at least $2 - D_v$, giving $T(v) \geq m(2 - D_v)$. The neighboring surpluses $F_w - \ell_w$ are nonnegative even for a three-child parent. Thus the same exact payment identity, now using $B_v/\beta \leq U_v - D_v$ and $F_v Z_v \geq 1$, gives

$$T(v) \geq p(F_v U_v Z_v - B_v/\beta - \alpha I_v) \geq p(D_v - \alpha(2 + d_0)).$$

All complementary edges have nonnegative expected credit. Multiply the good-mass bound by p and this bound by m , then add and divide by $p + m$. The terms in D_v cancel, giving $T(v) \geq A^{(0)}(u)$.

For a remaining one-sided near-cycle cut, let S' be the union of its component's non-root atoms. By [KKO22, Lemma 2.10], $x(\delta(S) \cap \delta(S')) \leq 1 + d_0$; every edge in the difference is bottom, so the bottom mass is at least $1 - d_0$. For a triangle parent Q with children a_1, a_2 , the cut identity

$$2x(E(a_1, a_2)) = x(\delta(a_1)) + x(\delta(a_2)) - x(\delta(Q))$$

gives bottom mass at least $1 - d_0/2 \geq 1 - d_0$. The bottom credit therefore gives $T(S) \geq B(u)$. These cases exhaust the one-sided classification. The root has almost-surely even boundary and is excluded. $\square$

Lemma 4.49 (Uniformization after mixing cutwise credit). *Fix $0 < u \leq H$, one geometric hierarchy for this threshold, and amplitude $\beta > 0$. For each row with $b_j(u) > 0$, use the preceding lemma on that hierarchy. Choose nonnegative weights w_j summing to one, supported on such rows, and put*

$$\begin{aligned} \bar{m} &:= \sum_j w_j m_j(u), & A_w &:= \sum_j w_j A_j(u), & B_w &:= \sum_j w_j B_j(u), \\ M_w &:= \min\{A_w, B_w\}, & C_w &:= \frac{M_w}{(1 - \bar{m})(2 + u) + M_w}, & \kappa_w &:= C_w - R_2(u)u. \end{aligned}$$

There is a random vector $z^{(u)}$ such that $z_e^{(u)} \geq -\beta x_e$ on every edge, $z^{(u)}(\delta(S)) \geq 0$ on every relevant odd u -near-minimum cut, and $\mathbb{E}[z_e^{(u)}] \leq -\kappa_w \beta x_e$.

Proof. The hierarchy construction in Lemma 4.42 uses only x , the distinguished edge, and the near-minimum threshold. Fix it, its polygon representations, and its cut classes before choosing

the parameter rows. In particular, a cut has the same degree-parent or remaining one-sided classification in every row. The good sets, matchings, and reduction events may differ. Realize their auxiliary laws conditionally on the same exact-law tree, preserving each row's synchronized pairs.

Put $W := \sum_j w_j W_j$, $M := 2 + u$, and

$$D := (1 - \bar{m})M + M_w, \quad C := M_w/D, \quad \lambda := M/D.$$

Every selected row has $0 < m_j < 1$ and positive cut credits. Thus $D > 0$, $0 < C < 1$, and

$$\lambda(1 - \bar{m}) + C = 1, \quad \lambda M_w = CM.$$

On a good edge in row j , $(W_j)_e - \mathbb{E}[(W_j)_e] \geq -(1 - m_j)\beta x_e$; on a bad edge both terms are zero, so this bound still holds. Averaging gives the bound $W_e - \mathbb{E}[W_e] \geq -(1 - \bar{m})\beta x_e$ without requiring the good sets to agree. Define

$$q := \lambda(W - \mathbb{E}[W]) - C\beta x, \quad z^{(u)} := q + r^{(2)}.$$

Then $q_e \geq -\beta x_e$ and $\mathbb{E}[q_e] = -C\beta x_e$.

On a relevant odd one-sided cut, each $W_j(\delta(S)) \geq 0$. The shared cut classification gives expected credit at least βA_w or βB_w , hence at least βM_w . Therefore

$$q(\delta(S)) \geq \beta(\lambda M_w - Cx(\delta(S))) \geq \beta(\lambda M_w - CM) = 0.$$

On a relevant odd two-sided cut, the coordinate bound loses at most $M\beta$. Add one two-sided repair from Lemma 4.47, with amplitude β , to supply this amount. Its construction depends only on the common hierarchy and tree, not on the chosen row; no separate copy is charged per row. It is nonnegative and costs at most $R_2(u)u\beta x_e$ in expectation. This proves all three assertions.

The expectations and weights are analytical quantities under the exact tree and auxiliary laws. The algorithm only samples a tree and computes its minimum join; it does not construct these centered witnesses. The implementation comparison is applied once after all layers are combined. $\square$

4.8 Layering and the final certificate

Lemma 4.50 (Finite-layer combination). *Let $0 = u_0 < u_1 < \dots < u_N = H$, and define*

$$\beta(u) := \frac{u}{4 + 2u}, \quad \Delta_i := \beta(u_i) - \beta(u_{i-1}).$$

Suppose that, on the same sampled tree, V_i satisfies $(V_i)_e \geq -\Delta_i x_e$ and $\mathbb{E}[(V_i)_e] \leq -\kappa_i \Delta_i x_e$. Assume also that $V_i(\delta(S)) \geq 0$ for every odd tree cut not containing e_0 with $x(\delta(S)) < 2 + u_i$. Then $Z := \sum_{i=1}^N V_i$ satisfies

$$Z_e \geq -\beta(H)x_e, \quad \mathbb{E}[Z_e] \leq -\sum_{i=1}^N \kappa_i \Delta_i x_e,$$

and $x(\delta(S))/2 + Z(\delta(S)) \geq 1$ for every such odd cut, without a restriction on its x -value.

Proof. The coordinate and expectation bounds follow by telescoping and linearity. For an odd cut not containing e_0 , put $\rho = x(\delta(S)) - 2 \geq 0$ and $j := \max\{i \in \{0, \dots, N\} : u_i \leq \rho\}$. Layers above j are nonnegative on this cut, while the other layers give

$$\frac{x(\delta(S))}{2} + Z(\delta(S)) \geq 1 + \frac{\rho}{2} - \beta(u_j)(2 + \rho) \geq 1,$$

because $\beta(u_j) \leq \beta(\rho)$. This includes grid endpoints and $\rho > H$. Auxiliary constructions can be jointly realized conditionally on the common tree. Neither independence between layers nor persistence of an edge's type is needed. $\square$

Lemma 4.51 (Mixed-row layered slack certificate). *Use Definition 4.2. There is a random vector Z with $Z_e \geq -\beta(H)x_e$ and*

$$\frac{x(\delta(S))}{2} + Z(\delta(S)) \geq 1$$

*for every cut avoiding e_0 with odd tree boundary. Moreover, $\mathbb{E}[Z_e] \leq -B_*x_e$, where*

$$B_* := \frac{40826275298683}{250000000025000000006250000000000} > 1.6330510103142689892774 \cdot 10^{-18}.$$

Proof. Set $u_i := iH/N$ for $0 \leq i \leq N$, with $H = 10^{-9}$ and $N = 50000$. The hierarchy domain holds because $H < 1.4 \cdot 10^{-9}$ and $14H < d_0$. Every local estimate in the four rows holds throughout $d \leq d_0$; no former row-specific absorption endpoint is required.

Use the following explicit support schedule. A singleton has weight one. For a pair $\{j, k\}$, put $\xi_j := A_j(u_i) - B_j(u_i)$ and use

$$w_j := \frac{\xi_k}{\xi_k - \xi_j}, \quad w_k := \frac{-\xi_j}{\xi_k - \xi_j}.$$

The exact checks give $\xi_j \xi_k < 0$ in every paired layer, so both weights are positive and their sum is one. They satisfy $A_w = B_w$. In every selected row $b_j(u_i) > 0$, and the coefficient $\kappa_i := \kappa_w(u_i)$ from Lemma 4.49 is positive. Define $k_i := \lfloor 10^{18} \kappa_i \rfloor$. The support schedule and exact block sums are

First layer	Last layer	Selected rows	Sum of k_i
1	24341	$\{1\}$	231676246734728
24342	34370	$\{1, 2\}$	58007995441884
34371	44430	$\{2, 3\}$	31434362696741
44431	49455	$\{3, 4\}$	5379274277092
49456	50000	$\{4\}$	112323239019

All computations use rational numbers, including the weights and floors; Section 4.9 specifies their evaluation. Adding the five blocks gives

$$\sum_{i=1}^{50000} k_i = 326610202389464.$$

No optimality assertion for the support schedule is needed.

Since $\beta'(u) = 1/(2+u)^2$, the layer amplitudes satisfy

$$\Delta_i := \beta(u_i) - \beta(u_{i-1}) \geq \frac{H}{N(2+H)^2}.$$

Consequently

$$\sum_i \kappa_i \Delta_i \geq \frac{10^{-18}H}{N(2+H)^2} \sum_i k_i = B_*. \quad (10)$$

In particular, the exact comparison gives the implementation reserve

$$B_* - 10^{-28} - 1.63305 \cdot 10^{-18} > 1.0102142 \cdot 10^{-24}.$$

At layer i , use amplitude Δ_i and mix the selected rows on one fixed hierarchy. Jointly realize all the finite auxiliary laws conditionally on the same sampled tree. Lemma 4.49 supplies the coordinate and odd near-minimum-cut bounds, so Lemma 4.50 proves all-cut feasibility and the asserted expectation bound. Different layers need not have nested hierarchies or persistent edge types. Since $\beta(H) < 1/2$, the join coordinates are nonnegative. The tour algorithm does not compute the mixtures, auxiliary events, or centered expectations. $\square$

4.9 Exact arithmetic details

We give the arithmetic rules, finite ranges, and intermediate margins needed to reproduce the numerical comparisons in the proof. The inputs are precisely the four rows of Definition 4.2. All subsequent quantities are computed from those rows and the formulas cited below; the displayed rounded margins are conclusions, not replacement inputs. The probability and conditioning arguments establish the inequalities to be checked. The arithmetic below verifies their scalar premises, not the underlying probabilistic assertions.

Rational arithmetic and exponential bounds. Represent a terminating decimal with k decimal places by its integer numerator divided by 10^k . Perform addition, multiplication, division, and comparisons over the rationals. Before dividing by a quantity whose sign is used, check its strict positivity. For integers a and $b > 0$, the rounding operations used in Definition 4.34 are exactly

$$\lfloor a/b \rfloor_{20} = \frac{\lfloor 10^{20}a/b \rfloor}{10^{20}}, \quad \lceil a/b \rceil_{20} = -\frac{\lfloor -10^{20}a/b \rfloor}{10^{20}}.$$

Thus downward and upward rounding enclose the input with width less than 10^{-20} , including when the input is negative.

For $0 \leq x < 42$, define

$$t_0 = 1, \quad t_j = xt_{j-1}/j \quad (1 \leq j \leq 40), \quad L(x) = \sum_{j=0}^{40} t_j, \quad U(x) = L(x) + \frac{xt_{40}}{41(1-x/42)}.$$

The tail terms after t_{41} have successive ratios at most $x/42$. Consequently

$$L(x) \leq e^x \leq U(x), \quad E_-(x) := 1/U(x) \leq e^{-x} \leq E_+(x) := 1/L(x).$$

These are rational functions, not floating-point exponential evaluations. Use E_- in a positive lower bound and E_+ in a positive upper bound; reverse the choices after a negative coefficient. For an interval $x \in [a, b] \subset [0, 42)$, use $e^{-x} \in [E_-(b), E_+(a)]$. For general rational interval operations, addition uses the endpoint sums, multiplication uses the minimum and maximum of the four endpoint products, and reciprocation sends $[a, b]$ to $[1/b, 1/a]$ when $a > 0$. These rules preserve inclusion. No intermediate rounding is required except the explicitly prescribed parity rounding and final floors. The definitions of T_{18} and the degree-16 parity initialization retain their stated shorter sums. Also, $c(x) = 1 - x + x^2/2 - x^3/6 \leq e^{-x}$ on $[0, 1]$.

Probability comparisons. For a binomial atom at an exact mean M , use

$$\mathcal{B}_{n,k}(M) := \binom{n}{k} (M/n)^k (1 - M/n)^{n-k}, \quad \mathcal{F}_{n,k}(M) := \sum_{j=0}^k \mathcal{B}_{n,j}(M).$$

Terms outside $0 \leq j \leq n$ are zero. The finite dimension ranges and mean endpoints are specified in Lemma 4.3; substitute them in these two expressions. The logarithmic inequalities in its proof cover

every larger dimension. In particular the CDF-at-two comparisons use $3 \leq n \leq 99$ at $M = 2.01$ and $3 \leq n \leq 9$ at $M = 2.5$, followed by the respective bounds valid for all $n \geq 100$ and $n \geq 10$. A finite dimension calculation is not used in place of those arguments.

Here are the remaining event evaluations, with all outer conditioning factors retained.

- (i) For the small non-half event in Lemma 4.25, enumerate $i = 0, \dots, 127$ on its displayed closed cover $[q_s, 0.04]$. Compute $T_i, \ell_i, \delta_i, v_i$ at the specified endpoints, then evaluate

$$o_s l_i \min\{0.015, F_i(v_i), F_i(0.1)\}.$$

Check

$$\begin{aligned} 0 < \delta_i < 0.02, \quad 0 < v_i < 0.1, \quad 1 - E_+(\ell_i) > 0.38, \\ \ell_i - 0.2 - T_{0.36}(0.1) > 0, \quad \Phi(0.1) > 0, \quad W_i(v_i) > 0. \end{aligned}$$

These checks and concavity justify the entire inner interval, not just its endpoints. Take the minimum with the displayed $q \geq 0.04$ bound.

- (ii) For the large non-half event in the same lemma, take the minimum of the two endpoint expressions on $[q_l, 0.025]$ and the bounds on $[0.025, 0.12]$ and $[0.12, 1]$. Replace each positive exponential factor by E_- . The derivative sign proved there excludes an interior minimum in the first range.
- (iii) For Lemma 4.21, evaluate exactly W_0, W_1, W_2 and all displayed derivative lower bounds with $Q = 0.04$, $Q_1 = 0.1$, $c_3 = 0.248$, and $\eta = 0.37$. For the direct mixed event use the final displayed expression in Lemma 4.25.
- (iv) For Lemma 4.31, perform exactly three separator iterations, checking that their denominators are positive and the iterates decrease. Evaluate the six conditioning probabilities and every one of the seven subset rows in each of its two tables. For each subset size $k = 1, 2, 3$, take the minima defining d_k, e_k , checking their positivity, and then the minimum of the four stated products. Multiply by $P_i(1 - u_i)$ and respectively $E_-(2)/2$ or $E_-(1)/2$. Neither a root finder nor a search over possible laws is used.
- (v) For the polygon event put $q_M = (\epsilon_M - 2d_0)/2$ and evaluate $\frac{27}{4}(0.94)(q_M - 10d_0)^2(1 - q_M)(1 - 3d_0/2)$, as in Lemma 4.25. Its target is $P = sp$, not p .

The following entries are strict lower bounds on the resulting event probability minus its target, in units of 10^{-9} . They are obtained by downward rounding only after the preceding exact calculations. The target is p_j except in the last row, where it is $s_j p_j$.

Event	Row 1	Row 2	Row 3	Row 4
Small non-half	2.42276	1.00656	1.00044	1.00853
Large non-half	1531.80871	1014.13875	538.05837	312.92870
Oriented window	483.19948	536.83280	587.67850	612.20426
Direct mixed	1.00880	1.00658	1.00047	1.00851
Paired, low branch	1.00900	1.00677	1.00053	1.00846
Paired, high branch	101841.53034	103346.45930	104710.10996	105347.83240
Polygon	1.01561	1.01163	1.02754	1.01507

All seven rows therefore satisfy the required thinning inequalities. The auxiliary bounds for good-half classification, competing bundles, and the polygon charge are the explicit endpoint comparisons in Lemmas 4.17, 4.18, 4.26, and 4.39, evaluated by the same rules.

Bad-endpoint parity. For each parameter row, first evaluate the eight stages $j = 0, \dots, 7$ in Definition 4.34. Next enumerate the 128 cells in Lemma 4.35. For each cell compute, in order, $D_i, \delta_i, L_i, q_i, z_i, \tilde{b}_i$ and its eight-stage bootstrap. Check the following before accepting that cell:

$$\begin{aligned} D_i > 0, \quad 0 < \delta_i < 1, \quad 2 - 2a_i + 2\rho + b_i\Delta_0 < 1.2, \quad 0 < q_i < 0.1, \\ E_-(1.2)/2 > \delta_i, \quad 1 - E_+(L_i) > 0.39, \quad L_i - 0.2 - T_{0.36}(0.1) > 0, \\ \Phi(0.1) > 0, \quad W_i(q_i) > 0, \quad z_i > 0, \quad \tilde{b}_i > 0. \end{aligned}$$

At every bootstrap stage check that the split argument belongs to $(0, 1)$, the rounded rank-three lower bound is positive, and the next B bound does not increase. Both the initial and cell bootstraps stop after the prescribed eighth stage; no convergence test is used. Finally take the minimum of $D_i - 2\tilde{B}_7 - \tilde{Q}_7$ over all 128 cells. As intermediate checks, the first three numerical columns below are upward bounds; the last is a strict downward bound, in units of 10^{-6} . The rounded columns are not fed back into the recurrences.

Row	B_7	Q_7	Δ_0	Contradiction margin
1	0.002796652030	0.069090439372	0.168459217591	1.00007
2	0.002813124012	0.069276493512	0.168591242545	1.00008
3	0.002827727366	0.069440901017	0.168708035849	1.00004
4	0.002834461747	0.069516548145	0.168761760561	0.99996

In both 128-cell covers, verify that the first left endpoint and last right endpoint are the prescribed endpoints and that each right endpoint equals the next left endpoint. The mean and tail enclosures used in a cell hold on that whole cell. These identities, together with the analytic rank estimates, provide coverage of the continuous domains.

Child parity, ancestor credit, and payment. Use the five values y_l in Lemma 4.38. With $e_{\text{ch}} = \epsilon_M/2 + 40d_0$, the precise upper enclosure is

$$\hat{Q}_2(y) = \frac{1 + (1 - y + e_{\text{ch}})^2 E_+(2(1 - y))}{2} + 14d_0.$$

Check the child-parity comparison in Lemma 4.37 using E_- on its right side. Then compute $c_y = 1 - s\hat{Q}_2(y)$ and $\kappa_l = \min\{\varepsilon_l(1 - \varepsilon_l), c_{y_l}\}$. Check $y_l > e_{\text{ch}}$, $\varepsilon_l^2 > 2d_0$, $0 < \kappa_1 \leq \dots \leq \kappa_5$, and the three last-level comparisons at $z = 0.35$ in that proof. The exact values give

$$10^6 \left(\sum_{l=1}^5 (\kappa_l - \kappa_{l-1})(0.35 - y_l - 2d_0) - \chi(1 + d_0) \right) > \begin{cases} 22.00457 & \text{in row 1,} \\ 192.96205 & \text{in row 2,} \\ 346.31936 & \text{in row 3,} \\ 417.51539 & \text{in row 4.} \end{cases}$$

Each κ_l is evaluated before any rounding. Monotonicity in z and d then covers $0.35 \leq z \leq 0.99$. The adjacent ranges use the endpoint expressions and the derivative of F_d already printed in Lemma 4.38. This is a finite sum of credit increments, so overlapping thresholds do not count the same credit twice.

The remaining matching and payment checks are direct substitutions in Lemmas 4.32, 4.38, and 4.40. In particular retain $q_{\text{bad}} < q_0$, $sq_0 < 1$, $a_{\text{bot}} > a > 0$, $1 - \sigma - 2d_0 > 0$, the comparisons

$J_1 > J_2, J_3$, and all row interfaces in Lemma 4.48. The proofs specify the worst endpoint whenever d varies in $[0, d_0]$; one must use those monotonicity arguments, rather than infer uniformity from a single numerical substitution.

Layer sum. Use the exact parameters in Definition 4.2, put $d = d_0$, and evaluate $u_i = iH/N$ for $1 \leq i \leq N$. First compute each row's $\chi_j, \sigma_j, \rho_{\text{sm},j}, a_j, a_{\text{bot},j}$ from that definition, and $a_j^{(4)}, V_j$ from Lemma 4.48. Check

$$1 - \sigma_j - 2d > 0, \quad 0 < a_j \leq a_j^{(4)}, \quad a_j < a_{\text{bot},j}, \quad V_j < 0.9.$$

Let S_i be the selected row set in the five-block table of Lemma 4.51. For each $j \in S_i$, evaluate

$$b_j(u_i), m_j(u_i), A_j^{(0)}(u_i), A_j^{(1)}(u_i), A_j(u_i), B_j(u_i)$$

in this order, using Lemma 4.48. Check $b_j(u_i) > 0$, $0 < m_j(u_i) < 1$, and $A_j(u_i), B_j(u_i) > 0$. The denominators $1 - u_i$ and $1 - 7u_i$ are positive since $u_i \leq H$.

Next evaluate the weights in Lemma 4.51, checking $\xi_j \xi_k < 0$ in every paired layer. Their sum is one and $A_w(u_i) = B_w(u_i)$ in that case. Compute $\bar{m}_i := \bar{m}$, $A_w(u_i), B_w(u_i), M_i := M_w(u_i), C_w(u_i)$, and $\kappa_i := \kappa_w(u_i)$ from Lemma 4.49. With

$$D_i := (1 - \bar{m}_i)(2 + u_i) + M_i, \quad R_i := \frac{5(2 + u_i)u_i}{1 - u_i},$$

one has $D_i > 0$ and $\kappa_i = (M_i - R_i D_i)/D_i$. Check positivity by the exact comparison $M_i > R_i D_i$.

Write $\kappa_i = n_i/d_i$ with positive integers n_i, d_i . Euclidean division determines the unique integers k_i, e_i satisfying

$$10^{18}n_i = k_id_i + e_i, \quad 0 \leq e_i < d_i.$$

Consequently

$$\frac{k_i}{10^{18}} \leq \kappa_i < \frac{k_i + 1}{10^{18}}, \quad k_i = \lfloor 10^{18}\kappa_i \rfloor.$$

This rule includes exact integer endpoints and does not require a reduced fraction. Starting from zero, add the k_i in increasing order of i within each block. All positivity, weight, and floor comparisons are checked before including a summand, with no rounding before the Euclidean division. This gives the five block sums printed in Lemma 4.51; no additional data are required. Their sum, Eq. (10), and clearing positive denominators give the stated B_* and implementation reserve.

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